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Understanding Machine Learning Vectors: From Confusion Matrices to Byzantine-Robust Federated Learning

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You ask me if an ordinary person—by studying hard—would get to be able to imagine these things like I imagine. Of course. I was an ordinary person who studied hard. There's no miracle people. It just happens they got interested in this thing, and they learned all this stuff. They're just people. There's no talent or special miracle ability to understand quantum mechanics or a miracle ability to imagine electromagnetic fields that comes without practice and reading and learning and study. So if you take an ordinary person who's willing to devote a great deal of time and study and work and thinking and mathematics, then he's become a scientist.

— Richard Feynman, *Fun to Imagine: Ways of Thinking*, BBC, August 1983

La découverte est le privilège de l'enfant. C'est du petit enfant que je veux parler, l'enfant qui n'a pas peur encore de se tromper, d'avoir l'air idiot, de ne pas faire sérieux, de ne pas faire comme tout le monde. Il n'a pas peur non plus que les choses qu'il regarde aient le mauvais goût d'être différentes de ce qu'il attend d'elles, de ce qu'elles devraient être, ou plutôt : de ce qu'il est bien entendu qu'elles sont.

— Alexandre Grothendieck, *Récoltes et Semailles*

La bonne méthode, c'est de partir pour faire un tour à pied et d'essayer de se débrouiller avec ce qu'on a, pour justement, en y réfléchissant, créer dans le cerveau ces images mentales. Et peu importe la complexité du problème. Peu importe le caractère impossible au départ de cette création, c'est en séchant, et justement, en s'appropriant progressivement des objets mentaux qui vont correspondre au problème qu'on va progresser.

— Alain Connes, *L'imagination à l'infini*, France Culture, juillet 2018

Abstract

This work studies classification in complex settings. The first part focuses on confusion matrices and aims to characterize model behavior from its outputs. The second part addresses Byzantine-robust federated learning, examining how to defend a distributed system based on local model updates. In both cases, we rely on machine learning vectors—model outputs and gradients—and investigate how to extract meaningful information from them.

Confusion matrix. The confusion matrix is a key tool for understanding, evaluating, and improving model behavior. It is a square matrix built from label–prediction pairs, using test labels and model predictions. Unlike aggregate metrics such as accuracy, it decomposes errors at the class level and provides a detailed view of model behavior.

In heterogeneous settings, where the label distributions of the training and test sets are imbalanced, the confusion matrix becomes harder to interpret. This imbalance biases the matrix entries and can obscure errors arising from similarities between classes.

Multi-label settings, where observations may belong to several classes, and soft-label settings, where observations are represented by non-negative vectors encoding degrees of membership or confidence, better capture real-world complexity. However, extending the confusion matrix to these settings is not straightforward.

As a result, several competing extensions coexist. Most existing works visualize these matrices on benchmark datasets to interpret model behavior or compare them qualitatively with alternative formulations. However, these comparisons lack a clear quantitative criterion, making it unclear which method should be preferred in practice.

Moreover, no existing approach provides a unified formulation of the confusion matrix for single-label, multi-label, and soft-label settings, limiting both the consistency and the scope of machine learning evaluation.

Contributions to the Confusion Matrix. Our main contributions are:

- Propose a principled normalization of the confusion matrix to better recover true class similarity in heterogeneous settings;
- Establish connections between normalization methods, importance sampling, and latent class representations;
- Introduce an optimal transport-based formulation extending confusion matrices to multi-label and soft-label settings, providing a unified framework and linking existing approaches;
- Introduce an experimental framework for evaluating whether confusion matrix extensions accurately recover true model confusion in complex classification settings.

These results are based on the published papers [32, 29] and the under-review paper [28].

Federated learning. Federated learning enables multiple data holders (workers) to collaboratively train a model without sharing their data. Each worker locally updates the global model and sends its update (typically a gradient) to a central server. The server aggregates these updates, updates the model, and broadcasts it back to the workers; this process is repeated at each training round.

In adversarial settings, faulty or malicious (Byzantine) workers may disrupt the training process. Byzantine-robust federated learning aims to mitigate their impact. A common strategy is to discard outlier gradients, based on the assumption that Byzantine gradients deviate more from honest gradients than honest gradients deviate from each other.

However, data heterogeneity—such as label skew, where workers observe different label distributions—also misaligns local updates. As a result, honest workers may appear as outliers, rendering these defense strategies ineffective.

Contributions to Federated Learning. The main contributions on Byzantine-robust federated learning under label skew are:

- Introduce a lightweight local sampling-based defense against Byzantine attacks that does not require validation data;
- Propose a validation-free Byzantine filter based on the observation that honest updates lie within the convex hull of class-wise gradients.

These results are based on the under-review papers [31, 30].

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Part I

Confusion Matrices

Chapter I

Preliminaries

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1 Overview

The confusion matrix is a central tool for understanding, evaluating, and improving classification models [61, 62, 41, 98]. Its origin is commonly traced to a 1904 lecture by Karl Pearson to the Draper’s Society [65, 5], where he introduced two-way tables to study categorical variables and their dependence. The confusion matrix is a particular type of contingency table that summarizes the empirical joint distribution of true and predicted labels, providing insight into classifier behavior beyond scalar performance measures [61, 62, 41].

Formally, the confusion matrix is a square matrix of size $C \times C$, where C denotes the number of classes. Each entry (i, j) counts the number of instances with true label i predicted as j , for all $i, j \in [C]$, where $[C] = 1, \dots, C$. It is typically computed on a test set [93].

We illustrate the confusion matrix on a three-class classification problem with classes **Cat**, **Dog**, and **Frog** (see Figure I.1). Diagonal entries correspond to correct predictions, while off-diagonal entries represent misclassifications. For instance, among samples with true label **Dog**, two are predicted as **Cat** and one as **Frog**. The more diagonal the matrix, the better the overall performance.

True / Pred	Cat	Dog	Frog
Cat	8	1	1
Dog	2	3	1
Frog	4	0	2

Figure I.1: Example of a confusion matrix. Rows correspond to true labels and columns to predicted labels.

2 Key Properties

2.1 Algorithmic Definition

Let $\{(x_k, y_k)\}_{k=1}^N$ be a dataset, where x_k denotes the k -th observation and $y_k \in [C]$ its corresponding class label. Let $\hat{y}_k \in [C]$ denote the predicted label for x_k , defined as

$$\hat{y}_k \in \operatorname{argmax}_{i=1}^C \Phi(x_k)_i,$$

where Φ is the model, and argmax returns the set of indices achieving the maximum value, from which $\hat{y}_k$ is selected.

Algorithm 1 describes the construction of the confusion matrix $M \in \mathbb{R}^{C \times C}$ from the dataset $\{(y_k, \hat{y}_k)\}_{k=1}^N$.

Algorithm 1 Construction of the Confusion Matrix

Require: Label–prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$

1: Initialize $M \in \mathbb{R}^{C \times C}$ with zeros

2: **for** $k = 1$ to N **do**

3: $M_{y_k, \hat{y}_k} \leftarrow M_{y_k, \hat{y}_k} + 1$

4: **end for**

5: **return** M

2.2 Scalar Evaluation Measures

Standard evaluation measures, including accuracy, precision, recall, and F1-score, can all be derived from the confusion matrix. For instance,

$$\text{Accuracy} = \frac{\sum_{i=1}^C M_{ii}}{M_{++}}, \quad \text{Precision}_k = \frac{M_{kk}}{M_{+k}}, \quad \text{Recall}_k = \frac{M_{kk}}{M_{k+}},$$

where M denotes the confusion matrix, with row, column, and matrix sums defined as

$$M_{i+} = \sum_j M_{ij}, \quad M_{+j} = \sum_i M_{ij}, \quad M_{++} = \sum_{i,j} M_{ij}.$$

These expressions highlight that the entries of the confusion matrix are the building blocks of widely used evaluation metrics, thereby providing a finer-grained view of classification errors and agreement.

2.3 Consistency with Label–Prediction Pairs

The diagonal entries M_{ii} count correct predictions for class i , while off-diagonal entries M_{ij} count misclassifications. This directly implies the following correspondences between label–prediction pairs and the matrix entries:

- **Diagonal & off-diagonal.** The sum of diagonal entries gives the number of correct predictions, while the sum of off-diagonal entries gives the number of errors. The total mass equals the number of label–prediction pairs: $M_{++} = N$.
- **Marginals.** Row and column sums, also referred to as marginals [98], describe the class distributions: M_{i+} gives the number of samples in true class i , while M_{+j} gives the number of samples predicted as class j . In Figure I.1, the label distribution is **Cat 10, Dog 6, and Frog 6**, while the prediction distribution is **Cat 14, Dog 4, and Frog 4**.

These properties ensure interpretability through consistency with the underlying set of label–prediction pairs.

2.4 Classifier Diagnostics

The confusion matrix exhibits the following properties:

- **Diagonality.** Higher diagonal values and lower off-diagonal values indicate better classifier performance. The matrix is diagonal if and only if the classifier correctly classifies all observations.
- **Class similarity.** A relatively large off-diagonal entry (i, j) , compared to other errors, indicates strong confusion: class j is frequently predicted when the true class is i . This suggests that, from the classifier's perspective, class i is hard to distinguish from class j .
- **Distribution bias.** A row i with globally large values indicates that class i is overrepresented in the true labels, which follows from marginal consistency. Consequently, when row marginals are imbalanced—for instance, if class k is dominant while class l is underrepresented—entries across rows are not directly comparable. Similarly, a column j with globally large values indicates that class j is overrepresented in the predictions, leading to the same limitation. In Figure I.1, this indicates that class **Cat** is overrepresented in both the labels and the predictions.
- **Under & over-prediction.** If the column marginal is much larger than the row marginal, $M_{k+} \ll M_{+k}$, then class k is overpredicted. Conversely, if the row marginal is much larger than the column marginal, $M_{+k} \ll M_{k+}$, then class k is underpredicted. These errors generally reflect systematic model behavior rather than similarity between classes.

Values in the confusion matrix arise from different factors, and their interpretation requires accounting for marginal distributions to make reliable classifier diagnostics. Normalization techniques can help mitigate distribution biases.

3 Normalization & Probabilistic Interpretation

Figure I.2 illustrates standard normalization methods and their effect on row and column sums:

- **Row normalization.** To mitigate label imbalance, a common approach is to apply row normalization:

$$\text{row}(M)_{ij} = \frac{M_{ij}}{M_{i+}},$$

which ensures that each row sums to one. This normalization provides an estimate of the conditional probability $\mathbb{P}(\hat{Y} = j \mid Y = i)$, where Y and $\hat{Y}$ denote the true and predicted label random variables, respectively [41].

- **Column normalization.** To mitigate prediction imbalance, one can apply column normalization:

$$\text{col}(M)_{ij} = \frac{M_{ij}}{M_{+j}},$$

which ensures that each column sums to one. This normalization provides an estimate of the conditional probability $\mathbb{P}(Y = i \mid \hat{Y} = j)$.

- **All normalization.** Finally, normalization by the total mass yields

$$\text{all}(M)_{ij} = \frac{M_{ij}}{M_{++}},$$

which ensures that the entire matrix sums to one. This normalization provides an estimate of the joint probability $\mathbb{P}(Y = i, \hat{Y} = j)$.

These normalizations reduce the impact of label and prediction imbalance on matrix values, making the matrix easier to interpret. They also provide complementary probabilistic interpretations of the confusion matrix, improving the analysis of classifier behavior.

True / Pred	Cat	Dog	Frog	M_{i+}
Cat	8	1	1	10
Dog	2	3	1	6
Frog	4	0	2	6
M_{+j}	14	4	4	22

(a) Raw

True / Pred	Cat	Dog	Frog	M_{i+}
Cat	0.80	0.10	0.10	1.00
Dog	0.33	0.50	0.17	1.00
Frog	0.67	0.00	0.33	1.00
M_{+j}	1.80	0.60	0.60	3.00

(b) Row-normalized

True / Pred	Cat	Dog	Frog	M_{i+}
Cat	0.57	0.25	0.25	1.07
Dog	0.14	0.75	0.25	1.14
Frog	0.29	0.00	0.50	0.79
M_{+j}	1.00	1.00	1.00	3.00

(c) Column-normalized

True / Pred	Cat	Dog	Frog	M_{i+}
Cat	0.36	0.05	0.05	0.46
Dog	0.09	0.14	0.05	0.28
Frog	0.18	0.00	0.09	0.27
M_{+j}	0.63	0.18	0.18	1.00

(d) Fully normalized

Figure I.2: Confusion matrices and their marginals under different normalizations. Row normalization fixes M_{i+} , column normalization fixes M_{+j} , and full normalization fixes M_{++} .

4 Conclusion

This overview illustrates how the confusion matrix provides insight into classifier behavior. Its simplicity and consistency provide a high degree of interpretability and enable reliable analysis.

Chapter II highlights the limitations of standard normalization methods in imbalanced settings, motivating the use of double normalization. It also provides additional perspectives on matrix normalization. Chapter III presents a generalization of the confusion matrix to broader classification settings based on optimal transport theory and the principle of maximum entropy. Finally, Chapter IV introduces a framework for evaluating confusion matrix extensions by measuring their ability to recover known latent confusion patterns from synthetic classifiers, enabling direct comparison between methods.

Chapter II

Normalization of Confusion Matrices: Methods and Geometric Interpretations

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This chapter is based on our publication [32].

In heterogeneous settings, confusion matrix values are shaped by two main factors: class similarity—how easily the model confuses two classes—and distribution bias, arising from skewed distributions in the training and test sets. However, confusion matrix values reflect a mix of both factors, making it difficult to disentangle their individual contributions.

To address this, we propose a generalization of row and column normalization based on Iterative Proportional Fitting, referred to as bi-normalization. Unlike standard normalization, this method recovers the underlying structure of class similarity. By disentangling error sources, it enables more accurate diagnosis of model behavior and supports more targeted improvements.

We also establish a correspondence between confusion-matrix normalizations, importance sampling, and the model’s internal class representations, thereby providing a clearer interpretation of normalization schemes and what they reveal about a classifier.

Contributions. We highlight key properties of bi-normalization and provide new perspectives on matrix normalizations:

- **Generalizing normalization.** We show that bi-normalization generalizes both row and column normalizations while satisfying key desirable properties.
- **Recovering class similarities.** We show that bistochastic normalization more reliably captures class similarities in heterogeneous settings compared to existing normalization schemes.
- **Importance sampling and class representation views.** We establish connections between confusion matrix normalizations—row, col, all, and bi—and both importance sampling and latent class representations, providing clearer interpretations of normalized confusion matrices.

Code: <https://github.com/JohanErbani/Bi-Normalization>

1 Introduction

While the confusion matrix is a widely used tool [84, 85, 78, 86], the meaning of its values becomes unclear in imbalanced settings.

Sources of confusion. Two main factors shape confusion matrix values: (i) class similarity and (ii) distribution bias [29], as illustrated in Figure II.1. (i) Class similarity refers to confusion between similar classes—the more alike two classes i and j are, the more frequently they are misclassified as each other, increasing values in entries (i, j) and/or (j, i) (Subfigure II.1c). While this source of error is intrinsic to the classes themselves, it also reflects inherent properties of the model: regardless of training, these similarities tend to persist. (ii) Distribution bias refers to errors caused by class imbalance in the predictions or test set. Models trained on imbalanced data tend to overpredict majority classes and underpredict minority ones [13, 69], inflating the columns of majority classes (Subfigure II.1b). Similarly, imbalance in the test set affects the rows: majority classes have higher counts, while minority classes have lower ones (Subfigure II.1a). This source of confusion is incidental and varies with class distributions.

Understanding the sources of classification errors enables targeted improvements. For example, errors from class similarity can be reduced by enhancing preprocessing to better distinguish similar classes or by augmenting the training set with more examples of those classes [40, 103, 101, 2]. Overprediction errors can be reduced by adjusting the loss function, such as reweighting class contributions to reduce the influence of frequent classes [39, 24, 15, 110]. However, confusion matrix values reflect a mix of both class similarity and distribution bias (Subfigure II.1d), making it difficult to disentangle their individual contributions.

Standard normalizations. In practice, machine learning workflows often apply row, column, or all normalization to confusion matrices:

$$\text{row}(M)_{ij} = \frac{M_{ij}}{M_{i+}}, \quad \text{col}(M)_{ij} = \frac{M_{ij}}{M_{+j}}, \quad \text{all}(M)_{ij} = \frac{M_{ij}}{M_{++}}.$$

Many papers present several normalized matrices, as each normalization highlights different aspects of the model behavior [41]. However, in imbalanced settings, these normalizations may fail to capture class similarities.

Bi-normalization. Class similarities in the confusion matrix become visible when all rows and columns have the same total, which typically occurs when both the test and training sets are balanced. When both the test set and predictions are imbalanced, a double normalization off rows and columns at the same time is needed to reveal these similarities. This normalization can be achieved using the Iterative Proportional

Figure II.1: How class similarity and distribution bias affect confusion matrices. The first three subfigures isolate each factor, while the last combines them: (a) Test set imbalance, with B as the majority class and C as the minority; (b) Prediction imbalance, with C over-predicted and B under-predicted; (c) Class similarities; (d) Combined influence of all factors. Darker colors indicate higher values.

(a) Test set: Imbalance, Predictions: Balance, Class similarity: No

	A	B	C
A			
B			
C			

(b) Test set: Balance, Predictions: Imbalance, Class similarity: No

	A	B	C
A			
B			
C			

(c) Test set: Balance, Predictions: Balance, Class similarity: Yes

	A	B	C
A			
B			
C			

(d) Test set: Imbalance, Predictions: Imbalance, Class similarity: Yes

	A	B	C
A			
B			
C			

Fitting (IPF) procedure—also known as the Sinkhorn-Knopp algorithm, biproportional fitting, the RAS method, or simply matrix scaling [51]. IPF is widely used in statistics, economics, and computer science [51]. We refer to the normalization produced by IPF as *bi-normalization*, denoted by bi . This normalization yields a confusion matrix in which each row and each column sums to 1: $\text{bi}(M)_{i+} = \text{bi}(M)_{+j} = 1$.

Additional perspectives. We also introduce two novel perspectives that provide further insight into matrix normalization:

- Importance sampling: Normalization can be viewed as applying an importance sampling strategy: it reweights each label–prediction pair to match a desired distribution—for instance, balancing labels via row normalization.
- Model representation: Normalization is also related to how the model encodes class representations. We show empirically that the overlap of class clusters in the latent space corresponds—depending on how it is measured—to a particular form of normalized confusion matrix.

Standard and bi normalizations can thus be interpreted through these perspectives, providing additional probabilistic and geometric meaning.

2 Related Work

The first part of this state-of-the-art reviews the application of IPF procedure to normalize contingency tables, while the second focuses on the impact of distribution bias on the confusion matrix.

2.1 Normalization Using Iterative Proportional Fitting

Deming and Stephan [25] proposed the use of the Iterative Proportional Fitting (IPF) procedure to estimate cell probabilities in a contingency table under given marginal constraints.

Ireland and Kullback [52] address the problem of estimating a new contingency table q from a given contingency table p , where each cell q_{ij} represents a probability, subject to known and fixed marginal probabilities $\sum_j q_{ij} = u_i$ and $\sum_i q_{ij} = v_j$. The authors introduce an algorithm minimizing the KL divergence from p to q . This procedure is equivalent to IPF [51].

In the fields of remote sensing and geographic modeling, disagreement between maps and reality is commonly displayed in a confusion matrix [20, 47]. When multiple classification or modeling methods are used, the resulting confusion matrices are

typically compared to assess significant differences [47]. Confusion matrix normalization using IPF is a standard analytical technique [19]. In this way, differences in sample sizes used to generate the matrices are eliminated and, therefore, individual cell values within the matrix are directly comparable [19].

For speaker verification system improvements, Nagineni and Hegde [82] use the IPF procedure to normalize confusion matrices and select a cohort set based on similarity modeling for each client speaker.

2.2 Causes of Errors

In the context of neural networks, Erhani et al. [29] examine how imbalanced training and test sets influence confusion matrix entries. They introduce the test–training ranking to distinguish sources of error. When some entries deviate from this criterion, it suggests strong similarity between the corresponding classes. However, class similarities can still impact errors even when the ranking is preserved. Moreover, deviations from the ranking are ambiguous—they do not indicate which entry is too high or too low [29].

2.3 Key Takeaways

Normalizing contingency tables—particularly confusion matrices—using IPF procedure is not a new idea; it is even considered a standard approach in some disciplines. However, to the best of our knowledge, no existing work in machine learning employs this normalization to separate the effects of distribution bias and class similarity within confusion matrixes. This decomposition provides deeper insights into the sources of errors, helping to assess and improve the classifier.

To the best of our knowledge, no prior work has connected confusion matrix normalization to importance sampling or to the structure of class representations in a model’s latent space. We introduce two perspectives—probabilistic and geometric—that offer deeper insight into classifier behavior.

3 Bi-Normalization

This section defines the bi-normalization, outline the key properties it should satisfy, and show how the IPF procedure can compute it accordingly.

Theoretical analysis is simpler with positive matrices. Since a non-negative confusion matrix M and its strictly positive counterpart $M + \epsilon$ (for small ϵ) exhibit similar model behavior, we propose using $M + \epsilon$ instead. We now assume that M is a positive confusion matrix.

3.1 Empirical Definition & Desirable Properties

Normalization approximates the confusion matrix that would arise under specific experimental conditions. Row normalization corresponds to the confusion matrix obtained from a balanced test set, whereas column normalization corresponds to that obtained from a model producing balanced predictions.

Empirical definition. Bi-normalization captures class similarities by approximating the confusion matrix under balanced label distributions in both the training and test sets. This is formalized by enforcing both row and column normalization: $\text{bi}(M)_{i+} = \text{bi}(M)_{+j} = 1$.

Standard normalization methods all, row, and col satisfy three properties described below: idempotence, class distribution invariance, and information preservation. By extension, bi-normalization is expected to satisfy these properties as well.

Idempotence. Normalization maps the matrix to specific experimental conditions. For example, the row normalization operator maps the matrix to a setting with balanced label distributions. Once these conditions are reached, further normalization should not modify the matrix, e.g., $\text{row} \circ \text{row}(M) = \text{row}(M)$, where $\circ$ denotes composition.

Class distribution invariance. If two confusion matrices differ only in the experimental conditions targeted by the normalization operator, then normalization yields similar matrices.

For instance, row normalization targets a balanced test set. Thus, regardless of the test set label distribution, a trained model yields similar row-normalized confusion matrices. Formally, let S and T be the confusion matrices obtained from a model Φ on two large test sets with different label distributions (without class extinction). Then, the error patterns across rows remain similar, i.e., $S_{i,:} \propto T_{i,:}$ for all i , where $\propto$ denotes equality up to a positive scaling factor¹. This implies that $\text{row}(S) \approx \text{row}(T)$, showing that the row operator is invariant to variations in the test set label distribution.

This invariance is formalized as invariance under left multiplication by a positive diagonal matrix: $\text{row}(AM) = \text{row}(M)$ for any diagonal matrix $A \in \mathbb{R}_{>0}^{C \times C}$. Similarly, column normalization is invariant under right multiplication by a positive diagonal matrix: $\text{col}(MB) = \text{col}(M)$ for any diagonal matrix $B \in \mathbb{R}_{>0}^{C \times C}$.

Bi-normalization should be invariant under both operations, as it combines row and column normalization, i.e., $\text{bi}(AMB) = \text{bi}(M)$.

Information preservation. Normalization infers, from the original confusion matrix, the one obtained under other experimental conditions. It should therefore preserve as much of the original information as possible, altering it only when necessary.

This requirement can be formalized as an optimization problem that promotes similarity between the original and normalized matrices. We use the Kullback–Leibler (KL) divergence as a dissimilarity measure. Given row and/or column constraints, the normalized matrix is defined as the one closest to the original in terms of KL divergence.

For instance, row normalization satisfies

$$\begin{aligned} \text{row}(M) \in \argmin_{P \in \mathbb{R}_{>0}^{C \times C}:} D_{\text{KL}}(P \| M), \\ P_{i+} = 1, P_{+j} = \sum_i \frac{M_{ij}}{M_{i+}} \forall i, j \end{aligned}$$

where $M \in \mathbb{R}_{>0}^{C \times C}$ is the original confusion matrix, and $D_{\text{KL}}(P \| M)$ denotes the KL divergence from P to M . Proofs and extensions to other normalization methods are provided in the Appendix.

Bi-normalization should minimize this optimization problem under the constraints that each row and each column sums to one.

3.2 Theoretical Definition, Properties & Estimation

This subsection provides the formal definition of the bi-normalization, its properties and how to estimate it.

The bi-normalization is defined as the unique solution of an optimization problem (see Appendix for the proof of existence and uniqueness).

Definition 1. $\text{bi}(M)$ is the unique minimizer of the following constrained problem:

$$\begin{aligned} \text{bi}(M) \in \argmin_{P \in \mathbb{R}_{>0}^{C \times C}:} D_{\text{KL}}(P \| M) \\ P_{i+} = P_{+j} = 1 \forall i, j \end{aligned}$$

We now establish key properties:

Proposition 1. $\text{bi}(M)$ satisfies idempotence, class distribution invariance, and information preservation as described in Subsection Empirical Definition & Desirable Properties.

¹For vectors or matrices U and V of the same size, $U \propto V$ means $U \approx aV$ for some $a \in \mathbb{R}_{>0}$.

The last property is satisfied by definition, while the two others follow directly from it, as shown in the appendix.

Although $\text{bi}(M)$ has no closed form, it can be efficiently estimated by the IPF algorithm (s). In our case, it consists of normalizing the matrix alternately by rows and columns until convergence²:

$$\text{col} \circ \text{row} \left(\dots \text{col} \circ \text{row} \left(\text{col} \circ \text{row} (M) \right) \dots \right) \longrightarrow \text{bi}(M)$$

The IPF procedure always converges when the input matrix M is positive [51]. Let $M_{i+} = u_i$ and $M_{+j} = v_j$ for all i and j , and denote by Q the limit of the IPF procedure³. Then, Q solves the following optimization problem [63, 51]:

$$\begin{aligned} Q \in & \underset{P \in \mathbb{R}_{>0}^{C \times C}}{\text{argmin}} \quad D_{\text{KL}}(P \| M) \\ & P_{i+} = u_i, P_{+j} = v_j \quad \forall i, j \end{aligned}$$

This directly implies the following:

Proposition 2. *$\text{bi}(M)$ can be approximated using the IPF procedure with row and column constraints set to 1. We have $\text{bi}(M) = Q \approx \hat{Q}$, where Q is the theoretical IPF limit and $\hat{Q}$ is its empirical estimate.*

Since $\text{bi}(M)$ is doubly stochastic⁴ and M is positive, the IPF procedure converges linearly [51].

4 Normalization as Importance Sampling

This section shows that normalization can be interpreted as an importance sampling strategy used to match a target distribution. It first shows that normalization can be expressed as a weighted sum, then interprets this weighted sum as an importance sampling process.

Let $\{(x_k, y_k)\}_{k=1}^N$ be a single-label dataset, and let $\hat{y}_k \in [C]$ be the model prediction for x_k , defined as

$$\hat{y}_k \in \underset{i=1}{\text{argmax}}^C \Phi(x_k)_i,$$

where Φ denotes the model. Let M be the confusion matrix constructed from the label-prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$.

4.1 Weighing Sum

By definition, the confusion matrix is $M = \sum_{k=1}^N E_{y_k \hat{y}_k}$, where E_{ij} denotes the $C \times C$ matrix with a 1 at entry (i, j) and 0 elsewhere. Each matrix $E_{y_k \hat{y}_k}$ is a confusion matrix constructed from the single pair $(y_k, \hat{y}_k)$, capturing its individual contribution to M .

Standard and bi normalizations can be obtained by multiplying M on the left and right by diagonal matrices. Specifically, let D^l and D^r denote diagonal matrices (with l for left and r for right), and let M be the original confusion matrix. Then, $D^l M D^r$ yields the normalized version of M . For instance, setting

$$D^l = \text{diag}\left(\frac{1}{M_{1+}}, \dots, \frac{1}{M_{C+}}\right) \text{ and } D^r = I$$

²Let $Q^{(t)}$ be the matrix produced by IPF at iteration t . Convergence is reached when $Q_{i+}^{(t)}$ and $Q_{+j}^{(t)}$ are sufficiently close to 1 under a fixed tolerance.

³Let $Q^{(t)}$ be the matrix produced by IPF at iteration t . Then $Q^{(t)} \rightarrow Q$ as $t \rightarrow \infty$.

⁴A doubly stochastic matrix is a square matrix with nonnegative entries such that each row and each column sums to 1.

yields the row-normalized matrix, where $\text{diag}(d_1, \dots, d_C)$ denotes a diagonal matrix with entries $d_1, \dots, d_C$ on its diagonal, and I the identity. Other normalizations are presented in the Appendix.

As a result, normalization can be expressed as a weighted sum of the individual contributions $E_{y_k \hat{y}_k}$:

$$D^l M D^r = \sum_{k=1}^N D_{y_k, y_k}^l D_{\hat{y}_k, \hat{y}_k}^r E_{y_k \hat{y}_k},$$

where each contribution $E_{y_k \hat{y}_k}$ is weighted by the product $D_{y_k, y_k}^l D_{\hat{y}_k, \hat{y}_k}^r$. For clarity, the weight can be expressed through a weight function ω . For instance, the weight function ω_r (with r for row) for row-normalization is

$$\omega_r : (y, \hat{y}) \mapsto \frac{1}{M_{y+}}, \quad \text{row}(M) = \sum_{k=1}^N \omega_r(y_k, \hat{y}_k) E_{y_k \hat{y}_k}.$$

When the dataset is viewed as a realization of random variables, this process corresponds to importance sampling.

4.2 Importance sampling

Let $\{(Y_k, \hat{Y}_k)\}_{k=1}^N$ be independent and identically distributed (i.i.d.) random pairs, representing label–prediction pairs, and f be their joint distribution $f(i, j) = \mathbb{P}(Y = i, \hat{Y} = j)$. The random counterpart of $\text{all}(M)$ converges almost surely (a.s.) to

$$\frac{1}{N} \sum_{k=1}^N E_{Y_k \hat{Y}_k} \xrightarrow[N \rightarrow \infty]{\text{a.s.}} \mathbb{E}[E_{Y \hat{Y}}],$$

where

$$\mathbb{E}[E_{Y \hat{Y}}]_{ij} = \mathbb{E}[\mathbb{1}_{Y=i, \hat{Y}=j}] = \mathbb{P}(Y = i, \hat{Y} = j) = f(i, j),$$

with $\mathbb{1}$ the indicator function.

Importance sampling refers to Monte Carlo methods that approximate expectations under a target distribution g by reweighting samples drawn from a proposal distribution f [104]. This approach is, for instance, useful when direct samples from g are unavailable. To approximate the expectation of $E_{Y \hat{Y}}$ when $(Y, \hat{Y})$ is distributed according to g rather than f , we use the estimator $\hat{\mu}_g$, defined as

$$\hat{\mu}_g = \frac{1}{N} \sum_{k=1}^N \frac{g(Y_k, \hat{Y}_k)}{f(Y_k, \hat{Y}_k)} E_{Y_k \hat{Y}_k} \xrightarrow[N \rightarrow \infty]{\text{a.s.}} \mathbb{E}\left[\frac{g(Y, \hat{Y})}{f(Y, \hat{Y})} E_{Y \hat{Y}}\right] = \mathbb{E}_g[E_{Y \hat{Y}}]$$

with

$$\mathbb{E}_g[E_{Y \hat{Y}}]_{ij} = \mathbb{E}_g[\mathbb{1}_{Y=i, \hat{Y}=j}] = \mathbb{P}_g(Y = i, \hat{Y} = j) = g(i, j),$$

where $\mathbb{E}_g$ and $\mathbb{P}_g$ denote expectation and probability assuming the variables follow the law g . For importance sampling to remain accurate, the variance of each entry of the estimator $\hat{\mu}_g$ must remain small, which occurs when f is approximately proportional to g [104]. By selecting different choices of g , standard normalizations arise naturally.

As an example, consider row normalization (see Appendix for other normalizations). Setting $g(i, j) = \mathbb{P}(\hat{Y} = j | Y = i)$ ⁵ recovers the row-normalized confusion matrix. In this case, the importance sampling ratio becomes

$$\frac{g(i, j)}{f(i, j)} = \frac{\mathbb{P}(\hat{Y} = j | Y = i)}{\mathbb{P}(Y = i, \hat{Y} = j)} = \frac{1}{\mathbb{P}(Y = i)} \approx \frac{N}{M_{i+}}.$$

Accordingly, the empirical estimator is

$$\frac{1}{N} \sum_{k=1}^N \frac{N}{M_{y_k+}} E_{y_k \hat{y}_k} = \sum_{k=1}^N \omega_r(y_k, \hat{y}_k) E_{y_k \hat{y}_k} = \text{row}(M).$$

⁵Note that g is not a normalized distribution, since $\sum_{i,j} g(i, j) = C$ rather than 1.

For row normalization to be reliable, the importance sampling ratio should remain controlled. In particular, this requires that each class i is sufficiently represented in the test set; otherwise, the estimator exhibits high variance and becomes unstable.

Key Takeaways. Normalization is an importance sampling procedure: each pair is reweighted to shift the empirical distribution toward a target one. Reliability requires that the two distributions are not too dissimilar.

5 Geometric View of Normalized Confusion Matrices

This section proposes to model class representations as volumes, and finally link the overlaps of these volumes to normalized confusion matrices.

5.1 Class Representations

This subsection introduces a geometric perspective on class representations in the model’s latent space. The key idea is to characterize class clusters as multidimensional histograms.

Representation space & dimensionality reduction. We use the output of the final pre-logit layer as the class representation space, a procedure already employed in previous work [9, 59, 121]. Let ε be the embedding function, so that $\Phi(x) = \text{Softmax} \circ \text{Logit} \circ \varepsilon(x)$, where $\varepsilon(x) \in \mathbb{R}^n$ for each (x, y) in the training set $\mathcal{D}$.

To facilitate histogram construction, we project the embeddings into a lower-dimensional space using Principal Component Analysis (PCA). Let P be the projection matrix with $m < n$, so that $P\varepsilon(x) \in \mathbb{R}^m$.

Class clusters & histograms. For each label i , we define the cluster of projected embedded points as $\mathcal{C}_i = \{P\varepsilon(x) : (x, y) \in \mathcal{D}, y = i\}$. Similarly, the cluster of points predicted as class j is $\hat{\mathcal{C}}_j = \{P\varepsilon(x) : (x, y) \in \mathcal{D}, \hat{y} = j\}$.

For each label and prediction clusters, we construct a multidimensional, non-overlapping histogram [102] (see Appendix for details). Each histogram is interpreted as a volume in $\mathbb{R}^{m+1}$, providing a geometric view of the clusters. Figure II.2 shows a toy example illustrating histograms of class clusters.

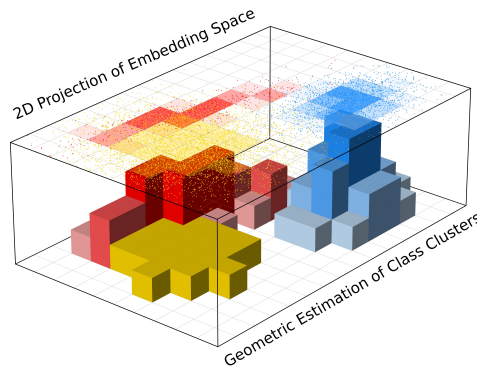


Figure II.2: Toy example showing a 2D projection of embedded observations forming class clusters (top map). Each cluster is associated with a histogram indicating the spatial distribution of points (bottom map).

Weighted histograms. Following the importance sampling perspective (Subsection 4), weighted histograms are constructed using the weight functions ω_a , ω_r , ω_c , and ω_b , corresponding respectively to all, row, col, and bi normalization. For a given bin, its height is defined not by the number of points it contains, but by the sum of their weights.

For instance, consider cluster $\mathcal{C}_i$ and the weighting ω_a . The height of a given bin b is $\sum_{(x,y) \in \mathcal{D}} \omega_a(y, \hat{y}) \mathbb{1}_{y=i} \mathbb{1}_{P_\varepsilon(x) \in b}$, whereas in the unweighted histogram it is $\sum_{(x,y) \in \mathcal{D}} \mathbb{1}_{y=i} \mathbb{1}_{P_\varepsilon(x) \in b}$.

We denote by $V(\mathcal{C}, \omega)$ the hypervolume in $\mathbb{R}^{m+1}$ of the weighted histogram built from cluster $\mathcal{C}$ using the weighting ω .

5.2 Geometric View of Normalized Confusion Matrices

This subsection introduces a new matrix, referred to as Geometric Confusion Matrix (GCM), which quantifies the overlap between the weighted histogram of label and prediction clusters:

$$\left(\text{GCM}_\omega \right)_{ij} = \lambda \left(V(\mathcal{C}_i, \omega) \cap V(\hat{\mathcal{C}}_j, \omega) \right),$$

for $i, j = 1, \dots, C$, where λ denotes the Lebesgue measure. The Lebesgue measure of $V(\mathcal{C}_i, \omega) \cap V(\hat{\mathcal{C}}_j, \omega)$ corresponds to the volume of the intersection between the ω -weighted histograms of $\mathcal{C}_i$ and the one of $\hat{\mathcal{C}}_j$. Unlike the standard confusion matrix, the GCM takes into account the spatial organization of embedded points (see Appendix for details).

We now provide geometric views of confusion matrix normalizations, all claims are supported by experimental results (see Section 7).

All-normalization. All-normalized matrix approximates the overlap between the unweighted histograms of label and prediction clusters:

$$\text{all}(M) \propto \text{GCM}_{\omega_a},$$

where $\propto$ means approximately equal up to a scaling factor⁶.

Row-normalization. Row-normalized matrix approximates the overlap between normalized label histograms and the resulting weighted prediction histograms:

$$\text{row}(M) \propto \text{GCM}_{\omega_r}.$$

More precisely, up to a known scaling factor, the label histograms are normalized in the sense that $\lambda(V(\mathcal{C}_i, \omega_r)) = 1$, whereas the volume of the prediction histograms, $\lambda(V(\hat{\mathcal{C}}_j, \omega_r))$ are not fixed a priori (see Appendix for details). Accordingly, label histograms are normalized, while prediction histograms are scaled according to ω_r .

Col-normalization. Similarly, column-normalized matrix approximates the overlap between normalized prediction histograms and resulting weighted label histograms:

$$\text{col}(M) \propto \text{GCM}_{\omega_c}.$$

Bi-normalization. Finally, bi-normalized matrix approximates the overlap between normalized histograms of both label and prediction clusters:

$$\text{bi}(M) \propto \text{GCM}_{\omega_b}.$$

These correspondences provide a geometric interpretation of normalized confusion matrices and illustrate how confusion matrices capture the organization of classes in the model's latent space.

6 Experimental Setup

This section describes our experimental setup.

⁶Let M and N be matrices of the same size. We write $M \propto N$ to indicate that $M \approx \alpha N$ for some scalar $\alpha \in \mathbb{R}_{>0}$.

6.1 Datasets & Heterogeneity

We conduct experiments on four datasets: MNIST [68], Fashion-MNIST [112], CIFAR-10 [58], and STL-10 [17].

Normalization methods differ mainly on non-diagonal matrices. For example, row, column, and bi normalizations yield the same result on diagonal matrices. To encourage non-diagonal confusion matrices, we make MNIST and Fashion-MNIST harder by applying random rotations of 0° , 90° , 180° , or 270° during preprocessing, applied to both the training and test images.

Bi-normalization is particularly useful under heterogeneous settings. To simulate data heterogeneity, we sample from the original datasets with a Dirichlet distribution of concentration α , following [4, 50, 33]. We consider five levels—very low, low, medium, high, and extreme—corresponding to $\alpha \in \{10, 3, 1, 0.3, 0.1\}$.

To avoid class extinction, we enforce a minimum representation of 20% of the original class size. For example, in a dataset with 1000 samples per class, at least 200 samples are retained for each class, while the remaining samples are allocated according to a Dirichlet distribution.

6.2 Models & Training

CNN models (see Appendix for details) are trained using stochastic gradient descent with cross-entropy loss, a batch size of 32, a learning rate of 10^{-3} , a momentum of 0.9, and a weight decay of 10^{-4} .

As mentioned above, the main differences between normalizations appear in the off-diagonal confusion matrices. Therefore, training is limited to a maximum of 10 epochs, and stops earlier if the classifier reaches 60% balanced test accuracy.

6.3 Metric

Let S and T be two confusion matrices in $\mathbb{R}_{\geq 0}^{C \times C}$, possibly already normalized.

To measure the similarity between confusion matrices, we use their overlap:

$$\text{Overlap}(S, T) = \sum_{ij} \min(\text{all}(S)_{ij}, \text{all}(T)_{ij}),$$

where $\min$ denotes the element-wise minimum (see Appendix for the rationale behind reusing all normalization). This score ranges from 0 to 1, with higher values indicating greater similarity. It reaches 1 if and only if $T = S$.

This metric is strictly equivalent to the L^1 distance, since $\|\text{all}(S) - \text{all}(T)\|_1 = 2 - 2 \text{Overlap}(S, T)$ (see Appendix for proof).

6.4 Baselines & Experiments

We compare bi-normalization with standard normalizations: row, col, and all. Each experiment is repeated over MNIST, Fashion-MNIST, CIFAR-10, and STL-10, with five heterogeneity levels and 30 random seeds.

Experiment 1. This experiment compares the confusion matrix obtained from balanced datasets (denoted as M_1) with normalized versions of the confusion matrix obtained from imbalanced datasets (denoted as M_2).

1. Initialize a model with a fixed random seed, then create two deep copies: model 1 and model 2.
2. Train model 1 on a balanced training set and compute its confusion matrix M_1 using a balanced test set.
3. Sample an imbalanced training set and test set, train model 2 on the imbalanced training set, and compute its confusion matrix M_2 on the imbalanced test set.
4. Apply normalization techniques to M_2 and evaluate their similarity to M_1 .

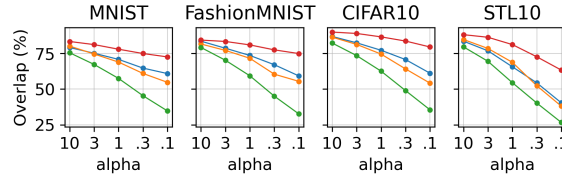
This experiment tests whether bi-normalization reveals class similarities under imbalanced data more effectively than other approaches.

Experiment 2. This experiment, conducted in an imbalanced setting, compares different versions of the GCM— $\text{GCM}\omega_a$, $\text{GCM}\omega_r$, $\text{GCM}\omega_c$, $\text{GCM}\omega_b$ —with normalized confusion matrices derived from M — $\text{row}(M)$, $\text{col}(M)$, $\text{all}(M)$, and $\text{bi}(M)$.

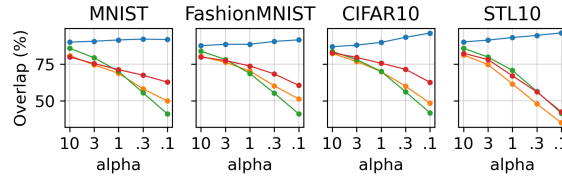
1. Initialize the model with a fixed seed.
2. Sample imbalanced training and test sets. Train the model. Compute both the GCM variants and the normalized confusion matrices using the test set.
3. Measure pairwise similarities.

This experiment evaluates how each normalization reveals the structure of class representations in the model’s latent space under a specific weighted histogram.

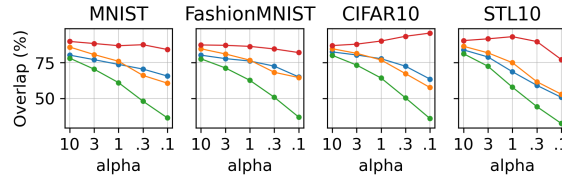
Details on the construction of multivariate histograms required for GCM are provided in the Appendix.



(a) Target: Matrix from balanced setting



(b) Target: $\text{GCM}\omega_r$



(c) Target: $\text{GCM}\omega_b$

Figure II.3: Overlap (%) between target matrices and normalized matrices. Legend: ● bi ● row ● col ● all

7 Empirical Results

Figure II.3 shows the results of Experiment 1 (Subfigure II.3a) and Experiment 2 (Subfigures II.3b and II.3c), described in Section 6.4.

Experiment 1. This experiment evaluates how well different normalizations recover class similarities. Starting from a confusion matrix obtained under an imbalanced setting, normalization aims to approximate the one obtained under a balanced setting.

Across datasets and heterogeneity levels, Subfigure II.3a shows that bi-normalization consistently achieves the highest overlap, outperforming other methods. As heterogeneity increases, (i) the gap between bi-normalization and the other methods tends to widen, and (ii) the ability to recover class similarity diminishes regardless of the normalization method.

This behavior is expected: the greater the imbalance between the training and test sets, the more distribution bias is introduced into the matrix. This bias tends to obscure class similarities, making row, col, and all normalizations unsuitable. Moreover,

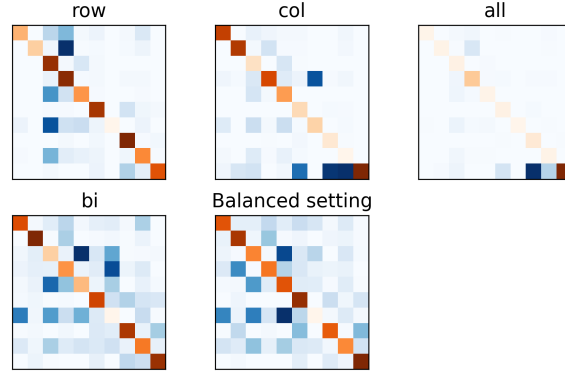


Figure II.4: One instance (seed 0) on Fashion-MNIST with $\alpha = 0.3$. Bi-normalization best recovers class relationships under distribution shifts. Diagonal variations are shown in orange; off-diagonal in blue, with darker shades indicating higher values.

higher heterogeneity leads to prediction and test class distributions that deviate further from a balanced setting, making normalization procedures less reliable, as discussed in Section 4.

Figure II.4 further illustrates that bi-normalization captures the class similarity patterns present in the balanced confusion matrix, whereas other methods often produce vertical or horizontal stripes.

By definition, bi-normalization is expected to achieve the lowest KL divergence. Figure 2 in the Appendix confirms this, providing further evidence of its ability to recover class similarities.

Experiment 2. This experiment evaluates the correspondence between the organization of class representations in the model’s latent space and normalized confusion matrices.

Subfigure II.3b demonstrates a clear correspondence between $\text{GCM}\omega_r$ and row normalization. Overlap increases as heterogeneity grows. We observe similar results for $\text{GCM}\omega_a$ with all normalizations, and for $\text{GCM}\omega_c$ with column normalization (see Appendix).

Subfigure II.3c shows that the correspondence between $\text{GCM}\omega_b$ and bi-normalization is weaker. This makes sense, as the granularity differs from the other approaches. For instance, in $\text{GCM}\omega_r$, all embedded points of the same label receive the same weight, while in $\text{GCM}\omega_b$, embedded points are weighted by both label and prediction. This indicates that spatial separation is finer under $\text{GCM}\omega_b$ weighting than with other weightings. Nonetheless, bi-normalization still exhibits higher correspondence than other methods.

This experiment confirms the validity of the geometric interpretation of normalizations described in Section 5.2.

8 Conclusion

We revisited confusion matrix normalization and showed that bi-normalization provides a principled way to isolate class similarity from distribution bias. While the use of IPF for contingency tables is well established in other fields, its value for interpreting classifier behavior in machine learning has been largely overlooked. We demonstrated that bi-normalization reveals class similarities more clearly than standard methods, especially in heterogeneous settings.

We also showed that normalization corresponds to an importance sampling strategy and introduced a geometric interpretation of class representations in latent space, offering deeper insight into what normalization reveals about model behavior.

Chapter III

Unifying Confusion Matrices from Single-Label to Soft-Label

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This chapter is based on our publication [29] and Section Case Study: EXIST on our collaboration [83].

The confusion matrix is well established in the single-label setting, where each observation belongs to exactly one class. While multi-label and soft-label settings better reflect the complexity of real-world data, extending the confusion matrix to these settings is not straightforward.

We introduce the *Transport-based Confusion Matrix* (TCM), a generalization of the confusion matrix applicable to single, multi, and soft-label settings. TCM preserves the familiar structure of the standard confusion matrix and coincides with it in the single-label case. It captures errors that were previously inaccessible, improving both the consistency and the scope of machine learning evaluation.

Contributions. We introduce the *Transport-based Confusion Matrix* (TCM), which offers the following advantages:

- **Universality.** A unified framework for analyzing model errors across single-label, multi-label, and soft-label settings. To the best of our knowledge, it is also the only method applicable to the soft-label setting.
- **Unification.** In the single-label setting, TCM coincides with the standard confusion matrix. It also preserves partial consistency with existing extensions, thereby bridging previously incompatible approaches.
- **Reliability and Interpretability.** By combining optimal transport theory with the principle of maximum entropy, TCM is both theoretically grounded and interpretable.

Code: <https://github.com/JohanErbani/TCM>

1 Introduction

No existing approach provides a unified formulation of the confusion matrix for single-label, multi-label, and soft-label settings, limiting both the consistency and the scope of machine learning evaluation.

Classification settings. The standard setting is *single-label*, where each observation belongs to exactly one class. For instance, an image may be labeled as either **Cat**, **Dog**, or **Frog**. In this case, the label space $\mathcal{Y}$ is defined as $\mathcal{Y} = [C]$, where C is the number of classes. Equivalently, it can be represented as the set of one-hot vectors:

$$\mathcal{Y} = \{\mathbf{y} \in \{0, 1\}^C : \|\mathbf{y}\|_1 = 1\}.$$

In *multi-label classification*, each observation may belong to several classes simultaneously. For example, an image can contain both a cat and a dog and thus be associated with both labels. In this case, $\mathcal{Y} = \{0, 1\}^C$. In the *soft-label* setting, each observation is associated with non-negative real values over classes, representing confidence or degree of membership. For example, a scientific paper may be labeled as 0.8 **Mathematics**, 0.4 **Physics**, and 0.2 **Computer Science**, reflecting how much it relates to each field. In this case, $\mathcal{Y} = \mathbb{R}_{\geq 0}^C$. Multi and soft-label frameworks are widely used in applications such as medical diagnosis, emotion recognition, and image or text classification [118, 108, 70, 117, 56, 75, 120, 18, 64, 22, 71]. Soft labels are particularly useful for capturing annotator disagreement in subjective tasks, and have been shown to improve generalization, robustness, and calibration [105, 18].

Problem & motivation. Error decomposition is straightforward in the single-label setting. For example, if the model predicts **Apple** instead of the true label **Pear**, the confusion is unambiguous. In multi and soft-label settings, however, the decomposition becomes unclear. For instance, if the model predicts **Apple**, **Lemon** while the true label is **Banana**, **Pear**, it is unclear whether **Banana** is confused with **Apple** and **Pear** with **Lemon**, whether the opposite assignment is more appropriate, or whether the confusion pattern is more complex. The problem becomes even harder when the numbers of predicted and true classes differ, or when labels are soft.

As a result, multiple extensions of the confusion matrix co-exist [61, 49, 62, 41, 10, 98, 29]. These definitions differ, and no formulation is designed to simultaneously handle single-label, multi-label, and soft-label settings, leading to inconsistencies in machine learning evaluation tools.

Transport-based Confusion Matrix. We introduce the Transport-based Confusion Matrix (TCM), a unified framework that extends the confusion matrix to multi and soft-label settings while preserving its structure and usability. Our approach is grounded in optimal transport theory and the principle of maximum entropy.

2 Related works

2.1 Multi-Label Matrices

The Scikit-learn library [87] provides a method for evaluating errors in multi-label classification, yielding one confusion matrix per class. However, this class-wise analysis is insufficient to fully capture a model’s behavior [49, 62].

Krstinić et al. [61] introduce four formulas to compute elementary confusion matrices, each tailored to a specific case (e.g., too many or too few predicted classes). Similarly, Heydarian, Doyle, and Samavi [49] propose three formulas that yield comparable experimental results [60]. Krstinić et al. [62] recover standard precision and recall by constructing two confusion matrices, one per metric. These approaches lack theoretical justification for their design choices.

Görtler et al. [41] propose a confusion matrix for hierarchical and multi-output labels. In the multi-label setting without hierarchy, each class is represented by a binary variable (present/absent), yielding a $2C \times 2C$ matrix. Each pair of true class i and predicted class j is associated with a 2×2 submatrix:

$$\frac{\mathbb{P}(Y^i = 1, \hat{Y}^j = 1) \mid \mathbb{P}(Y^i = 1, \hat{Y}^j = 0)}{\mathbb{P}(Y^i = 0, \hat{Y}^j = 1) \mid \mathbb{P}(Y^i = 0, \hat{Y}^j = 0)}$$

where Y^i and $\hat{Y}^j$ denote binary variables indicating the presence of true class i and predicted class j , respectively. This formulation departs from the standard $C \times C$ confusion matrix and is less directly interpretable.

2.2 Soft-Label on the Simplex Matrices

Binaghi et al. [10] and Silván-Cárdenas and Wang [98] propose confusion matrices for restricted soft-label scenarios, where both labels and predictions are simplex points (i.e., non-negative vectors summing to 1).

Binaghi et al. [10] introduce a matrix based on fuzzy set theory. However, this approach loses the familiar behavior of the confusion matrix: even perfect predictions may yield a non-diagonal matrix [91].

In geographic information science, Silván-Cárdenas and Wang [98] propose a method to compare terrestrial images. Each pixel is associated with multiple classes; one image serves as the reference and the other as the approximation. Each entry (i, j) is defined as an interval, where the lower (resp. upper) bound corresponds to the minimum (resp. maximum) sub-pixel class overlap.

2.3 Key Takeaways

Existing solutions often lack theoretical justification, deviate from the familiar behavior of the confusion matrix, and appear mutually incompatible. Moreover, no method simultaneously integrates single-label, multi-label, and soft-label frameworks, leading to inconsistencies across evaluation settings.

To address these limitations, we propose a unifying approach: the Transport-based Confusion Matrix. Grounded in optimal transport and the principle of maximum entropy, this framework preserves the behavior of the classical confusion matrix, accommodates all labeling settings simultaneously, and reconciles approaches that previously appeared incompatible. To the best of our knowledge, this perspective has not yet been explored and provides a more comprehensive and interpretable evaluation of model performance.

The main state-of-the-art approaches discussed in this paper are summarized in Table III.1.

Table III.1: Overview of state-of-the-art confusion matrices and TCM in single-label, multi-label, soft-label frameworks restricted to simplex point labels and predictions (Simplex), and unrestricted soft-label frameworks.

Matrix	Single	Multi	Simplex	Soft	Description
Confusion Matrix (CM)	✓	✗	✗	✗	Each entry (i, j) counts how often i is predicted as j .
Multi-Label Confusion Matrix (MLCM) [61]	✓	✓	✗	✗	The confusion matrix relies on four distinct formulas, each corresponding to a specific case.
Multi-Label Confusion Tensor (MLCT) [62]	✓	✓	✗	✗	Two matrices whose diagonals match classical recall and precision scores. Interpretation of off-diagonal entries is unclear.
Sub-pixel Confusion Matrix (SCM) [98]	✓	✗	✓	✗	It computes the minimum and maximum sub-pixel class overlap, yielding interval-valued entries.
Transport-based Confusion Matrix (TCM)	✓	✓	✓	✓	Grounded in optimal transport and maximum entropy.

3 Background

This section introduces the concepts required to define TCM constructions. The first subsection describes how a confusion matrix can be decomposed into elementary matrices. The second subsection presents an interpretation of these elementary matrices that allows us to generalize the confusion matrix. This perspective connects naturally to optimal transport (Subsection 3.4), where the notion of measures is central (Subsection 3.3).

3.1 Single-Label Confusion Matrix

Let $\{(x_k, y_k)\}_{k=1}^N$ be a single-label dataset, and let $\hat{y}_k \in [C]$ be the model prediction for x_k , defined as

$$\hat{y}_k \in \operatorname{argmax}_{i=1}^C \Phi(x_k)_i,$$

where Φ denotes the model. Let M be the confusion matrix constructed from the label-prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$.

The confusion matrix can be written as a sum of elementary matrices, each corresponding to a label–prediction pair:

$$M = \sum_{k=1}^N E_{y_k \hat{y}_k}, \quad (\text{III.1})$$

where E_{ij} is the $C \times C$ matrix with a 1 at position (i, j) and 0 elsewhere. Each $E_{y_k \hat{y}_k}$ is itself a confusion matrix constructed from a single label–prediction pair $(y_k, \hat{y}_k)$. We refer to such matrices as *elementary matrices*.

Equation (III.1) shows that extending the confusion matrix to more general settings reduces to extending the definition of the elementary matrix. Summing these generalized elementary matrices over the dataset then yields a generalized confusion matrix.

3.2 How to Transform Predictions into Labels

In the single-label setting, each elementary matrix $E_{y_k \hat{y}_k}$ can be interpreted as a set of instructions that transforms the prediction $\hat{y}_k$ into the label y_k . To illustrate this

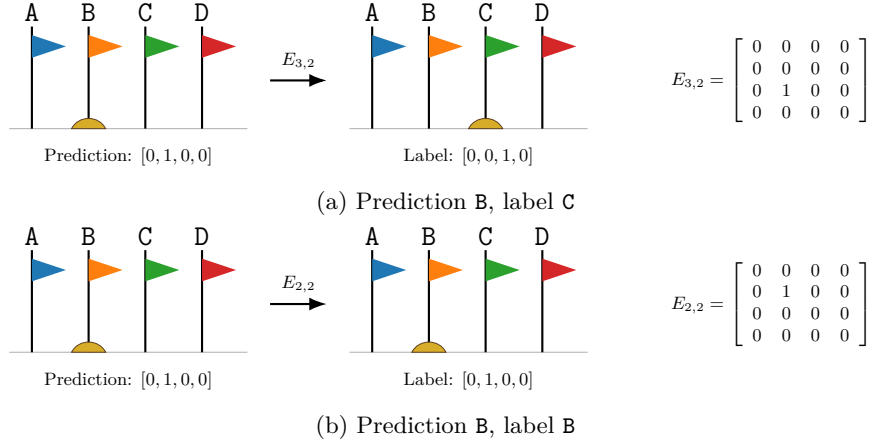


Figure III.1: Illustration of elementary matrices as transport instructions. For a misclassification where class B is predicted instead of class C, the mass is transported from class B to class C. For a correct prediction in class B, the mass stays at class B, so no transport is needed.

idea, we introduce the following analogy, depicted in Figure III.1.

Consider an area with marker flags, each representing a class. For a classification task with classes A, B, C, and D, there are four flags. At the base of the flag corresponding to the predicted class, we place one kilogram of sand. This represents the value 1, while the other flags, with no sand, represent 0. For example, if the model predicts B, the one-hot vector is $[0, 1, 0, 0]$, which corresponds to one kilogram of sand at flag B.

The elementary matrix specifies how to move this sand to match the true label. Entry (i, j) indicates the amount of sand to be moved from flag j to flag i . For instance, if the true label is C, the associated elementary matrix is $E_{3,2}$, with entry 1 at $(3, 2)$ and 0 elsewhere; this indicates that one kilogram of sand must be transported from flag B to flag C. After the transfer, all the sand lies under flag C, giving the one-hot vector $[0, 0, 1, 0]$, which matches the label. In contrast, if the prediction is correct, the elementary matrix is $E_{2,2}$. This corresponds to moving one kilogram of sand from flag B to flag B, so no change is needed.

This interpretation naturally connects to optimal transport, where each elementary matrix can be viewed as a transport plan. We first introduce the notion of a measure, which provides the appropriate framework for representing mass distributions in optimal transport.

3.3 Informal Introduction to Measures

We briefly introduce measures; see [99] for a formal treatment.

A measure assigns a non-negative value to certain sets, called *measurable sets*. These sets are subsets of a space denoted by X . This collection of measurable sets is called a σ -algebra on X . Intuitively, a measure acts like a weighing scale: it assigns a weight to each measurable set, while the σ -algebra specifies which sets can be measured. A set not included in the σ -algebra cannot be assigned a weight. Let μ be a measure, and let A and B be two measurable sets, with $\emptyset$ denoting the empty set. A measure satisfies the following natural properties: $\mu(\emptyset) = 0$; for any disjoint sets A and B , $\mu(A \cup B) = \mu(A) + \mu(B)$; and for any sets $A \subseteq B$, then $\mu(A) \leq \mu(B)$.

A probability measure is a measure with total mass equal to one, i.e., $\mu(X) = 1$. We focus on discrete probability measures on a finite space, where the σ -algebra is the power set of X , that is, the collection of all subsets of X , denoted by $\mathcal{P}(X)$.

For instance, consider a coin flip experiment with $X = \{\text{heads}, \text{tails}\}$. Then

$$\mathcal{P}(X) = \{\emptyset, \{\text{heads}\}, \{\text{tails}\}, X\}$$

If the coin always lands on heads, it can be modeled by the Dirac measure δ_{heads} . A Dirac measure at x assigns mass to a set depending solely on whether the set contains

x or not. For any measurable set A ,

$$\delta_x(A) = \begin{cases} 0 & \text{if } x \notin A \\ 1 & \text{if } x \in A \end{cases}$$

If the coin lands on heads with probability $p_{\text{heads}} \in [0, 1]$ and on tails with probability $p_{\text{tails}} = 1 - p_{\text{heads}}$, then μ is a Bernoulli measure given by

$$\mu = p_{\text{heads}}\delta_{\text{heads}} + p_{\text{tails}}\delta_{\text{tails}}.$$

More generally, a discrete probability measure [80] takes the form

$$\mu = \sum_{x \in X} p_x \delta_x, \quad \text{with } p_x \geq 0 \text{ and } \sum_{x \in X} p_x = 1.$$

For any $A \subseteq X$,

$$\mu(A) = \sum_{x \in A} p_x$$

3.4 Informal Introduction to Optimal Transport

We give a short introduction to optimal transport adapted to our needs; see [106] for a formal treatment.

Optimal transport studies how to transform a measure $\hat{\mu}$ into another μ while minimizing a transportation cost.

We consider discrete probability measures of the form

$$\mu = \sum_{i=1}^C p_i \delta_{a_i}, \quad \hat{\mu} = \sum_{j=1}^C q_j \delta_{b_j},$$

where $p_i, q_j \geq 0$ and $\sum_i p_i = \sum_j q_j = 1$.

A transport plan is a matrix $\pi \in \mathbb{R}_{\geq 0}^{C \times C}$, where π_{ij} represents the amount of mass transported from b_j to a_i . The set of admissible transport plans is

$$T(\mu, \hat{\mu}) = \left\{ \pi \in \mathbb{R}_{\geq 0}^{C \times C} : \sum_{j=1}^C \pi_{ij} = p_i, \sum_{i=1}^C \pi_{ij} = q_j \right\}. \quad (\text{III.2})$$

These constraints ensure that all the mass from the source measure is transported to match the target measure, and that transported mass remains non-negative.

To break ties between admissible transport plans, a cost function

$$c : \{a_i\}_{i=1}^C \times \{b_j\}_{j=1}^C \rightarrow \mathbb{R}_{\geq 0}$$

is needed. It assigns a cost to transporting one unit of mass from b_j to a_i [106]. A low cost encourages mass transfer between two locations, whereas a high cost discourages it. For instance, in a physical setting, one typically seeks to move mass over minimal distances, making the Euclidean distance a natural choice for the cost function.

The optimal transport problem, known as the Kantorovich problem, consists in finding a transport plan minimizing the total cost:

$$\operatorname{argmin}_{\pi \in T(\mu, \hat{\mu})} \sum_{i,j=1}^C c(a_i, b_j) \pi_{ij}. \quad (\text{III.3})$$

Solutions are called *optimal transport plans*.

Consider again the sand-and-flags analogy from Subsection 3.2. Suppose the model predicts **B** while the true label is **C**. We represent the label and the prediction by the measures μ and $\hat{\mu}$, respectively:

$$\mu = \delta_{\mathbf{C}}, \quad \text{and} \quad \hat{\mu} = \delta_{\mathbf{B}},$$

Algorithm 2 Transport-Based Confusion Matrix (TCM)

Require: Label-prediction pairs $\{(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)})\}_{k=1}^N$, with $(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)}) \in \mathbb{R}_{\geq 0}^C \times \mathbb{R}_{\geq 0}^C$, and a weighting function $w : \mathbb{R}_{\geq 0}^C \times \mathbb{R}_{\geq 0}^C \rightarrow \mathbb{R}_{\geq 0}$

- 1: Initialize $M \leftarrow 0 \in \mathbb{R}^{C \times C}$
- 2: **for** each label-prediction pair $(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)})$ **do**
- 3: $\mathbf{u} \leftarrow \frac{\mathbf{y}^{(k)}}{\|\mathbf{y}^{(k)}\|_1}, \quad \hat{\mathbf{u}} \leftarrow \frac{\hat{\mathbf{y}}^{(k)}}{\|\hat{\mathbf{y}}^{(k)}\|_1}$
- 4: $\mathbf{v} \leftarrow \min(\mathbf{u}, \hat{\mathbf{u}})$ {element-wise minimum}
- 5: **if** $\mathbf{u} = \hat{\mathbf{u}}$ **then**
- 6: $\pi^{(k)} \leftarrow \text{diag}(\mathbf{v})$
- 7: **else**
- 8: $\pi^{(k)} \leftarrow \text{diag}(\mathbf{v}) + \frac{(\mathbf{u} - \mathbf{v}) \otimes (\hat{\mathbf{u}} - \mathbf{v})}{\|\mathbf{u} - \mathbf{v}\|_1}$
- 9: **end if**
- 10: $M \leftarrow M + w(\mathbf{y}^{(k)}, \hat{\mathbf{y}}^{(k)})\pi^{(k)}$
- 11: **end for**
- 12: **return** M

where the probability space is $X = \{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}$, and the σ -algebra $\mathcal{P}(X)$. The goal is to transform $\hat{\mu}$ into μ . In the sand-and-flags analogy (see Subsection 3.2), this corresponds to moving the sand to the appropriate flags. The set of admissible transport plans $T(\mu, \hat{\mu})$ consists of all plans that transfer sand from the prediction configuration to the label configuration. In this setting, there exists a unique admissible plan, namely $T(\mu, \hat{\mu}) = \{E_{3,2}\}$. Consequently, the solution to the Kantorovich problem is $E_{3,2}$, independently of the chosen cost function.

4 Transport-based Confusion Matrix

This section details how to compute TCM, which is summarized in Algorithm 2. We begin with the Kantorovich problem relevant to our setting and define the associated cost function, whose solution yields a closed-form expression for TCM. We then interpret the formulas underlying TCM and explain the role of the weighting factor.

Let $\mathbf{y}$ and $\hat{\mathbf{y}}$ denote two vectors in $\mathbb{R}^C$ representing a ground-truth label and its associated prediction, respectively. For instance, in a three-class setting, a single-label annotation $y = 2$ can be encoded as $\mathbf{y} = [0, 1, 0]$. In a multi-label setting, if the model predicts classes 1 and 2, this corresponds to $\hat{\mathbf{y}} = [1, 1, 0]$. In a soft-label setting, the label may take non-negative real values, e.g., $\mathbf{y} = [0.8, 0.5, 0]$.

4.1 Problem Formalization

Adopting an optimal transport perspective on elementary matrices provides a natural way to extend them to multi-label and soft-label settings. To make optimal transport applicable, we interpret each label-prediction pair as a pair of measures and solve the associated Kantorovich problem:

$$\mathbf{y} \text{ is mapped to } \sum_{k=1}^C \mathbf{y}_k \delta_k, \quad \text{and } \hat{\mathbf{y}} \text{ is mapped to } \sum_{k=1}^C \hat{\mathbf{y}}_k \delta_k,$$

where the underlying space is $[C]$ equipped with the σ -algebra $\mathcal{P}([C])$.

For the set of admissible transport plans to be non-empty, the total mass of the measures associated with $\mathbf{y}$ and $\hat{\mathbf{y}}$ must be equal, i.e.,

$$\|\mathbf{y}\|_1 = \|\hat{\mathbf{y}}\|_1.$$

This condition is generally not satisfied in multi or soft-label settings, where the prediction may assign too much or too little mass. To address this, we compare

normalized vectors, $\mathbf{y}/\|\mathbf{y}\|_1$ and $\hat{\mathbf{y}}/\|\hat{\mathbf{y}}\|_1$, which ensures equal total mass (equal to 1) and yields the following probability measures:

$$\frac{\mathbf{y}}{\|\mathbf{y}\|_1} \text{ is mapped to } \sum_{k=1}^C \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} \delta_k, \quad \text{and} \quad \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} \text{ is mapped to } \sum_{k=1}^C \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} \delta_k$$

The Kantorovich problem is then formulated as

$$\arg \min_{\pi \in T(\mathbf{y}, \hat{\mathbf{y}})} \sum_{i,j=1}^C c(i, j) \pi_{ij}, \quad (\text{III.4})$$

where $T(\mathbf{y}, \hat{\mathbf{y}})$ denotes the set of admissible transport plans associated with the normalized label and prediction:

$$T(\mathbf{y}, \hat{\mathbf{y}}) = \left\{ \pi \in \mathbb{R}_{\geq 0}^{C \times C} : \pi_{i+} = \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \pi_{+j} = \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} \right\}. \quad (\text{III.5})$$

To complete the formulation, it remains to specify the cost function.

4.2 Cost Function

The cost function indicates the cost of transporting a unit of mass from one location to another [106].

In our setting, it reflects how difficult it is for the model to confuse class i with class j . For instance, if a classifier systematically confuses A and B, but never A and C, a suitable cost function would satisfy $c(\mathbf{A}, \mathbf{B}) = 0$ and $c(\mathbf{A}, \mathbf{C}) = \infty$. However, such information is not available a priori and is precisely what we aim to infer through the confusion matrix.

We therefore adopt a minimal assumption: a class is identical to itself and distinct from all others. This idea is encoded by the following cost function, which assigns 0 when the classes are identical and 1 otherwise:

$$c(i, j) = \begin{cases} 0 & \text{if } i = j \\ 1 & \text{if } i \neq j \end{cases}$$

This cost function is known as the *discrete metric* [80, 106].

4.3 Solutions

The following proposition characterizes the solutions to the Kantorovich problem (Equation III.4):

Proposition 3. *A matrix $\pi \in T(\mathbf{y}, \hat{\mathbf{y}})$ is an optimal transport plan if and only if*

$$\text{diag}(\pi) = \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right), \quad (\text{III.6})$$

where $\text{diag}(\pi)$ denotes the diagonal of π , and $\min(\mathbf{y}, \hat{\mathbf{y}})$ is understood element-wise.

According to Proposition 3, the set of optimal transport plans, denoted $T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$, is defined as

$$T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}}) = \left\{ \pi \in T(\mathbf{y}, \hat{\mathbf{y}}) : \text{diag}(\pi) = \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\} \quad (\text{III.7})$$

The next proposition shows that $T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ typically contains infinitely many solutions:

Proposition 4. *The set $T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ contains a unique optimal transport plan if and only if either the prediction is exact (i.e., $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$), or there exists at most one overestimated class (i.e., $\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} < \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}$), or there exists at most one underestimated class (i.e., $\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} < \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}$). Otherwise, it contains infinitely many optimal solutions.*

According to Propositions 3 and 4, the following hold for optimal plans:

- (i) There is a unique optimal plan when the number of classes satisfies $C \leq 3$;
- (ii) Any surplus mass from overestimated classes is transported only to underestimated classes;
- (iii) A plan is diagonal if and only if $\mathbf{y} = \hat{\mathbf{y}}$.

These statements are demonstrated in Appendix.

Because the cost function relies on minimal assumptions—namely, that a class is identical to itself and distinct from all others—excess mass can be redistributed to any underestimated class without affecting the cost. In particular, the transport cost from a class i to any other class $j \neq i$ is identical. As a result, there are typically infinitely many optimal transport plans.

To break ties between optimal plans, we adopt the principle of maximum entropy: when only partial information is available, one should select the probability distribution that maximizes entropy while satisfying the known constraints [45]. This yields the least biased estimate consistent with the available information. This principle is widely used in physics and has been applied in fields such as ecology, spectroscopy, imaging, and neuroscience [6]. The following proposition shows that there exists a unique entropy maximizer among optimal plans and provides its formula.

Proposition 5. *Let f be the function defined by*

$$\begin{aligned} f : \Delta^{C-1} \times \Delta^{C-1} &\rightarrow \mathbb{R}_{\geq 0}^{C \times C} \\ (u, \hat{u}) &\mapsto \text{diag}(\min(u, \hat{u})) + \frac{(u - \min(u, \hat{u})) \otimes (\hat{u} - \min(u, \hat{u}))}{\|u - \min(u, \hat{u})\|_1}, \end{aligned} \quad (\text{III.8})$$

where $\Delta^{C-1} = \{u \in \mathbb{R}_{\geq 0}^C : \sum_{i=1}^C u_i = 1\}$ denotes the probability simplex, and the minimum is taken elementwise. By continuous extension, f is also well defined at $u = \hat{u}$.

Then, there exists a unique entropy-maximizing transport plan in $T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$, denoted by $\pi(\mathbf{y}, \hat{\mathbf{y}})$, and it is given by

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = f\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)$$

Proposition 5 establishes a unified closed-form expression for the elementary matrix, valid across single, multi, and soft-label settings.

Remark. Considering the formula of f in Proposition 5, when $u = \hat{u}$, the denominator $\|u - \min(u, \hat{u})\|_1$ becomes zero. In the Appendix, we show that as $\|u - \hat{u}\|_1 \rightarrow 0$,

$$\frac{(u - \min(u, \hat{u})) \otimes (\hat{u} - \min(u, \hat{u}))}{\|u - \min(u, \hat{u})\|_1} \rightarrow 0 \in \mathbb{R}^{C \times C}.$$

Thus, we can rewrite

$$f = \begin{cases} \text{diag}(\min(u, \hat{u})) & \text{if } u = \hat{u}, \\ \text{diag}(\min(u, \hat{u})) + \frac{(u - \min(u, \hat{u})) \otimes (\hat{u} - \min(u, \hat{u}))}{\|u - \min(u, \hat{u})\|_1} & \text{otherwise.} \end{cases} \quad (\text{III.9})$$

These equalities are formalized through the concept of continuous extension.

4.4 Key Takeaways on the Elementary Matrix

To better understand the behavior of elementary matrices, we summarize here the key foundational and behavioral properties that characterize them. An interactive demonstration is also available at:

https://johanerbani.github.io/Elementary_matrix_of_TCM/

Within the TCM framework, elementary matrices satisfy the following principles:

- (i) Labels and predictions are compared *relatively* rather than absolutely. This is ensured by normalizing both the label and prediction vectors.
- (ii) A normalized label can be obtained from the normalized prediction by transporting units of predicted mass toward the appropriate classes. This transformation is formalized using optimal transport theory combined with the principle of maximum entropy, and can be intuitively understood through the sand-and-flag analogy (see Subsection 3.2).
- (iii) Elementary matrices define transport plans: entry (i, j) quantifies the mass transported from class j to class i to transform the normalized prediction into the normalized label. In particular, row sums match the normalized label distribution, column sums match the normalized prediction distribution, and the total sum equals one.
- (iv) The quantity that remains in place after alignment with the label distribution is considered *correct*. These quantities are recorded on the matrix diagonal: entry (i, i) represents the common mass at component i between the normalized label and prediction vectors.
- (v) The quantity that is moved to achieve alignment with the label distribution is considered *incorrect*. Excess mass from overestimated classes (i.e., where the prediction exceeds the label) is transported to underestimated classes (i.e., where the prediction is lower than the label). These quantities are recorded off-diagonal: entry (i, j) is nonzero only when class i is underestimated and class j is overestimated. In this case, it is proportional to both the deficit in class i and the excess mass in class j , i.e.,

$$\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1} \right) \left(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \frac{\mathbf{y}_j}{\|\mathbf{y}\|_1} \right).$$

- (vi) Elementary matrices follow specific value patterns. They are diagonal if and only if the prediction is exact, in which case

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \text{diag}\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}\right) = \text{diag}\left(\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)$$

When the prediction is not exact, only the rows corresponding to underestimated classes and the columns corresponding to overestimated classes contain nonzero values. The greater the underestimation of a class, the larger the values in the corresponding row tend to be; similarly, the greater the overestimation, the larger the values in the corresponding column. Moreover, if an entry (i, j) is nonzero, then the entry (j, i) is zero.

The above points are illustrated below:

$$\begin{array}{ccc}
 \pi(\mathbf{y}, \hat{\mathbf{y}}) = & \text{Correctly positioned mass} & + \quad \text{Mispositioned mass} \\
 & \downarrow & \downarrow \\
 & \text{Diag. matrix} & \text{Matrix with zero diag.} \\
 & \downarrow & \downarrow \\
 & \begin{array}{|c|c|c|c|} \hline \color{lightblue}1 & & & \\ \hline & \color{lightblue}1 & & \\ \hline & & \color{lightblue}1 & \\ \hline & & & \color{lightblue}1 \\ \hline \end{array} & \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \color{lightblue}1 & & & \color{lightblue}1 \\ \hline & \color{lightblue}1 & & \\ \hline & & \color{lightblue}1 & \\ \hline \end{array} \\
 & \underbrace{\hspace{10em}}_{\pi(\mathbf{y}, \hat{\mathbf{y}})_{ii}} & \underbrace{\hspace{10em}}_{\pi(\mathbf{y}, \hat{\mathbf{y}})_{ij}} \\
 & \parallel & \parallel \\
 & \left(\text{Common mass in } i \right) & \frac{\left(\text{Deficit in } i \right) \left(\text{Excess in } j \right)}{\left(\text{Total mispositioned mass} \right)} \\
 & \parallel & \parallel \\
 & \min\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}\right) & \frac{\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}\right)\right) \left(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}\right)\right)}{\left\| \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1}
 \end{array}$$

The elementary matrix interprets prediction errors as mass transport from predictions to labels. By separating correctly positioned mass from redistributed mass, while preserving the marginal distributions, it provides a concise and interpretable representation of discrepancies between predictions and labels.

4.5 Confusion Matrix

Let $\{(\mathbf{y}_k, \hat{\mathbf{y}}_k)\}_{k=1}^N$ denote the label–prediction vectors used to build the confusion matrix. Here, the subscript k indexes the data points, not the vector components.

The Transport-based Confusion Matrix (TCM) is defined as

$$\text{TCM} = \sum_{k=1}^N w(\mathbf{y}_k, \hat{\mathbf{y}}_k) \pi(\mathbf{y}_k, \hat{\mathbf{y}}_k), \quad (\text{III.10})$$

where $\pi(\mathbf{y}_k, \hat{\mathbf{y}}_k)$ is the optimal transport plan maximizing the entropy (see Proposition 5), and $w : \mathbb{R}_{\geq 0}^C \times \mathbb{R}_{\geq 0}^C \rightarrow \mathbb{R}_{\geq 0}$ is a weighting function. The scalar $w(\mathbf{y}_k, \hat{\mathbf{y}}_k)$ weights the contribution of $\pi(\mathbf{y}_k, \hat{\mathbf{y}}_k)$.

In multi or soft-label settings, the total mass (or number of active classes) may vary across label–prediction pairs, motivating the use of weighting schemes. Examples include:

- $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto 1$, where all contributions are equally weighted;
- $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto \|\mathbf{y}\|_1$, where the weight increases with the number of active classes in the label. For instance, an observation labeled **Apple**, **Lemon** contributes twice as much as one labeled **Apple**;
- $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto \frac{1}{\|\mathbf{y}\|_1}$, where the weight decreases with the number of active classes in the label. This can help break ties between class confusions, as label vectors with many active classes tend to distribute mass more evenly across the elementary matrix.

The choice of weighting is left to the user and may depend on the application. In practice, comparing multiple weightings can provide additional insight into model behavior.

Overall, the TCM aggregates elementary transport plans, providing a global representation of model behavior across the dataset. By summing these plans, it captures both correctly positioned mass and systematic misallocations, while preserving the underlying label and prediction distributions. Through weighting, it allows the matrix to adapt to specific needs. This construction extends the confusion matrix to multi and soft-label settings in a unified manner.

5 Comparison with the State of the Art

This section shows that TCM unifies several state-of-the-art approaches (Subsection 5.1) and is axiomatically more consistent (Subsection 5.2).

5.1 Unification

TCM follows the same underlying principles as existing confusion matrices and unifies several formulations that previously appeared incompatible.

To generalize the single-label confusion matrix, any extension to multi or soft-label settings should coincide with it in the single-label case.

Proposition 6. *TCM and the single-label matrix are identical in the single-label framework.*

Other state-of-the-art matrices, such as the MLCM and SCM, also exhibit this property.

In the MLCM framework, multi-label elementary matrices are defined through four ad hoc formulas, one for each case:

- (i) Exact prediction, $\mathbf{y} = \hat{\mathbf{y}}$;
- (ii) The label is a subset of the prediction, $\mathbf{y} \leq \hat{\mathbf{y}}$ (elementwise) and $\mathbf{y} \neq \hat{\mathbf{y}}$;
- (iii) The prediction is a subset of the label, $\hat{\mathbf{y}} \leq \mathbf{y}$ (elementwise) and $\hat{\mathbf{y}} \neq \mathbf{y}$;
- (iv) None of the above configurations, i.e., the general case.

Proposition 7. *In case (i), (ii), and (iii) MLCM and TCM weighted by $w : (\mathbf{y}, \hat{\mathbf{y}}) \mapsto \|\mathbf{y}\|_1$ produce identical elementary matrices.*

Our formula for case (iv) follows from a single, theoretically grounded expression. In contrast, the formula in [61] is ad hoc, constructed specifically for case (iv) without theoretical justification.

In the soft-label setting restricted to labels and predictions on the simplex (i.e., non-negative vectors summing to 1), Silván-Cárdenas and Wang [98] introduce SCM, a matrix with interval-valued entries. Although optimal transport is not explicitly used, their elementary matrix is closely related to ours. In particular, our approach recovers the contributions of SCM, as shown in the following proposition:

Proposition 8. *Elementary matrices from SCM framework can be computed as follows: for each entry (i, j) , select $\underline{\pi}$ and $\bar{\pi}$ in $T^{opt}(\mathbf{y}, \hat{\mathbf{y}})$ such that $\underline{\pi}_{ij} \leq \pi_{ij} \leq \bar{\pi}_{ij}$ for all $\pi \in T^{opt}(\mathbf{y}, \hat{\mathbf{y}})$, then define entry (i, j) as $[\underline{\pi}_{ij}, \bar{\pi}_{ij}]$.*

For a given label–prediction pair on the simplex, each entry of the TCM contribution lies within the corresponding SCM interval. While SCM entries (i, j) capture worst and best-case confusion scenarios, providing a pairwise description of errors, the TCM instead offers a global description that accounts for all classes simultaneously and preserves both marginal distributions and total mass. In particular, row sums match

the label distribution, column sums match the prediction distribution, and the total sum equals one.

Overall, TCM provides a unified framework for confusion matrices across single-label, multi-label, and soft-label settings. It recovers key properties of existing approaches while replacing ad hoc constructions with a single formulation based on optimal transport.

5.2 Axiomatic Requirements

To extend the confusion matrix to soft classifications, Silván-Cárdenas and Wang [98] propose that an elementary matrix should satisfy two key properties. First, *diagonalization*: the matrix is diagonal if and only if the prediction is exact. Second, *marginal consistency*: the row sums match the label vector, while the column sums match the prediction vector used to construct the elementary matrix.

Diagonalization allows clear identification of perfect matches. Marginal consistency ensures that performance metrics derived from the confusion matrix, such as accuracy or F1 score, remain consistent with the label and prediction vectors, thereby improving interpretability [98].

Proposition 9. *In the TCM framework, the elementary matrices satisfy both diagonalization and marginal consistency.*

In contrast, several state-of-the-art approaches do not meet these criteria. In particular, SCM [98], due to its interval-valued entries, does not satisfy marginal consistency, whether one considers the lower bound, the upper bound, or the midpoint of the intervals.

Overall, among existing approaches, TCM is the only method that simultaneously satisfies the proposed axiomatic properties while preserving the standard confusion matrix structure, making it the most consistent and interpretable framework.

6 Case Study: EXIST

This section presents a short case study illustrating the use of TCM.

EXIST is a series of scientific events and shared tasks on sexism identification in social networks [90]. One dataset contains several thousand annotated tweets for studying sexism detection and characterization. The benchmark defines three tasks described Table III.2: (i) sexism detection, (ii) intention classification, and (iii) fine-grained sexism categorization. Each tweet is labeled by six annotators with diverse profiles to reduce bias and capture uncertainty. In the soft-label setting, labels are computed as the average of annotator votes. We analyze a model from the EXIST 2025; see [83] for details on the architecture and training.

Figure III.2 presents the TCMs for the three tasks. The main conclusions are:

- **Task 1.** The matrix is strongly diagonal, indicating high classification performance with limited confusion between classes.
- **Task 2.** Diagonal values remain dominant, confirming good overall performance. However, **Direct** and **Non-sexist** show both high diagonals and strong columns, indicating over-prediction. Rows for **Judgemental** and **Reported** have weaker diagonals and are frequently confused with **Direct** and **Non-sexist**.
- **Task 3.** The diagonal structure is preserved, and **Non-sexist** remains over-predicted. **Sexual Violence** has a weak column, indicating it is rarely predicted. Rows for **Misogyny Non-Sexual Violence** and **Sexual Violence** show weak diagonals and higher off-diagonal mass, indicating frequent errors. Notable confusions include **Misogyny Non-Sexual Violence** being predicted as **Non-sexist**, and **Sexual Violence** being confused with **Misogyny Non-Sexual Violence**, **Stereotyping Dominance**, and **Non-sexist**.

Task	Setting	Class	Definition
1	Simplex	Non-sexist	Not sexist
		Sexist	Sexist or about sexism (e.g., denouncing a sexist act)
2	Simplex	Non-sexist	Not sexist
		Direct	Explicitly sexist
		Reported	Reports a sexist situation
		Judgemental	Denounces injustice against women
3	Soft	Non-sexist	Not sexist
		Ideological Inequality	Rejects gender equality or discredits feminism
		Stereotyping Dominance	Promotes stereotypes or male superiority
		Objectification	Reduces women to objects
		Sexual Violence	Sexual harassment or sexual content Expresses hatred or violence toward women
		Misogyny Non-Sexual Violence	

Table III.2: EXIST 2025 tweet classification tasks. Simplex denotes the soft-label setting, where labels lie on the simplex (i.e., non-negative vectors summing to 1).

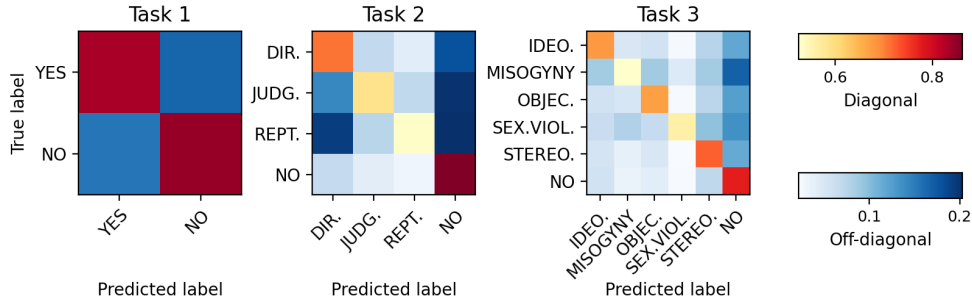


Figure III.2: Row-normalized TCMs with weights $w(\mathbf{y}, \hat{\mathbf{y}}) = 1$, illustrating model performance on the soft-label versions of Tasks 1, 2 and 3 from EXIST 2025. Since diagonal values are significantly larger than off-diagonal ones, two shared color scales are used to better highlight variations within each group.

Overall, all tasks show a clear diagonal structure, indicating good performance. However, the confusion matrices exhibit vertical stripes, reflecting under and over-prediction. Some errors are more pronounced, notably between **Reported** and **Direct**, and between **Misogyny Non-Sexual Violence** and **Non-sexist**.

7 Conclusion

This work addresses key limitations of existing confusion matrix extensions, including limited theoretical grounding, restricted applicability, and reduced interpretability.

We introduced the Transport-based Confusion Matrix (TCM), a unified framework based on optimal transport and the principle of maximum entropy. TCM extends

the standard confusion matrix to multi-label and soft-label settings while remaining identical to it in the single-label case. It provides a closed-form solution, preserves the familiar matrix structure, and satisfies key consistency properties.

We also showed that existing approaches share common principles that can be recovered within the TCM framework. In this sense, TCM clarifies the relationships between previous methods while replacing ad hoc constructions with a single theoretically grounded formulation.

Overall, TCM offers a consistent and interpretable tool for analyzing model behavior across a broad range of classification settings.

Chapter IV

Benchmarking Confusion Matrices Beyond Single-Label Classification

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This chapter is based on our preprint [28].

While multi-label and soft-label settings better reflect the complexity of real-world data, they also give rise to multiple competing definitions of confusion matrices, with no principled way to compare them. Existing work typically evaluates these formulations on benchmark datasets, but does not assess whether they faithfully represent the intrinsic behavior of the underlying model.

We introduce CONFUSIONBENCH, a benchmark for evaluating confusion matrix extensions in multi-label and soft-label classification settings. The benchmark relies on stochastic classifiers with controlled latent confusion structures, providing a ground-truth confusion matrix for assessing the recovery ability of confusion matrix extensions. Our results reveal substantial differences between existing matrices that prior evaluation practices failed to expose.

Contributions. Our contributions are as follows:

- **ConfusionBench.** We introduce CONFUSIONBENCH, a benchmark to evaluate how well confusion matrix extensions recover ground-truth confusions in multi-label and soft-label settings.
- **Empirical comparison.** We provide a systematic empirical comparison of existing confusion matrix extensions across controlled scenarios varying the number of classes, the complexity of confusion patterns, and the type of label representation.

1 Introduction

In multi-label and soft-label settings, several confusion matrix extensions coexist [61, 62, 41, 49, 29]. This leads to inconsistent model behavior interpretations and provides no clear guidance on which method should be used.

Problem & motivation. Multi-label settings, where observations may belong to several classes, and soft-label settings, where observations are represented by non-negative vectors encoding degrees of membership or confidence, better capture real-world complexity [107, 76, 74, 100, 21, 75, 16, 109, 105, 18]. However, extending the confusion matrix to these settings is not straightforward.

As a result, multiple extensions of confusion matrices have been proposed. Most existing works [61, 49, 62, 41, 10, 98, 29] either visualize their matrices on benchmark datasets to interpret model behavior or compare them qualitatively with alternative formulations. However, because no ground-truth confusion matrix characterizing the intrinsic behavior of a model is available, these comparisons lack a clear quantitative criterion, making it unclear which method should be preferred in practice.

ConfusionBench. We introduce CONFUSIONBENCH, a framework for evaluating confusion matrices based on their ability to recover known confusion patterns. Our approach reverses the standard pipeline: instead of estimating confusion from a trained model, we construct stochastic classifiers with controlled, known confusion structures, and generate synthetic label-prediction pairs accordingly. This provides a ground-truth reference matrix, enabling direct and quantitative comparison between methods.

2 Related Work

2.1 Evaluation Methods

Existing works evaluate confusion matrix extensions using four main approaches.

Qualitative analysis. Matrices are visualized on benchmark datasets to interpret model behavior [61, 49, 62, 41, 10, 98, 29]. This mainly illustrates how to read the proposed representation.

Relative comparison. Methods are compared against existing formulations to highlight similarities and differences [62, 98, 29, 10, 60]. While informative, this only provides relative comparisons and does not assess absolute quality.

Metric recovery. Some works evaluate whether the proposed matrix can recover scalar performance metrics [62, 98, 29, 10].

Axiomatic validity. Some works define desirable properties that a confusion matrix extension should satisfy in order to remain consistent with the classical single-label case [98, 29]. For example, an extension may be required to reduce to the standard confusion matrix for single-label predictions, or to be diagonal under perfect predictions. Such criteria are useful for ruling out invalid formulations, but they do not quantify how well a valid method captures the underlying confusion behavior.

Algorithm 3 ConfusionBench

Require: Number of classes C , number of samples N , concentration parameter α , set of confusion matrix extensions $\mathcal{A}$

Ensure: Evaluation scores for all methods in $\mathcal{A}$

- 1: Generate a latent row-stochastic confusion matrix $Q \in \mathbb{R}_{\geq 0}^{C \times C}$ with concentration parameter α
- 2: **for** $k = 1$ to N **do**
- 3: Sample a label vector y_k
- 4: Generate a prediction vector $\hat{y}_k$ conditionally on y_k using Q
- 5: Apply dropout noise by randomly removing active entries from y_k or $\hat{y}_k$
- 6: **end for**
- 7: **for** each method $A \in \mathcal{A}$ **do**
- 8: Compute the estimated confusion matrix M_A from $\{(y_k, \hat{y}_k)\}_{k=1}^N$
- 9: Compare $\text{row}(M_A)$ with Q using Overlap and Jensen–Shannon distance
- 10: **end for**
- 11: **return** Evaluation scores for all methods in $\mathcal{A}$

While useful, these approaches do not provide a ground-truth confusion matrix describing the true behavior of the model. As a result, they cannot determine which method best captures the underlying confusion structure.

2.2 Unavailable Ground-Truth & Synthetic Evaluation

The absence of ground truth is a common challenge in machine learning. To address this issue, synthetic generative settings are widely used in several areas of machine learning to provide controlled ground-truth structures for evaluation.

In causal inference, synthetic data are commonly used to evaluate the recovery of known causal structures [34, 38, 66, 81]. Similarly, synthetic benchmarks in disentanglement analysis, explainable artificial intelligence, and visual question answering [77, 57, 53, 79, 72] have been introduced to provide controlled, reproducible, and unbiased ground-truth structures, enabling precise evaluation of specific reasoning capabilities.

2.3 Our Contribution

We introduce CONFUSIONBENCH, a synthetic benchmark for evaluating confusion matrix extensions in multi-label and soft-label settings. By constructing stochastic classifiers with controlled confusion patterns, CONFUSIONBENCH provides a ground-truth reference matrix that enables direct and quantitative comparison between methods.

3 ConfusionBench

CONFUSIONBENCH evaluates confusion matrix extensions by comparing the estimated confusion matrix with a known target confusion structure. The closer a method is to the target matrix, the more faithfully it captures the intrinsic behavior of the classifier. The CONFUSIONBENCH procedure is detailed below and summarized in Algorithm 3.

3.1 Classifiers from Confusion Matrices

In the single-label setting, the normalized confusion matrix admits a well-known probabilistic interpretation [41] including:

$$\text{all}(P)_{ij} = \frac{P_{ij}}{P_{++}} \approx \mathbb{P}(Y = i, \hat{Y} = j), \quad \text{row}(P)_{ij} = \frac{P_{ij}}{P_{i+}} \approx \mathbb{P}(\hat{Y} = j \mid Y = i),$$

where $P_{++} = \sum_{i,j} P_{ij}$ and $P_{i+} = \sum_j P_{ij}$. Therefore, the row-normalized confusion matrix characterizes the conditional prediction behavior of the classifier given the true class.

Building on this observation, CONFUSIONBENCH constructs stochastic classifiers whose behavior is fully determined by a row-stochastic matrix Q (i.e., $Q_{ij} \geq 0$ and $\sum_{j=1}^C Q_{ij} = 1$ for all i), representing its latent confusion patterns. Given a sample (x, y) with label $y \in [C]$, the classifier produces a prediction $\hat{y} \in [C]$ sampled from the distribution defined by the y -th row of Q , that is, $\hat{y} = j$ with probability Q_{yj} .

In contrast to standard classifiers, which produce predictions based on inputs x and are typically deterministic at inference time, this synthetic classifier is stochastic and conditioned only on the labels. This construction allows us to control and know exactly the intrinsic confusion pattern of the classifier, thereby providing a target confusion matrix to recover.

3.2 Latent Confusion Matrices

To generate model behaviors with controlled confusion patterns, we construct a random row-stochastic matrix $Q \in \mathbb{R}_{\geq 0}^{C \times C}$, where each row i defines the classifier behavior conditioned on the true class: the classifier predicts class j with probability Q_{ij} when the true label is i .

The matrix Q is generated according to the following procedure, designed to reproduce common confusion patterns observed in classifiers:

1. **Row-stochasticity.** Each row is constructed by averaging samples drawn from a Dirichlet distribution $\text{Dir}(\alpha)$, ensuring that every row lies on the probability simplex. The concentration parameter α controls the sparsity of the resulting distributions: smaller values ($\alpha < 1$) produce more heterogeneous vectors with a few dominant entries, whereas larger values yield more diffuse distributions.
2. **Structured confusion.** Each row may contain multiple dominant components, corresponding to classes that are frequently confused. Let K denote the target number of dominant entries. When α is sufficiently small relative to C , averaging K independent Dirichlet samples produces rows exhibiting approximately K prominent modes.
3. **Diagonal dominance.** To model classifiers that perform better than random guessing, we enforce diagonal dominance by ensuring that, for each class i , the diagonal entry Q_{ii} is the largest coefficient in row i . In practice, this is achieved by swapping the diagonal entry Q_{ii} with the maximum value in row i .
4. **Confusion dispersion.** The parameter α also governs the global dispersion of confusion mass across classes. Small values of α concentrate probability mass on a limited subset of classes, while larger values induce more uniformly distributed confusion patterns.

Algorithm 4 summarizes the construction of Q , and Figure IV.1 presents heatmaps of the resulting latent confusion matrices.

3.3 Synthetic Label-Prediction Pairs

A dataset of label-prediction pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$ is required to derive confusion matrices. We detail below the procedure used to generate these pairs, together with the generation process interpretation. The data generation procedures is detailed in Algorithm 5.

Observation-level interpretation. For simplicity, our framework generates classifier predictions $\hat{y}_k$ directly from the labels y_k , without explicitly modeling the observations x_k . Nevertheless, this procedure admits a natural interpretation.

Each observation x_k is composed of n_k components $\{q_l\}_{l=1}^{n_k}$. Each component is associated with at most one class, independently of the labeling setting considered, and may also be associated with no class. Labels y_k and predictions $\hat{y}_k$ are obtained by independently processing these components and aggregating the resulting class information.

Algorithm 4 Sampling a Latent Confusion Matrix

Require: Number of classes C , Dirichlet concentration $\alpha > 0$, maximum number of modes n

Ensure: A row-stochastic matrix $Q \in \mathbb{R}_{\geq 0}^{C \times C}$ with diagonal dominance

1: **for** $i = 1$ **to** C **do**

2: Sample the number of active modes

$$K_i \sim \text{Unif}\{1, \dots, n\}.$$

3: Initialize $q_i \leftarrow \mathbf{0}_C$

4: **for** $k = 1$ **to** K_i **do**

5: Sample

$$z_k \sim \text{Dir}(\alpha \mathbf{1}_C)$$

6: Update $q_i \leftarrow q_i + z_k$

7: **end for**

8: Average the sampled modes:

$$q_i \leftarrow \frac{1}{K_i} q_i$$

9: Find the largest entry in the row:

$$j^* \leftarrow \arg \max_{j \in \{1, \dots, C\}} q_{ij}$$

10: Swap q_{ii} and q_{ij^*}

11: Set the i -th row of Q to q_i

12: **end for**

13: **return** Q

The nature of the components and the aggregation rule depend on the classification setting. In single-label classification, each observation contains a single component, i.e., $n_k = 1$ for all k . This component represents an object. For instance, it may correspond to a **Dog**. Aggregation then simply activates the predicted class. In multi-label classification, components also represent objects, but their number may vary across observations and be larger than one, i.e., $n_k \geq 1$. Aggregation activates every class predicted by at least one component. In soft-label classification, observations are decomposed into many components representing local patterns, such as tokens associated with **Physics** topics. Class membership is measured by the number of times each class is predicted.

The classifier processes these components independently and assigns each of them either to a class or to no class:

- **Labeled-predicted.** The component belongs to class i . The classifier detects it and predicts class j with probability Q_{ij} .
- **Unlabeled-predicted.** The component is assigned to no class, but the classifier interprets it as close to class i . It then predicts a class according to the latent confusion pattern $Q_{i,:}$.
- **Labeled-unpredicted.** The component belongs to a class but does not trigger any prediction.

The unlabeled-unpredicted case is ignored, since it has no effect.

This interpretation is formalized in Algorithm 5, even though the observation x_k is not explicitly modeled. Labels are assumed to be directly induced by the observation. Prediction generation only requires the amount of components associated with each class, which is encoded in the label. No-class components are not explicitly represented, but are mimicked through dropout: removing label classes mimics the unlabeled-predicted case, while removing prediction classes mimics the labeled-unpredicted case.

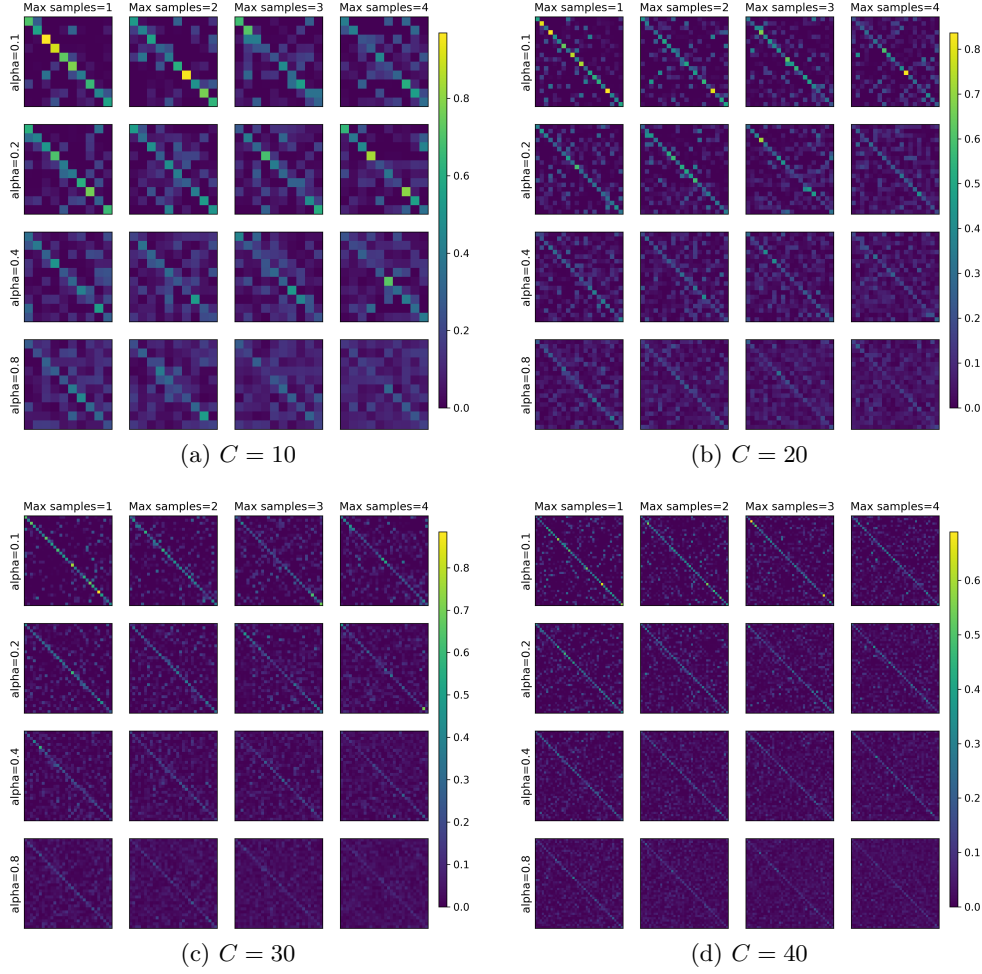


Figure IV.1: Latent confusion matrices (run 0) across different numbers of classes, ranging from simple confusion patterns (Max samples = 1) to more complex ones (Max samples = 4), and from concentrated patterns ($\alpha = 0.1$) to more dispersed ones ($\alpha = 0.8$).

Finally, Algorithm 5 keeps only nonzero pairs: the label must contain at least one active class, and the model must predict at least one class.

Single-label dataset. The single-label setting is obtained from Algorithm 5 by setting $L = S = 1$. Each sample therefore contains one active class and one component. We initialize y_k and $\hat{y}_k$ as zero vectors, select a class i , set $[y_k]_i = 1$, sample j from $Q_{i,:}$, and set $[\hat{y}_k]_j = 1$.

Multi-label dataset. The multi-label setting is obtained by setting $L > 1$ and $S = 1$. Each sample may contain several active classes, with one component per active class. After initializing y_k and $\hat{y}_k$, we select l active classes in y_k . For each active class i , we sample a predicted class j from $Q_{i,:}$ and activate j in $\hat{y}_k$.

At this stage, labels and predictions contain the same number of active classes, which is unrealistic in practice. We therefore introduce a dropout mechanism that randomly removes active labels or predictions, modeling missing labels and missing predictions.

Soft-label dataset. The soft-label setting is obtained by setting $L, S > 1$. Each active class may then be supported by several components. For each active class i , we sample an integer $m_i \in \{1, \dots, S\}$, representing the number of components supporting class i , and set

$$[y_k]_i \leftarrow m_i.$$

Each component is then predicted independently according to $Q_{i,:}$. Thus, for $r = 1, \dots, m_i$, we sample $j \sim Q_{i,:}$ and update

$$[\hat{y}_k]_j \leftarrow [\hat{y}_k]_j + 1.$$

Repeating this process over all active classes yields soft-label vectors $(y_k, \hat{y}_k)$, whose entries count class-level evidence.

As in the multi-label case, dropout is applied to create differences in support and total mass between labels and predictions.

3.4 Ground-Truth Justification

This subsection gives two independent arguments for why any row-normalized confusion matrix extension should be consistent with the latent matrix Q .

Consistency across classification settings. The stochastic classifier defined by Q has the same confusion pattern regardless of the classification setting. Therefore, a confusion matrix extension should recover the same matrix across single-label, multi-label, and soft-label settings.

Using Algorithm 5 with $L = 1$ generates a single-label dataset $\{(y_k, \hat{y}_k)\}_{k=1}^N$. Let M be the standard confusion matrix computed from this dataset. Since predictions are sampled from the rows of Q , the row-normalized confusion matrix converges almost surely to Q :

$$\text{row}(M) \xrightarrow[N \rightarrow \infty]{} Q \quad \text{almost surely.}$$

Moreover, valid confusion matrix extensions reduce to the standard confusion matrix in the single-label setting. Hence, they are consistent estimators of Q in this regime.

This provides a natural requirement: if the underlying classifier behavior is still governed by Q , then a confusion matrix extension should recover Q regardless of the classification setting.

Beyond co-occurrence. The single-label confusion matrix M estimates label-prediction co-occurrence:

$$\text{all}(M)_{ij} \xrightarrow[N \rightarrow \infty]{} \mathbb{P}(Y = i, \hat{Y} = j) \quad \text{almost surely.}$$

However, co-occurrence is not the intended object in general. We view a confusion matrix as aiming to capture a more specific relation: class i is present in the observation and is recognized as class j . The target is therefore

$$\mathbb{P}(\text{class } i \text{ is recognized as class } j).$$

This relation is stricter than co-occurrence. For example, suppose that x contains a **Dog** component that produces no prediction, and a no-class component predicted as **Wolf**. Then the label **Dog** and the prediction **Wolf** co-occur, but the model has not confused **Dog** with **Wolf**.

In real data, this component-level relation is generally unobserved. In our synthetic framework, it is available because the decision process is known. We can therefore define an ideal matrix $\hat{Q}$ in any classification setting: initialize $\hat{Q}$ to zero, and increment $\hat{Q}_{ij}$ each time a component of class i is predicted as class j . Components with no class, or predictions with no class, do not update the matrix, since they do not encode a class-to-class confusion.

After row-normalization, this ideal matrix converges independently of the classification setting:

$$\text{row}(\hat{Q})_{ij} \xrightarrow[N \rightarrow \infty]{} Q_{ij} = \mathbb{P}(\text{class } i \text{ is recognized as class } j \mid i \text{ is present}) \quad \text{almost surely.}$$

Thus, any confusion matrix extension that aims to capture this component-level relation should recover Q after row-normalization.

In the single-label setting of our framework, each observation contains one labeled component and produces one predicted class. Hence, component-level recognition

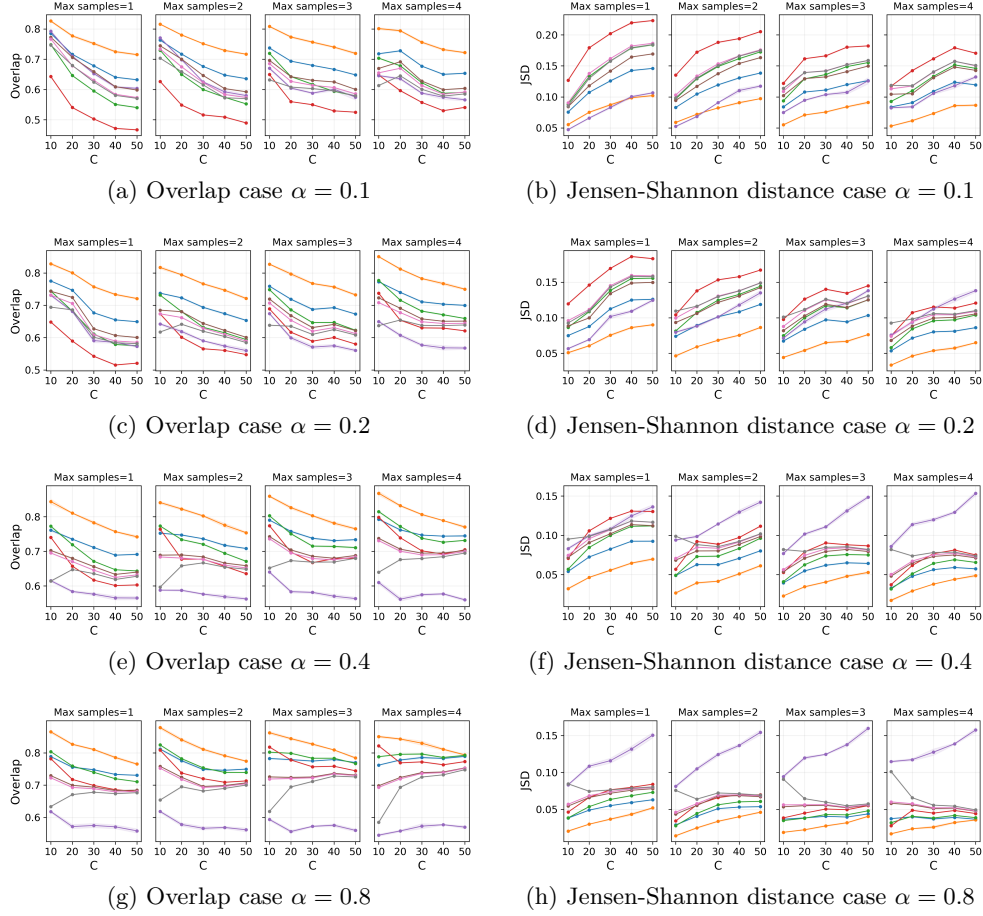


Figure IV.2: Agreement in the multi-label setting between ground-truth and empirical confusion matrices under varying noise levels and class cardinalities, averaged over 5 seeds ($\pm$ standard deviation). Max samples controls the maximum number of modes per row, while α controls the confusion dispersion: higher values produce more uniform distributions. Legend: $\bullet$ TCM^{one} $\bullet$ $\text{TCM}^{\text{inv-lab}}$ $\bullet$ SCM^{mid} $\bullet$ SCM^{max} $\bullet$ SCM^{min} $\bullet$ MLCM $\bullet$ MLCT^{r} $\bullet$ MLCT^{p}

coincides with label–prediction co-occurrence. The standard confusion matrix therefore estimates the intended quantity in this setting.

Based on these arguments, consistency with Q is a desirable property for any confusion matrix extension M . The closer $\text{row}(M)$ is to Q , the more faithfully M captures the underlying classifier behavior.

4 Experimental Setup

4.1 Scenarios & Hyperparameters

We consider 16 scenarios in both the multi-label and soft-label settings, ranging from simple confusion patterns (Max samples = 1) to more complex ones (Max samples = 4), and from concentrated patterns ($\alpha = 0.1$) to more dispersed ones ($\alpha = 0.8$). Specifically, Max samples takes values in $\{1, 2, 3, 4\}$ and $\alpha \in \{0.1, 0.2, 0.4, 0.8\}$. We additionally vary the number of classes, with $C \in \{10, 20, 30, 40, 50\}$.

For each experiment, we generate $N = 15000$ label–prediction pairs and estimate the corresponding confusion matrices. We use a dropout probability $p = 0.25$ and limit the number of active classes per label vector to at most 10. Each experiment is repeated over 5 random seeds.

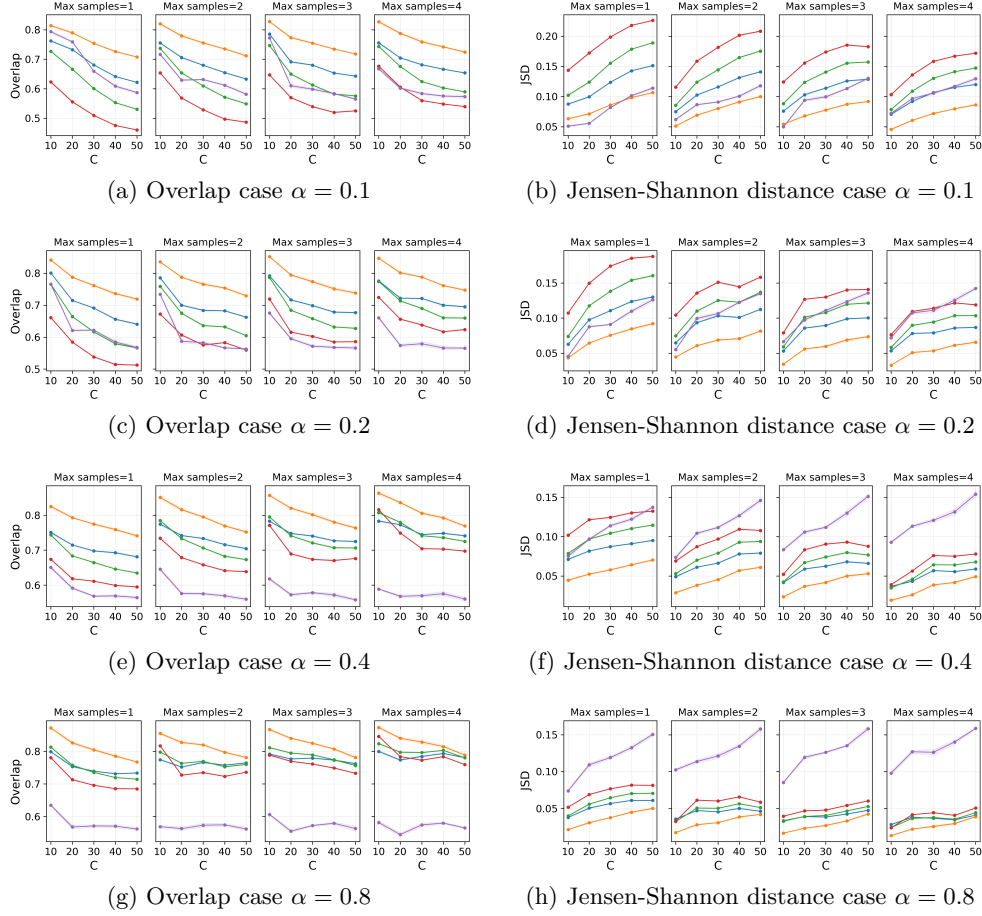


Figure IV.3: Agreement in the soft-label setting between ground-truth and empirical confusion matrices under varying noise levels and class cardinalities, averaged over 5 seeds ($\pm$ standard deviation). Max samples controls the maximum number of modes per row, while α controls the confusion dispersion: higher values produce more uniform distributions. Legend: $\bullet$ TCM^{one} $\bullet$ $\text{TCM}^{\text{inv-lab}}$ $\bullet$ SCM^{mid} $\bullet$ SCM^{max} $\bullet$ SCM^{min}

4.2 Baselines

We consider confusion matrices satisfying the following requirements: being square matrices of size C and being diagonal under perfect predictions.

We compare the following methods. The Multi-Label Confusion Matrix (MLCM) [61], and the two confusion tensors proposed by Krstinić et al. [62]: the Multi-Label Confusion Tensor Recall (MLCT^r) and Precision (MLCT^p).

The Sub-pixel Confusion Matrix (SCM) [98] is designed for label–observation pairs lying on the simplex and produces interval-valued matrices. Following the authors’ representations, we consider three variants: SCM^{min} (lower bound), SCM^{mid} (mid-point), and SCM^{max} (upper bound). We additionally evaluate SCM using normalized labels and normalized predictions.

The Transport-based Confusion Matrix (TCM) [29] includes a weighting factor w that must be specified according to the application context. We consider two variants: a constant weighting factor $w : (y, \hat{y}) \mapsto 1$, denoted TCM^{one} , and an inverse-label weighting factor $w : (y, \hat{y}) \mapsto 1/|y|$, denoted $\text{TCM}^{\text{inv-lab}}$. In TCM^{one} , all label–prediction pairs contribute equally. In $\text{TCM}^{\text{inv-lab}}$, pairs associated with labels containing more classes contribute less to the final matrix.

In the multi-label setting, we compare all aforementioned methods. In the soft-label setting, we only compare SCM and TCM variants, as the other methods are not designed for soft labels.

4.3 Metric

To quantify the agreement between estimated and latent confusion matrices, we consider the overlap metric [32], with a variant in which M is first row-normalized to match the row-stochastic structure of Q :

$$\text{Overlap}(M) = \sum_{i,j} \min(\text{all} \circ \text{row}(M)_{ij}, \text{all}(Q)_{ij}) \in [0, 1].$$

The overlap quantifies the amount of shared probability mass between the distributions $\text{all} \circ \text{row}(M)$ and $\text{all}(Q)$.

We additionally report the Jensen-Shannon divergence, which provides a symmetric and bounded measure of discrepancy between distributions.

5 Evaluation

5.1 Multi-label case

Figure IV.2 reports the agreement between the latent confusion matrix Q and the confusion matrices estimated by each method in the multi-label setting. We evaluate this agreement using both the overlap metric, where higher values indicate better recovery, and the Jensen-Shannon divergence, where lower values indicate better recovery.

Overall, the transport-based methods provide the most faithful recovery of the latent confusion structure. In particular, $\text{TCM}^{\text{inv-lab}}$ consistently achieves the highest overlap and the lowest Jensen-Shannon divergence across all scenarios. This result is expected in our experimental setting: the classifier behavior is defined independently of the number of active classes in the label, while labels with fewer active classes induce simpler confusion patterns that are easier to recover. As a result, weighting each label-prediction pair by the inverse number of active labels provides a more faithful representation of the classifier behavior. By contrast, TCM^{one} also performs well: it outperforms the non-TCM methods when confusion patterns are concentrated, i.e., for $\alpha < 0.8$, and remains comparable to the best non-TCM methods, such as SCM, when $\alpha = 0.8$.

The SCM variants exhibit intermediate performance. Among them, SCM^{mid} consistently outperforms SCM^{min} and SCM^{max} . Although it generally performs worse than TCM^{one} , except when $\alpha = 0.8$, it still achieves better overall performance than the multi-label-specific baselines.

MLCM , MLCT^r , and MLCT^p are the least competitive methods. They generally achieve lower overlap and higher Jensen-Shannon divergence, suggesting that, despite being specifically designed for multi-label classification, they do not accurately recover the underlying row-wise confusion behavior of the synthetic classifier.

We also observe that the number of classes C has a systematic effect on recovery quality. As C increases, performance tends to decrease. This is expected: when the number of classes grows, the confusion mass is distributed over a larger matrix, making the recovery problem more difficult for a fixed number of samples.

5.2 Soft-label case

Figure IV.3 reports the same evaluation in the soft-label setting. Since MLCM and MLCT are not designed for soft labels, we compare only the SCM and TCM variants. The soft-label setting yields results analogous to those observed in the multi-label setting.

The transport-based methods again recover the latent confusion matrix more accurately than the SCM variants, with larger performance gaps for small values of α . The ranking between methods remains similar: $\text{TCM}^{\text{inv-lab}}$ consistently outperforms the other approaches, TCM^{one} performs better than SCM^{mid} , and SCM^{min} and SCM^{max} are the least reliable SCM variants.

As in the multi-label case, increasing the number of classes tends to make the recovery task harder. Nevertheless, the ranking of methods remains stable across class cardinalities, supporting the robustness of the observed trends.

These results show that CONFUSIONBENCH can distinguish between confusion matrix extensions that may otherwise all appear valid according to standard axiomatic criteria.

6 Limitations

CONFUSIONBENCH has several limitations, which we discuss below.

Use of labels by the classifier. The classifier does not follow the standard prediction mechanism of a learned model: predictions are sampled conditionally on labels rather than inferred from observations. However, as discussed in Section 3.3, this construction has a natural interpretation in terms of elementary components of the observation.

Class independence. The classifier treats classes independently. This simplification ignores potentially complex dependencies between classes, which a learned model may capture in real-world settings. Consequently, the ground-truth confusion structure may be easier to recover than in practice, potentially leading to optimistic performance estimates.

Specific latent patterns. The confusion patterns are generated according to a specific stochastic process, which may not capture the full diversity of behaviors observed in practical classifiers. However, the assumptions underlying this process are intentionally minimal and are designed to reflect common confusion patterns, such as diagonal dominance, sparse or diffuse errors, and varying numbers of dominant confusions per class.

Synthetic benchmark. CONFUSIONBENCH relies on synthetic data rather than predictions produced by models trained on real datasets. This choice is central to the proposed evaluation protocol, since it makes the latent confusion structure observable and enables direct quantitative comparison between confusion matrix extensions. However, synthetic benchmarks necessarily simplify real-world conditions.

Overall, the results reported in CONFUSIONBENCH should be interpreted as a controlled analysis of confusion matrix extensions rather than as a universal ranking of these methods. Under different assumptions, such as classifiers that are not better than random, or class-dependent prediction mechanisms, the ranking of methods may differ.

7 Conclusion

We introduced CONFUSIONBENCH, a synthetic benchmark for evaluating confusion matrix extensions beyond the single-label setting. Instead of estimating confusion from an unknown classifier, CONFUSIONBENCH constructs stochastic classifiers with known latent confusion structures, enabling direct quantitative evaluation of how accurately generalized confusion matrices recover the underlying confusion behavior.

Our experiments in both multi-label and soft-label settings show that existing confusion matrix extensions differ substantially in their ability to recover the latent confusion behavior. Across a wide range of controlled scenarios, the transport-based confusion matrices are the most faithful to the underlying structure, with the inverse-label weighting variant usually obtaining the best agreement. SCM-based methods provide a reasonable but less accurate alternative, while multi-label specific methods such as MLCM and MLCT are less effective.

These findings highlight the importance of evaluating confusion matrices not only through qualitative visualizations, metric recovery, or axiomatic consistency, but also through their faithfulness to the intrinsic behavior of the classifier. CONFUSIONBENCH provides a principled way to perform this evaluation by making the target confusion structure observable.

Overall, CONFUSIONBENCH provides both a diagnostic tool for comparing confusion matrix extensions and practical guidance for selecting faithful representations of model behavior in complex classification settings.

Algorithm 5 Synthetic Data Generation

Require: Number of samples N , number of classes C , latent confusion matrix $Q \in \mathbb{R}_{\geq 0}^{C \times C}$, maximum number of active classes L , dropout probability p , maximum number of components per class S

Ensure: Soft-label count pairs $\{(y_k, \hat{y}_k)\}_{k=1}^N$, with $y_k, \hat{y}_k \in \mathbb{N}_0^C$

```

1: Initialize  $k \leftarrow 1$ 
2: while  $k \leq N$  do
3:   // Initialize a candidate label-prediction pair
4:   Initialize  $y \leftarrow \mathbf{0}_C$  and  $\hat{y} \leftarrow \mathbf{0}_C$ 
5:   // Sample the active true classes in the observation
6:   Sample the number of active true classes:
        $\ell \sim \text{Unif}\{1, \dots, L\}$ 
7:   Sample a set  $A \subseteq \{1, \dots, C\}$  of  $\ell$  distinct classes uniformly without replacement
8:   // Generate the components
9:   for each  $i \in A$  do
10:    Sample the number of components supporting class  $i$ :
         $m_i \sim \text{Unif}\{1, \dots, S\}$ 
11:    Set
         $[y]_i \leftarrow m_i$ 
12:  end for
13:  // Generate component predictions from the latent confusion matrix
14:  for each  $i \in A$  do
15:    for  $r = 1$  to  $[y]_i$  do
16:      Sample a predicted class:
         $j \sim \text{Categorical}(Q_{i,:})$ 
17:      Update the prediction:
         $[\hat{y}]_j \leftarrow [\hat{y}]_j + 1$ 
18:    end for
19:  end for
20:  // Apply dropout to create missing labels or missing predictions
21:  if  $k$  is even then
22:    // Remove some label components
23:    for each  $i$  such that  $[y]_i > 0$  do
24:      Sample the number of dropped label components:
         $b_i \sim \text{Binomial}([y]_i, p)$ 
25:      Remove them from the label:
         $[y]_i \leftarrow [y]_i - b_i$ 
26:    end for
27:  else
28:    // Remove some prediction components
29:    for each  $j$  such that  $[\hat{y}]_j > 0$  do
30:      Sample the number of dropped prediction components:
         $b_j \sim \text{Binomial}([\hat{y}]_j, p)$ 
31:      Remove them from the prediction:
         $[\hat{y}]_j \leftarrow [\hat{y}]_j - b_j$ 
32:    end for
33:  end if
34:  // Keep only valid pairs with at least one label and one prediction
35:  if  $y \neq \mathbf{0}_C$  and  $\hat{y} \neq \mathbf{0}_C$  then
36:    Accept the pair:
         $(y_k, \hat{y}_k) \leftarrow (y, \hat{y})$ 
37:     $k \leftarrow k + 1$ 
38:  end if
39: end while
40: return  $\{(y_k, \hat{y}_k)\}_{k=1}^N$ 

```

Part II

Byzantine-Robust Federated Learning under Label Skew

Chapter V

Preliminaries

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1 Overview

Federated learning (FL) enables a set of distributed workers to train a shared model while keeping their raw data local [43, 42]. At each round, a trusted server broadcasts the current global model. Each worker computes an update from its local data and sends it back to the server, which aggregates the received updates to form the next global model. We focus on the synchronous gradient-based setting, where honest worker updates are stochastic gradients computed on local minibatches.

Under ideal conditions, the server aggregates honest gradients by averaging. However, some workers may be faulty or malicious and may send arbitrary corrupted updates, called Byzantine updates. Byzantine-robust training aims to limit the influence of such workers [42, 43]. A standard strategy replaces averaging with a robust aggregation rule [43, 42]. These rules are designed to reduce the influence of abnormal updates, but their guarantees often rely on honest gradients being sufficiently concentrated around a common direction [42, 4, 88, 7]. This assumption often fails under data heterogeneity, a central challenge in federated learning [4, 54, 35].

Data heterogeneity increases the dispersion of honest gradients, making Byzantine updates harder to distinguish from legitimate ones [4, 43, 54]. This work focuses on label skew, a common form of heterogeneity where workers have different label distributions but share the same class-conditional feature distributions. For example, in a federated medical project, hospitals may observe different disease prevalences because of their specialization. One hospital may treat many pneumonia cases, while another rarely does. This corresponds to label skew when $P_i(Y)$ varies across hospitals, while $P_i(X | Y = c)$ remains identical across hospitals for every class c . Under label skew, each worker optimizes a local objective biased toward its own label distribution [54, 119, 7]. As a result, honest gradients can differ substantially even in the absence of Byzantine behavior.

2 Background

2.1 Federated Learning Setting

We consider a federated learning setting with n workers and a trusted central server. The goal is to train a parametrized classifier Φ_θ over C classes without centralizing the workers' data. Each honest worker i owns a local dataset $\mathcal{D}_i$ of observation-label pairs $(x, y) \in \mathcal{X} \times [C]$, where $\mathcal{X}$ is the input space and $[C] = \{1, \dots, C\}$.

At each communication round, the server broadcasts the current parameter vector θ_t to the workers. Each worker computes a local update from its data, typically a stochastic gradient estimated on a mini-batch, and sends it back to the server. The server aggregates the received updates to produce the next global model.

2.2 Threat Model

We study Byzantine-robust federated learning with $n = h + f$ workers, where h workers are honest and $f < h$ may be Byzantine. The set of honest workers is denoted by $\mathcal{H} \subset [n]$. The server is assumed to be trusted.

Byzantine workers may send arbitrary updates to the server. Following the standard threat model in [4, 54, 7], we assume that Byzantine workers may have full knowledge of the honest workers' updates. The objective of the server is therefore to compute an aggregate update close to the one that would have been obtained from honest workers only.

2.3 Data & Label-Skew Model

Gathering all local datasets $\mathcal{D}_i$, let $\mathcal{D} = \cup_{i \in \mathcal{H}} \mathcal{D}_i = \{(x_k, y_k)\}_{k=1}^N$ denote the global training set of size N . The dataset $\mathcal{D}$ contains only samples from honest workers. Although Byzantine workers may also possess data, these samples are excluded as they are not used in our analysis.

The label-skew setting is defined by two key assumptions. First, the label distributions vary across workers, meaning that the class proportions in the local dataset $\mathcal{D}_i$ may differ from one worker to another. Second, for any class $c \in [C]$, the conditional feature distribution is shared across all workers. In other words, samples belonging to class c are drawn from a common distribution L^c , independently of the worker that holds them.

3 Related Work

3.1 Robustness Strategies with Homogeneous Data

Robust aggregation methods mitigate the impact of Byzantine updates during training [89, 43]. Examples include coordinate-wise median (CWMed) [116], coordinate-wise trimmed mean (CwTM) [116], Krum [11], Minimum Diameter Averaging [44], and Comparative Gradient Elimination [46]. These methods strongly rely on the assumption that honest workers have homogeneous data. Their robustness under data heterogeneity remains unclear [42, 4, 88].

Beyond robust aggregators, additional strategies enhance robustness in homogeneous settings by leveraging a clean dataset held by the server to evaluate and filter worker updates [114, 14, 36, 92].

3.2 Robustness Strategies Under Data Heterogeneity

To reduce gradient variance caused by data heterogeneity, a common approach applies a pre-aggregation step before the main aggregation rule, leading to a two-step process. Examples include Bucketing [54], and Nearest-Neighbor Mixing (NNM) [4]. For instance, when combining Bucketing with CWMed, the server first applies Bucketing

to the received gradients to reduce heterogeneity, then applies CWMed to produce the final robust update.

Bucketing [54] randomly groups inputs, averages each group, and feeds these averages to the aggregation rule. This reduces the variance of honest updates in expectation, but in some iterations it may fail to reduce heterogeneity, allowing Byzantine workers to disrupt training [42]. In contrast, NNM [4] deterministically replaces each gradient with the average of its $n - f$ nearest neighbors. Although it provides optimal probabilistic robustness guarantees, its high computational cost limits scalability in large distributed systems [42].

Other strategies adopt filtering mechanisms that remove suspected Byzantine clients prior to aggregation, as in FedREDefense [115] and Gradient Splitting (GAS) [73]. FedREDefense [115] checks whether a client update is consistent with a genuine training process by reconstructing it through training on optimized synthetic data. However, updates obtained from any valid training procedure can bypass this detection, for instance in label-flipping attacks [115]. GAS [73] mitigates the curse of dimensionality by decomposing high-dimensional gradients into low-dimensional subvectors and applying a robust aggregation rule to compute reliability scores for each client. Gradients with high scores are identified as honest and aggregated to form the global update, which also alleviates gradient heterogeneity across clients.

Finally, some strategies rely on access to a server-side dataset, such as BOBA [7] and RAGE [23], an assumption that may be unrealistic in many federated learning scenarios.

4 Manuscript organization

Chapter IV proposes a lightweight defense mechanism that operates on the worker side, in contrast to standard server-side approaches. It aligns honest gradients from the start through a weighted loss function. Chapter V then introduces a server-side mechanism designed to filter Byzantine updates. Finally, Chapter VI presents an attack highlighting weaknesses in the state-of-the-art pre-aggregator NNM.

Chapter VI

A Local Sampling Strategy

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This chapter is based on our publication [31].

Under label skew, honest updates can differ substantially from one another, making Byzantine updates harder to distinguish.

To address this challenge, we introduce *NorthStar*, a simple worker-side sampling strategy that aligns local mini-batch class proportions across honest workers. By replacing worker-specific label proportions with a shared reference distribution, NorthStar reduces a key source of gradient dissimilarity under label skew.

Contributions. We propose a lightweight, principled defense mechanism against Byzantine workers under label skew:

- **Lightweight Defense Mechanism.** We propose *NorthStar*, a worker-side sampling strategy that improves Byzantine robustness by reducing honest-gradient dissimilarity;
- **Theoretical Results.** We provide asymptotic arguments explaining the effectiveness of NorthStar. In particular, we show that standard sampling induces a label-distribution mismatch term in honest-gradient dissimilarity, whereas NorthStar removes this term;
- **Experimental Results.** We empirically show that NorthStar consistently improves robustness across multiple models, datasets, aggregation rules, and Byzantine attack scenarios.

Code: <https://github.com/JohanErbani/NorthStar>

Algorithm 6 NorthStar Local Training at Worker k

Require: Local dataset $\mathcal{D}_k = \{\mathcal{D}_k^{(c)}\}_{c=1}^C$ partitioned into label-wise subsets, positions $\{s_k^c\}_{c=1}^C$ initialized to 1 at the first training round, reference distribution $\mathbf{p} = (p_1, \dots, p_C)$, mini-batch size B , model Φ_θ with parameter θ_t , loss function ℓ

Ensure: Local stochastic gradient g_k

- 1: $\mathcal{B} \leftarrow \emptyset$
- 2: **for** $c = 1$ **to** C **do**
- 3: $B_c \leftarrow \lceil Bp_c \rceil$
- 4: $n_c \leftarrow |\mathcal{D}_k^{(c)}|$
- 5: **if** $n_c > 0$ **then**
- 6: Construct $\mathcal{B}_c$ by taking the next B_c samples from $\mathcal{D}_k^{(c)}$ in cyclic order, starting from index s_k^c

$$\mathcal{B}_c \leftarrow \left\{ (x, y) \in \mathcal{D}_k^{(c)} : \text{indexed by } s_k^c, s_k^c + 1 \pmod{n_c}, \dots, s_k^c + B_c - 1 \pmod{n_c} \right\}$$

- 7: $\mathcal{B} \leftarrow \mathcal{B} \cup \mathcal{B}_c$
- 8: Update s_k^c to the next position after the selected block

$$s_k^c \leftarrow s_k^c + B_c - 1 \pmod{n_c} + 1$$

- 9: **end if**
- 10: **end for**
- 11: $g_k \leftarrow \nabla_\theta \left(\frac{1}{|\mathcal{B}|} \sum_{(x,y) \in \mathcal{B}} \ell(y, \Phi_\theta(x)) \right) \Big|_{\theta=\theta_t}$
- 12: **return** g_k

1 NorthStar

NorthStar is a worker-side sampling procedure in which each worker locally constructs its mini-batches according to a reference label distribution shared across workers. This reference distribution is broadcast once by the server before training begins.

This section first presents the intuition behind the design of NorthStar, then introduces the reference distribution and provides its algorithm.

1.1 Intuition behind NorthStar

Figure VI.1 illustrates the intuition behind NorthStar sampling.

Centralized learning. Consider a binary classification problem with classes **Dog** and **Elephant**. Let a mini-batch of size 128 be composed almost entirely of **Dog** images, with only a single **Elephant** example. The resulting gradient is then dominated by the **Dog** class, as it mainly reflects the direction that minimizes the loss over the majority of samples. Consequently, the **Elephant** example has negligible influence on the update.

A natural way to mitigate this imbalance is to partition the local dataset into label-wise subsets and perform balanced sampling. For instance, one may sample with replacement 64 images from the **Dog** subset and 64 from the **Elephant** subset. This leads to two immediate consequences: (i) the gradient is no longer dominated by **Dog**, as both classes contribute equally; and (ii) if the local dataset contains only a single **Elephant** example, it is repeated 64 times. While such repetition may introduce bias and overfitting in centralized settings, in federated learning it helps each worker account for classes in the appropriate proportion.

Federated learning. Now consider a federated learning setting with two honest workers. Before training, the server broadcasts to workers a reference class distribution, e.g., $[0.75, 0.25]$, indicating that mini-batches should contain 75% **Dog** and

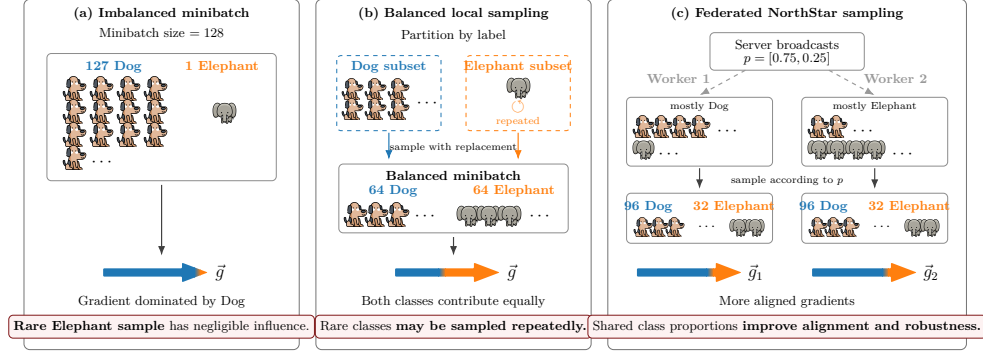


Figure VI.1: Intuition behind NorthStar; (a) An imbalanced minibatch yields a gradient dominated by the majority class; (b) Balanced sampling with replacement reduces majority-class domination; (c) In federated learning, NorthStar sampling encourages workers to align their local gradients with a common reference distribution that is broadcast once by the server.

25% **Elephant**. This imbalance can reflect intrinsic class difficulty: **Dog** exhibits high intra-class variability (e.g., many breeds), whereas **Elephant** is more homogeneous.

Assume the first worker predominantly observes **Dog** images, while the second mostly observes **Elephant** images. Each worker partitions its local dataset by label and samples with replacement according to the reference distribution. In practice, each mini-batch contains $96 = 128 \times 0.75$ **Dog** images and $32 = 128 \times 0.25$ **Elephant** images. As a result, both workers assign comparable importance to each class, leading to more aligned gradients and improved robustness at the system level.

Notably, this procedure induces repeated sampling: the first worker repeatedly reuses its few **Elephant** examples, while the second oversamples **Dog** images throughout training. Despite this, our experiments suggest that the alignment effect across clients outweighs the local redundancy.

1.2 Reference Distribution

NorthStar relies on a *reference distribution* shared across workers to guide the local sampling process. This reference distribution is represented by a simplex vector $\mathbf{p} \in \mathbb{R}_{\geq 0}^C$, i.e., a non-negative vector of size C such that $\sum_{i=1}^C p_i = 1$. The vector $\mathbf{p}$ defines the target class proportions enforced during local training. In practice, a larger value of p_i increases the weight assigned to errors on class i .

We describe below three scenarios for specifying $\mathbf{p}$.

Shared local distributions. In a Byzantine-free setting, each worker sends its local label distribution to the server once before training. The server aggregates these distributions and broadcasts the resulting global distribution $\mathbf{p}$, where

$$p_i = \frac{|\mathcal{D}^{(i)}|}{|\mathcal{D}|},$$

$|\mathcal{D}^{(i)}|$ denotes the total number of samples belonging to class i across all workers, and $|\mathcal{D}|$ is the total number of samples in the system. This approach has two limitations: (i) local label distributions may reveal private information about the workers' data, and (ii) in the presence of Byzantine workers, malicious participants can bias the reference distribution by reporting false distributions.

To mitigate these issues, we compute the coordinate-wise median of the local label distributions using a secure multi-party computation (MPC) protocol [1]. This preserves privacy and limits the ability of Byzantine workers to exploit label statistics. In our setting, the protocol introduces negligible overhead: the sharing step is performed only once before training and involves only C coordinates, where C is the number of classes and is typically small.

Byzantine workers may still send arbitrary distributions. We therefore evaluate this setting under worst-case assumptions in our experiments.

Prior knowledge. The global label distribution is assumed to be known in advance from expert knowledge or historical data. For instance, in tumor detection from MRI scans or crop type classification from satellite imagery, national statistics can provide a reliable estimate of the label distribution. In this case, we expect

$$p_i \approx \frac{|\mathcal{D}^{(i)}|}{|\mathcal{D}|}.$$

A priori. All workers may agree on a predefined reference distribution, independent of the true label distribution. For example, in fair hiring systems, classes can be weighted equally to mitigate algorithmic bias, i.e.,

$$p_i = \frac{1}{C}.$$

1.3 Algorithm

Algorithm 6 summarizes the local training procedure used by NorthStar at worker k . The method assumes that, before training begins, the server broadcasts a reference label distribution $\mathbf{p} = (p_1, \dots, p_C)$ to all workers. Each worker then partitions its local dataset $\mathcal{D}_k$ into label-wise subsets $\mathcal{D}_k^{(c)}$, one for each class $c \in \{1, \dots, C\}$.

At each local training step, worker k constructs a mini-batch by allocating approximately Bp_c samples to class c , where B is the mini-batch size. More precisely, the worker sets $B_c = \lceil Bp_c \rceil$ and selects B_c samples from the corresponding subset $\mathcal{D}_k^{(c)}$. For each class c , samples are selected cyclically from $\mathcal{D}_k^{(c)}$. This makes the worker traverse the available examples of class c before reusing them, thereby minimizing unnecessary repetitions.

If a class is absent from the local dataset, it is skipped. The final mini-batch is obtained by aggregating the selected samples across all available classes. Worker k then computes the stochastic gradient on this NorthStar mini-batch and sends it to the server, as in standard federated learning. Thus, NorthStar only modifies the local sampling procedure and does not require additional communication after the initial broadcast of $\mathbf{p}$.

2 Theoretical Arguments

This section provides theoretical arguments supporting the effectiveness of NorthStar. All proofs are provided in the appendix.

We analyze a single local training step and do not model the cumulative effect of repeated sampling over the full training trajectory. The results are asymptotic with respect to the number of honest workers, i.e., they describe the limit as $|\mathcal{H}| \rightarrow \infty$.

We consider a label-skew setting where, for each class c , observations follow a client-independent conditional distribution $\mathcal{L}_c$. Worker i owns $M_i^{(c)}$ samples of class c , denoted by $x_{i,1}^{(c)}, \dots, x_{i,M_i^{(c)}}^{(c)}$, which form its class-specific dataset $\mathcal{D}_i^{(c)}$. These samples are i.i.d. from $\mathcal{L}_c$. The quantities $M_i^{(c)}$ may vary across workers. For each fixed class c , they are assumed to be i.i.d. across workers, independent of the class- c observations, and strictly positive, i.e., $M_i^{(c)} > 0$, for analytical simplicity.

Robust aggregation relies on the similarity of honest gradients. We therefore study the honest-gradient dissimilarity

$$\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \|g_i - g^*\|^2,$$

where g_i is the gradient of honest worker i and g^* is the expected honest gradient, i.e., $\mathbb{E}[\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} g_i]$.

2.1 Standard Gradients

We first consider standard local gradients, without NorthStar sampling.

For analytical simplicity, we assume stratified mini-batches: worker i samples classes according to its local label distribution $\mathbf{q}_i = (q_{i,1}, \dots, q_{i,C})$ where

$$q_{i,c} = \frac{M_i^{(c)}}{\sum_{d=1}^C M_i^{(d)}}.$$

Let $\mathbf{q} = (q_1, \dots, q_C)$ denote the expected honest label distribution, with $q_c = \mathbb{E}[q_{i,c}]$. The standard local stochastic gradient is

$$g_i = \sum_{c=1}^C q_{i,c} g_i^{(c)}, \quad g_i^{(c)} = \frac{1}{B_i^{(c)}} \sum_{\ell=1}^{B_i^{(c)}} \nabla_{\theta} \ell(\Phi_{\theta}(X_{i,\ell}^{(c)}), c), \quad B_i^{(c)} = q_{i,c} b.$$

Here, $X_{i,\ell}^{(c)}$ is the ℓ -th sampled class- c example in the mini-batch. For simplicity, we ignore integer-rounding effects and assume that $B_i^{(c)} = q_{i,c} b$ is an integer. The expected honest gradient is

$$g^{\star} = \sum_{c=1}^C q_c g^{(c)}, \quad g^{(c)} = \mathbb{E}_{x \sim \mathcal{L}_c} [\nabla_{\theta} \ell(\Phi_{\theta}(x), c)].$$

Asymptotically, gradient dissimilarity splits into two terms: (i) a class-wise estimation error; and (ii) a label-distribution mismatch term.

Proposition 10. *Assume that honest workers are i.i.d. and that, for each class c , $\mathbb{E}[\|g_i^{(c)}\|^2] < \infty$.*

Then

$$\begin{aligned} & \frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \|g_i - g^{\star}\|^2 \\ & \xrightarrow[|\mathcal{H}| \rightarrow \infty]{\text{a.s.}} \underbrace{\mathbb{E} \left[\left\| \sum_{c=1}^C q_{i,c} (g_i^{(c)} - g^{(c)}) \right\|^2 \right]}_{\text{class-wise estimation error}} + \underbrace{\mathbb{E} \left[\left\| \sum_{c=1}^C (q_{i,c} - q_c) g^{(c)} \right\|^2 \right]}_{\text{label-distribution mismatch}} \end{aligned}$$

Thus, under label skew, standard gradients can remain dissimilar even when class-wise gradients are accurately estimated, because workers combine classes with different weights.

2.2 Gradients under NorthStar

We now analyze a simplified NorthStar step based on random sampling with replacement. In practice, NorthStar uses cyclic sampling, which reduces repetitions and is therefore expected to be less detrimental to convergence.

At each local step, worker i samples B_c examples from $\mathcal{D}_i^{(c)}$, where $B_c = p_c b > 0$ is shared across workers and determined by the broadcasted reference distribution $\mathbf{p} = (p_1, \dots, p_C)$. We assume $p_c > 0$ for every class c and ignore integer-rounding effects, so that B_c is treated as an integer. Let $u_{i,k}^{(c)}$ be the number of times sample $x_{i,k}^{(c)}$ is selected. Conditionally on $M_i^{(c)}$,

$$(u_{i,1}^{(c)}, \dots, u_{i,M_i^{(c)}}^{(c)}) \sim \text{Multinomial} \left(B_c; \frac{1}{M_i^{(c)}}, \dots, \frac{1}{M_i^{(c)}} \right).$$

The NorthStar gradient is

$$g_{\text{NS } i} = \sum_{c=1}^C p_c g_{\text{NS } i}^{(c)}, \quad g_{\text{NS } i}^{(c)} = \frac{1}{B_c} \sum_{k=1}^{M_i^{(c)}} u_{i,k}^{(c)} \nabla_{\theta} \ell(\Phi_{\theta}(x_{i,k}^{(c)}), c).$$

The expected honest gradient is

$$g_{\text{NS}}^* = \sum_{c=1}^C p_c g_{\text{NS}}^{(c)}, \quad g_{\text{NS}}^{(c)} = g^{(c)} = \mathbb{E}_{x \sim \mathcal{L}_c} [\nabla_{\theta} \ell(\Phi_{\theta}(x), c)].$$

Asymptotically, gradient dissimilarity contains only one term: a class-wise estimation error.

Proposition 11. *Assume that honest workers are i.i.d. and that, for each class c , $\mathbb{E}[\|g_{\text{NS } i}^{(c)}\|^2] < \infty$.
Then*

$$\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \|g_{\text{NS } i} - g_{\text{NS}}^*\|^2 \xrightarrow[|\mathcal{H}| \rightarrow \infty]{\text{a.s.}} \underbrace{\mathbb{E} \left[\left\| \sum_{c=1}^C p_c \left(g_{\text{NS } i}^{(c)} - g^{(c)} \right) \right\|^2 \right]}_{\text{class-wise estimation error under NorthStar}}$$

NorthStar gradient dissimilarity remains nonzero due to class-wise stochastic estimation error, but not because of worker-specific class proportions.

2.3 NorthStar versus Standard Gradients

Gradient dissimilarity. Under standard sampling, asymptotic honest-gradient dissimilarity decomposes as

$$\begin{aligned} & \left(\text{gradient dissimilarity without NorthStar} \right) \\ &= \\ & \left(\text{class-wise estimation error} \right) + \left(\text{label-distribution mismatch} \right) \end{aligned}$$

With NorthStar, the corresponding decomposition becomes

$$\begin{aligned} & \left(\text{NorthStar gradient dissimilarity} \right) \\ &= \\ & \left(\text{class-wise estimation error} \right) \end{aligned}$$

Thus, standard sampling contains an additional term induced by worker-specific label proportions. NorthStar removes this source of dissimilarity by replacing local class weights with a shared reference distribution.

Since the remaining class-wise estimation terms have the same nature in both decompositions, they are expected to have comparable magnitude. In contrast, the label-distribution mismatch term can dominate under strong label skew and appears only in the standard procedure. This explains why NorthStar reduces honest-gradient dissimilarity, as observed empirically.

Target model. The next proposition shows that, when $\mathbf{p} = \mathbf{q}$, NorthStar preserves the expected target of the standard procedure.

Proposition 12. *If the broadcasted reference distribution satisfies $\mathbf{p} = \mathbf{q}$, then NorthStar preserves the expected honest gradient:*

$$g_{\text{NS}}^* = g^*.$$

This provides insight into the effectiveness of NorthStar: it aligns honest gradients by replacing worker-specific class weights with a shared reference distribution, while preserving the target model when $\mathbf{p} = \mathbf{q}$.

3 Experimental Setup

3.1 Datasets

Our experiments are conducted on the following datasets : CIFAR-10 [58], MNIST [67], and Fashion-MNIST [111]. We divide training data among the workers using a Dirichlet distribution to generate non-IID local datasets, following the approach of [4, 50], with concentration parameters $\alpha \in \{0.1, 1, 10\}$. For each dataset, the test set follows the same class distribution as its corresponding training set.

3.2 Attacks & Baselines

We consider the following attacks : A Little Is Enough (ALIE) [8], Fall Of Empire (FOE) [113], Mimic [54], Min-Sum [97] and Label Flipping (LF) [3]. We use the following robust aggregators : Coordinate-Wise Median (CWMed) [116], Coordinate-Wise Trimmed Mean (CwTM) [11] and RFA [89]. We also compare performance with and without pre-aggregation, using Nearest Neighbor Mixing (NNM) [4]. We use the CrossEntropy loss as a baseline to evaluate our approach.

3.3 FL settings

We consider 60 workers, with $f \in \{2, 8, 14, 20, 26\}$ Byzantine workers. We train a four-layer CNN for CIFAR-10, and a two-layer CNN for MNIST and Fashion-MNIST (model in Appendix). The models are trained with batch size 128 and momentum $\beta = 0.9$. Momentum has shown strong robustness against Byzantine behavior in both heterogeneous [4] and homogeneous settings [27, 55, 37]. Model performance is evaluated by computing the accuracy averaged over the last 200 training steps.

4 Results

This section evaluates the impact of NorthStar on Byzantine-robust federated learning under label skew. We first analyze the results obtained with RFA, then discuss the aggregated results across robust aggregators reported in the appendix.

4.1 Effect of Label Skew

Tables VI.3, VI.2, and VI.1 report the results obtained with RFA on CIFAR-10, Fashion-MNIST, and MNIST. Across the three datasets, the benefit of NorthStar increases as label skew becomes stronger.

When $\alpha = 10$, local label distributions are close to homogeneous, and both CE and NS usually reach similar performance. In this regime, NorthStar often yields small gains or nearly neutral changes, which is expected because the local class proportions are already close across workers.

The effect becomes clearer when $\alpha = 1$ and especially when $\alpha = 0.1$. In these heterogeneous regimes, standard training often suffers from large drops in accuracy as the number of Byzantine workers increases. NorthStar mitigates this degradation in many settings. For instance, on CIFAR-10 with $\alpha = 0.1$, NorthStar improves RFA under almost all attacks and Byzantine counts, with large gains. Similar trends are observed on Fashion-MNIST and MNIST, where NorthStar is particularly effective when the baseline collapses under strong attacks and large values of f . Figure VI.2 illustrates this behavior.

These results support the theoretical intuition developed in Section 1: under label skew, honest gradients differ partly because workers use different class proportions.

By enforcing a shared reference distribution in local mini-batches, NorthStar reduces this source of dissimilarity and makes robust aggregation more effective.

4.2 Dataset-Level Trends

The effect of NorthStar depends on dataset difficulty and on how close the baseline is to saturation. On MNIST, baseline accuracies are often close to 99% when heterogeneity is mild or moderate. In this regime, there is little room for improvement, and NorthStar mostly leads to negligible variations, sometimes slightly negative. This is expected: when the baseline already reaches near-optimal accuracy, enforcing a shared sampling distribution cannot improve much and may introduce small fluctuations due to repeated examples. However, when the baseline collapses under strong heterogeneity or large Byzantine counts, NorthStar can produce large recovery effects. For instance, with RFA under ALIE at $\alpha = 0.1$ and $f = 20$, accuracy increases from 18.16% to 87.05%, corresponding to a gain of 68.89 percentage points.

On CIFAR-10 and Fashion-MNIST, which are harder datasets, the gains are more gradual and more consistently visible across attacks. Under moderate heterogeneity, $\alpha = 1$, NorthStar often improves accuracy by several percentage points, especially when the number of Byzantine workers increases. On CIFAR-10, for example, RFA gains 9.05 points under ALIE with $f = 14$, 7.85 points under ALIE with $f = 20$, and 29.86 points under MinSum with $f = 14$. On Fashion-MNIST, the recovery can be even stronger: with RFA under ALIE at $\alpha = 1$ and $f = 26$, NorthStar improves accuracy by 43.67 points.

The trend becomes clearer under strong heterogeneity, $\alpha = 0.1$. In this setting, standard training often suffers from severe performance degradation, while NorthStar substantially improves robustness in most configurations. On CIFAR-10, gains with RFA are positive for almost all attacks and Byzantine counts, often exceeding 5 points and reaching more than 20 points in some cases. On Fashion-MNIST, the gains are also large and systematic, with improvements such as 40.91 points under ALIE with $f = 20$, 28.41 points under MinSum with $f = 26$, and 44.66 points under FOE with $f = 26$. These results support the main intuition behind NorthStar: its benefit is limited when the baseline already performs well, but becomes substantial when label skew increases honest-gradient dissimilarity and causes robust aggregation to fail.

4.3 Interaction with NNM

NorthStar is compatible with pre-aggregation methods such as NNM. Since NNM already reduces the dispersion of honest updates before aggregation, the additional benefit of NorthStar is usually smaller when heterogeneity is mild or when NNM already stabilizes training. This is visible for $\alpha = 10$, where NNM+RFA often reaches high accuracy and NorthStar only induces small variations.

Under stronger label skew, however, the combination becomes more beneficial. On CIFAR-10 with $\alpha = 0.1$, NNM+NorthStar clearly improves over NNM+RFA under ALIE and Mimic. For instance, under ALIE, the gain reaches 23.19 points with RFA at $f = 26$, and reaches 24.82 points when averaging over the three aggregators. Similar trends appear on Fashion-MNIST, where NNM+NorthStar improves accuracy by 15.56 points under ALIE at $f = 26$, and by 22.70 points in the averaged-aggregator setting.

On MNIST, the effect is more mixed because NNM+RFA is often already close to saturation. In easy configurations, NorthStar brings little improvement and may slightly decrease performance. However, when NNM alone fails under strong heterogeneity, NorthStar can still recover large gains. For example, with $\alpha = 0.1$ under ALIE, NNM+NorthStar improves RFA by 14.03 points at $f = 20$ and by 23.40 points at $f = 26$.

Overall, these results show that NorthStar and NNM address complementary aspects of the problem. NNM reduces dispersion after gradients have been computed, whereas NorthStar reduces label-induced dissimilarity at the sampling stage.

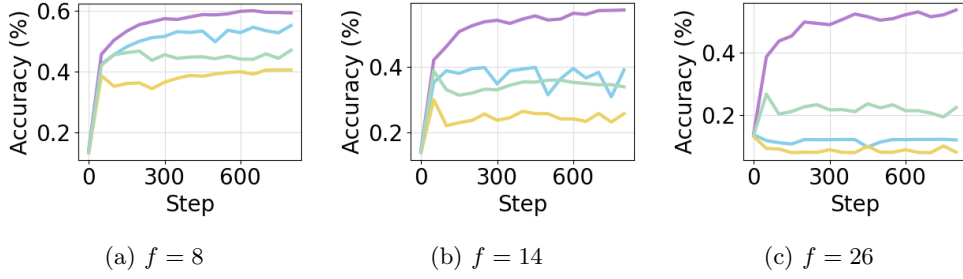


Figure VI.2: CIFAR-10 test accuracy under ALIE with RFA and label-skew parameter $\alpha = 0.1$. Legend: • NorthStar, • NNM+NorthStar, • RFA, • NNM+RFA.

4.4 Results Averaged over Aggregators

The appendix reports results averaged over CWMed, CwTM, and RFA, both with and without NNM. These aggregated results confirm the trends observed with RFA alone. NorthStar provides the largest gains under severe label skew, while its effect is smaller under mild heterogeneity. This shows that the method is not tied to a specific aggregation rule. Instead, it acts as a complementary worker-side mechanism that improves the quality and alignment of honest updates before they reach the server.

Overall, the experiments show that NorthStar is most effective in the regimes for which it was designed: heterogeneous federated learning with label skew and Byzantine workers.

5 Conclusion

We introduced *NorthStar*, a worker-side sampling strategy for Byzantine-robust federated learning under label skew. Instead of modifying the server-side aggregation rule, NorthStar changes how honest workers construct their local mini-batches. By sampling according to a shared reference label distribution, it reduces the gradient dissimilarity caused by worker-specific class proportions.

Our theoretical analysis shows that standard sampling introduces a label-distribution mismatch term in honest-gradient dissimilarity, whereas NorthStar removes this term by enforcing a shared sampling distribution across workers. When the reference distribution matches the expected honest label distribution, NorthStar also preserves the expected training target.

Empirically, NorthStar improves robustness across datasets, attacks, aggregation rules, and Byzantine counts, with the largest gains observed under severe label skew. The method is simple, local, and compatible with existing robust aggregation rules and pre-aggregation defenses. These results suggest that improving Byzantine robustness does not only require stronger server-side aggregation: it can also be achieved by reducing honest-gradient heterogeneity at its source.

Table VI.1: MNIST, RFA. Mean accuracy (%) over the last 200 steps. We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

		ALIE					Mimic					MinSum					FOE					LF				
		2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
		gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain
10	CE	98.73	98.76	98.45	98.21	95.49	98.72	98.61	98.47	98.25	98.06	98.55	98.04	97.08	97.03	69.86	98.50	97.94	97.57	97.51	95.80	98.74	98.82	98.73	98.78	98.71
	NS	98.69	98.59	98.36	98.04	95.01	98.68	98.62	98.45	98.22	98.14	98.66	98.02	96.97	95.00	92.28	98.59	97.92	96.95	95.88	94.70	98.73	98.83	98.76	98.72	98.55
	RFA	-0.04	-0.17	-0.09	-0.17	-0.48	-0.04	0.01	-0.02	-0.03	0.08	0.11	-0.02	-0.11	-2.03	22.42	0.09	-0.02	-0.62	-1.63	-1.10	-0.01	0.01	0.03	-0.06	-0.16
	CE	98.71	98.74	98.65	98.57	98.46	98.73	98.72	98.58	98.27	97.89	98.50	98.41	97.98	97.76	96.43	98.50	98.41	97.98	97.76	96.43	NaN	NaN	NaN	NaN	NaN
	NNM+RFA	98.70	98.77	98.72	98.52	98.34	98.73	98.72	98.58	98.27	97.89	98.59	98.46	98.28	97.76	96.07	98.59	98.46	98.28	97.76	96.07	NaN	NaN	NaN	NaN	NaN
	gain	-0.01	0.03	0.07	-0.05	-0.12	NaN	NaN	NaN	NaN	NaN	0.09	0.05	0.30	0.00	-0.36	0.09	0.05	0.30	0.00	-0.36	NaN	NaN	NaN	NaN	NaN
1	CE	98.65	98.67	98.28	97.41	67.35	98.64	98.50	98.21	97.87	97.08	98.49	98.21	97.70	96.76	69.17	98.50	98.26	97.92	97.28	87.77	98.69	98.75	98.63	98.59	98.59
	NS	98.61	98.52	98.37	97.54	92.00	98.56	98.38	98.19	98.05	97.58	98.63	98.23	97.51	96.39	93.15	98.48	98.12	97.41	95.99	94.65	98.64	98.66	98.58	98.57	98.60
	RFA	-0.04	-0.15	0.09	0.13	24.65	-0.08	-0.12	-0.02	0.18	0.50	0.14	0.02	-0.19	-0.37	23.98	-0.02	-0.14	-0.51	-1.29	6.88	-0.05	-0.09	-0.05	-0.02	0.01
	CE	98.73	98.72	98.63	98.51	98.50	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	98.37	97.75	97.49	93.60	90.86	NaN	NaN	NaN	NaN	NaN
	NNM+RFA	98.55	98.68	98.59	98.42	98.36	98.60	98.51	98.40	98.17	97.52	NaN	NaN	NaN	NaN	NaN	98.18	98.22	97.90	97.85	96.92	NaN	NaN	NaN	NaN	NaN
	gain	-0.18	-0.04	-0.04	-0.09	-0.14	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.19	0.47	0.41	4.25	6.06	NaN	NaN	NaN	NaN	NaN
0.1	CE	98.58	98.21	96.22	18.16	8.65	98.65	97.90	96.75	93.46	82.56	98.35	95.17	92.71	55.54	21.97	98.35	96.36	94.74	76.42	19.66	98.65	98.70	98.63	98.30	97.16
	NS	98.46	98.16	97.48	87.05	18.01	98.30	98.02	97.60	96.93	91.87	98.18	97.34	94.46	86.47	47.41	98.20	97.17	94.80	88.13	67.85	98.39	98.37	98.24	98.33	98.15
	RFA	-0.12	-0.05	1.26	68.89	9.36	-0.35	0.12	0.85	3.47	9.31	-0.17	2.17	1.75	30.93	25.44	-0.15	0.81	0.06	11.71	48.19	-0.26	-0.33	-0.39	0.03	0.99
	CE	98.66	98.40	94.15	84.11	74.89	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	98.09	88.63	83.17	69.17	39.30	NaN	NaN	NaN	NaN	NaN
	NNM+RFA	98.52	98.25	98.13	98.14	98.29	98.52	98.33	98.10	97.16	96.33	NaN	NaN	NaN	NaN	NaN	97.92	97.43	61.17	45.26	41.23	NaN	NaN	NaN	NaN	NaN
	gain	-0.14	-0.15	3.98	14.03	23.40	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.17	8.80	-22.00	-23.91	1.93	NaN	NaN	NaN	NaN	NaN

Table VI.2: Fashion-MNIST, RFA. Mean accuracy (%) over the last 200 steps. We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

		ALIE					Mimic					MinSum					FOE					LF				
		2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26
		gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain	gain
10	CE	86.86	86.96	84.40	81.16	74.62	86.98	87.17	86.32	85.42	81.83	86.22	83.17	82.18	78.63	57.67	86.24	84.29	83.14	79.80	71.52	86.91	87.15	86.98	86.50	86.28
	NS	87.17	86.84	85.81	82.83	75.65	87.11	87.18	85.97	85.22	83.05	86.49	84.59	81.73	76.10	74.16	86.59	83.67	81.39	78.19	74.81	87.02	87.08	86.86	86.72	86.65
	RFA	0.31	-0.12	1.41	1.67	1.03	0.13	0.01	-0.35	-0.20	1.22	0.27	1.42	-0.45	-2.53	16.49	0.35	-0.62	-1.75	-1.61	3.29	0.11	-0.07	-0.12	0.22	0.37
	gain	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37
1	CE	86.87	86.71	86.63	87.04	87.11	86.98	87.17	86.32	85.42	81.83	86.22	83.17	82.18	78.63	57.67	86.24	84.29	83.14	79.80	71.52	86.91	87.15	86.98	86.50	86.28
	NS	87.16	87.05	87.16	87.00	86.74	87.00	87.06	86.40	85.85	83.48	86.30	85.38	84.73	81.94	78.29	86.30	85.38	84.73	81.94	78.29	86.30	85.38	84.73	81.94	78.29
	RFA	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37
	gain	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37	0.29	0.34	0.53	-0.04	-0.37
0.1	CE	86.64	85.97	85.39	84.88	84.73	86.91	86.77	86.22	85.67	83.96	86.91	86.77	86.22	85.67	83.96	86.91	86.77	86.22	85.67	83.96	86.91	86.77	86.22	85.67	83.96
	NS	87.02	86.71	86.58	86.29	85.67	86.91	86.77	86.22	85.67	83.96	86.91	86.77	86.22	85.67	83.96	86.91	86.77	86.22	85.67	83.96	86.91	86.77	86.22	85.67	83.96
	RFA	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94
	gain	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94	0.38	0.74	1.19	1.41	0.94
0.1	CE	85.57	81.24	73.42	29.06	8.65	85.52	85.58	82.44	71.60	55.93	82.16	69.54	65.35	53.13	10.50	81.82	70.75	67.16	64.74	17.31	86.02	86.20	86.68	84.60	77.06
	NS	86.52	83.31	78.85	69.97	18.01	86.71	86.37	85.82	82.25	69.87	85.53	81.83	80.48	74.95	38.91	85.33	79.77	79.41	76.36	61.97	86.33	86.32	87.12	85.79	83.98
	RFA	0.95	2.07	5.43	40.91	9.36	1.19	0.79	3.38	10.65	13.94	3.37	12.29	15.13	21.82	28.41	3.51	9.02	12.25	11.62	44.66	0.31	0.12	0.44	1.19	6.92
	gain	0.95	2.07	5.43	40.91	9.36	1.19	0.79	3.38	10.65	13.94	3.37	12.29	15.13	21.82	28.41	3.51	9.02	12.25	11.62	44.66	0.31	0.12	0.44	1.19	6.92
0.1	CE	86.16	81.67	80.17	77.77	70.56	86.35	86.53	86.35	76.91	73.38	86.35	86.53	86.35	76.91	73.38	86.35	86.53	86.35	76.91	73.38	86.35	86.53	86.35	76.91	73.38
	NS	86.08	84.58	81.64	81.38	86.12	86.35	86.53	86.35	76.91	73.38	86.35	86.53	86.35	76.91	73.38	86.35	86.53	86.35	76.91	73.38	86.35	86.53	86.35	76.91	73.38
	RFA	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56
	gain	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56	-0.08	2.91	1.47	3.61	15.56

Table VI.3: CIFAR10, RFA. Mean accuracy (%) over the last 200 steps. We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

		ALIE					Mimic					MinSum					FOE					LF					
		2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	
10	RFA	CE	72.68	62.74	49.95	41.00	32.99	73.95	72.89	70.41	66.71	58.21	73.45	64.03	53.72	25.60	19.44	69.79	59.46	53.02	45.19	31.86	74.11	73.19	70.89	67.89	62.21
		NS	73.18	61.63	50.59	36.17	32.04	74.83	73.38	71.35	66.90	59.70	73.80	65.80	56.96	48.04	28.10	70.23	59.49	55.94	47.00	34.11	74.91	73.36	71.97	67.60	62.89
		gain	0.50	-1.11	0.64	-4.83	-0.95	0.88	0.49	0.94	0.19	1.49	0.35	1.77	3.24	22.44	8.66	0.44	0.03	2.92	1.81	2.25	0.80	0.17	1.08	-0.29	0.68
	NNM+RFA	CE	73.88	73.12	71.52	69.09	65.57	74.16	72.83	70.96	69.54	63.96	74.16	72.83	70.96	69.54	63.96	71.29	67.83	64.68	59.28	49.91	NaN	NaN	NaN	NaN	NaN
		NS	74.94	73.09	70.61	66.29	60.77	74.84	73.91	72.20	69.78	63.85	74.84	73.91	72.20	69.78	63.85	72.12	68.55	64.72	58.18	46.62	NaN	NaN	NaN	NaN	NaN
		gain	1.06	-0.03	-0.91	-2.80	-4.80	0.68	1.08	1.24	0.24	-0.11	0.68	1.08	1.24	0.24	-0.11	0.83	0.72	0.04	-1.10	-3.29	NaN	NaN	NaN	NaN	NaN
1	RFA	CE	69.82	55.85	37.55	29.29	19.66	72.89	71.13	66.53	57.71	49.16	67.64	54.87	21.16	19.39	19.96	65.89	56.54	45.41	30.42	18.27	72.58	70.24	67.33	62.88	53.88
		NS	70.87	56.85	46.60	37.14	23.78	72.90	71.40	70.01	64.72	54.29	72.19	60.13	51.02	34.67	23.06	67.32	58.62	51.04	39.54	26.91	73.35	71.61	70.40	67.75	62.69
		gain	1.05	1.00	9.05	7.85	4.12	0.01	0.27	3.48	7.01	5.13	4.55	5.26	29.86	15.28	3.10	1.43	2.08	5.63	9.12	8.64	0.77	1.37	3.07	4.87	8.81
	NNM+RFA	CE	72.02	66.89	59.66	56.87	65.23	73.02	71.51	68.90	65.75	56.03	NaN	NaN	NaN	NaN	NaN	68.53	64.05	62.61	55.29	44.93	NaN	NaN	NaN	NaN	NaN
		NS	72.30	71.77	70.37	65.87	64.24	72.97	71.66	71.02	68.56	61.49	NaN	NaN	NaN	NaN	NaN	69.21	67.91	66.73	61.25	51.67	NaN	NaN	NaN	NaN	NaN
		gain	0.28	4.88	10.71	9.00	-0.99	-0.05	0.15	2.12	2.81	5.46	NaN	NaN	NaN	NaN	NaN	0.68	3.86	4.12	5.96	6.74	NaN	NaN	NaN	NaN	NaN
0.1	RFA	CE	48.17	27.44	20.39	18.53	11.57	58.83	56.32	47.15	28.94	28.99	54.23	26.90	9.97	6.13	18.69	52.83	33.04	14.33	9.25	11.33	57.56	54.30	45.86	42.31	38.75
		NS	60.87	37.18	29.12	23.56	11.65	63.38	63.01	61.46	49.35	37.24	59.96	38.20	24.50	21.23	25.78	57.72	43.12	25.76	13.36	9.57	62.89	60.24	57.87	52.28	46.72
		gain	12.70	9.74	8.73	5.03	0.08	4.55	6.69	14.31	20.41	8.25	5.73	11.30	14.53	15.10	7.09	4.89	10.08	11.43	4.11	-1.76	5.33	5.94	12.01	9.97	7.97
	NNM+RFA	CE	56.13	46.13	34.20	30.19	23.08	59.30	61.27	59.41	54.07	29.42	NaN	NaN	NaN	NaN	NaN	56.29	53.76	48.30	40.50	26.69	NaN	NaN	NaN	NaN	NaN
		NS	64.00	58.57	54.55	45.97	46.27	63.31	63.28	63.33	58.35	48.39	NaN	NaN	NaN	NaN	NaN	61.07	57.28	53.32	46.91	8.78	NaN	NaN	NaN	NaN	NaN
		gain	7.87	12.44	20.35	15.78	23.19	4.01	2.01	3.92	4.28	18.97	NaN	NaN	NaN	NaN	NaN	4.78	3.52	5.02	6.41	-17.91	NaN	NaN	NaN	NaN	NaN

Chapter VII

Inside the Convex Hull of Gradients

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This chapter is based on our publication [30].

Despite label-skew heterogeneity, honest updates share a key structural property: they are convex combinations of class-wise gradients. Leveraging this observation, we introduce HullGuard, a Byzantine filter designed to provide strong robustness under label skew. In our experiments, HullGuard filters out most Byzantine workers during training and outperforms existing strategies. It can also be combined with existing robust aggregation methods and does not require an additional validation set at the server side. We complement these empirical results with a theoretical analysis that provides insight into the effectiveness of the method.

Contributions. We propose a principled defense mechanism against Byzantine workers under label skew:

- **Structural Property.** We show that honest updates satisfy a structural property under label skew: each honest gradient can be written as a noisy convex combination of global class-wise gradients. This property helps distinguish heterogeneity caused by label skew from deviations induced by Byzantine behavior.
- **Highly Efficient Defense Mechanism.** We introduce HullGuard, a Byzantine filter that exploits this structure by selecting workers whose gradients are most aligned with the global class-wise gradients.
- **Theoretical Results.** We provide theoretical argument providing insight into the effectiveness of the method, with full proofs in the appendix.
- **Experimental Results.** We validate the approach under increasingly challenging label-skew and Byzantine settings. The results show that HullGuard filters out most Byzantine workers, outperforms competing methods, and maintains strong performance across all difficulty levels, even when existing approaches collapse.

Code: <https://github.com/JohanErbani/HullGuard>

1 HullGuard Filter

This section introduces HullGuard, a robust Byzantine filtering method designed to address the label-skew setting. The first part presents an analogy to help understand the mechanism behind HullGuard. The second part provides an overview of the algorithm. The final part presents the algorithmic details and theoretical arguments.

1.1 Analogy for HullGuard

Imagine a university recruiter selecting rowers for the university rowing team. Among n candidates, the recruiter must select the best h athletes. To compare them, K time trials are organized. In each trial, a team of h rowers is randomly sampled and attempts to achieve the best possible time. Let $A_1, \dots, A_K$ denote the sampled teams and $t(A_k)$ the time obtained by team A_k .

The recruiter would like to assess the contribution of each candidate to the team's performance. For each candidate i , the recruiter estimates the probability that a team performs better when candidate i is present than when the candidate is absent, yielding a score $s(i)$:

$$s(i) = \frac{|\{(k,l): t(A_k) < t(A_l) \text{ such that } i \in A_k, i \notin A_l\}|}{|\{k: i \in A_k\}| |\{l: i \notin A_l\}|} \approx \mathbb{P}(t(A) < t(A') \mid i \in A, i \notin A')$$

The recruiter then selects the h candidates with the highest scores. This strategy can be theoretically grounded under simple assumptions (see Proposition 19 in Appendix).

1.2 HullGuard Algorithm

We now present an overview of HullGuard. The description is intentionally informal; italicized terms will be formally defined in the following subsections. Algorithm 7 summarizes the procedure. At each round, workers send their update to the server along with the label distribution of the data used to compute it.

The goal of HullGuard is to identify honest workers (analogous to good rowers) and distinguish them from Byzantine workers (bad rowers). The algorithm begins by sampling K groups (analogous to teams) $A_1, \dots, A_K$ of size h . Each group is evaluated using a performance score. In the recruiter story this score was the time $t(A)$. In

HullGuard, the score is the *empirical Frobenius variance of the global class-wise gradient estimator* (i.e., the variance of an estimator), denoted $\hat{\sigma}^2(A)$.

To compute $\hat{\sigma}^2(A)$, we proceed as follows. For each group A , we randomly sample subgroups $B_1, \dots, B_{K'}$ such that $B_k \subset A$ for $k = 1, \dots, K'$. For each subgroup, we compute an estimate of the *global class-wise gradients* (i.e., a matrix per subgroup), producing $\hat{\mathcal{U}}_1, \dots, \hat{\mathcal{U}}_{K'} \in \mathbb{R}^{C \times D}$. We then compute the *empirical Frobenius variance* based on these matrices, denoted $\hat{\sigma}^2(A)$. This quantity measures how much the *global class-wise gradients* vary within A .

The key observation is that, under label skew, honest updates are a noisy convex combinations¹ of vectors called *global class-wise gradients*. These vectors are identical across workers. As a result, a group A composed mostly of honest workers produces similar gradient estimates across the subgroups B_k , leading to a small group score $\hat{\sigma}^2(A)$ (analogous to a small time: the lower, the better).

In contrast, Byzantine updates tend to exhibit *class-wise gradients* that differ from those of honest workers in order to be disruptive. Consequently, when a group A contains Byzantine workers, their presence increases the variability of the estimates, i.e., the matrices $\hat{\mathcal{U}}_1, \dots, \hat{\mathcal{U}}_{K'}$ differ more. In summary, the more honest workers a group A contains, the smaller its *empirical Frobenius variance* $\hat{\sigma}^2(A)$.

Following the same process as in the recruiter, HullGuard assigns a score to each worker based on pairwise comparisons between group scores $\hat{\sigma}^2(A_1), \dots, \hat{\sigma}^2(A_K)$ and selects the h workers with the highest scores, thereby filtering out Byzantine workers.

Finally, a user-chosen robust aggregation step is applied. It mitigates two issues: some Byzantine workers may remain after HullGuard filtering, and honest gradients may still be heterogeneous due to data distribution differences across workers, causing the update to deviate from the honest mean gradient. For instance, the coordinate-wise median can be used.

1.3 HullGuard in Detail

This subsection introduces the notion of class-wise gradients, explains why honest updates are noisy convex combinations of global class-wise gradients, describes how class-wise gradients are estimated, and shows why the variance of these estimators within a group can act as a Byzantine detector.

Local gradient as a convex combination of local class-wise gradients

For each worker i , the local gradient can be written as a convex combination of local class-wise gradients:

$$\underbrace{\nabla \mathcal{L}_i = \sum_{c=1}^C \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} \nabla \mathcal{L}_i^c}_{\text{local update}}, \quad \text{with } \underbrace{\nabla \mathcal{L}_i^c = \frac{1}{|\mathcal{D}_i^c|} \sum_{(x,y) \in \mathcal{D}_i^c} \nabla \ell(c, \Phi(x))}_{\text{local class-wise gradient}}, \quad (\text{VII.1})$$

where $\mathcal{D}_i^c \subset \mathcal{D}_i$ is the subset of samples belonging to class c . The class-wise gradient $\nabla \mathcal{L}_i^c$ is defined as the zero vector if class c is not present in $\mathcal{D}_i$.

Equation (VII.1) shows that the local gradient is a convex combination of class-wise gradients, with weights given by the local class proportions.

¹A vector u is a convex combination of vectors $v_1, \dots, v_C$ if there exist coefficients $\alpha_1, \dots, \alpha_C$ such that $\alpha_c \geq 0$, $\sum_{c=1}^C \alpha_c = 1$, and $u = \sum_{c=1}^C \alpha_c v_c$.

Local gradient as a noisy combination of global class-wise gradients

Local gradient $\nabla \mathcal{L}_i$ can be written with global (not local) class-wise gradients $\nabla \mathcal{L}^c$ and an error term ε_i :

$$\nabla \mathcal{L}_i = \sum_{c=1}^C \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} \nabla \mathcal{L}^c + \varepsilon_i \quad (\text{VII.2})$$

with $\underbrace{\nabla \mathcal{L}^c = \frac{1}{|\mathcal{D}^c|} \sum_{(x,y) \in \mathcal{D}^c} \nabla \ell(c, \Phi(x))}_{\text{global class-wise gradient (not worker-specific)}}, \quad \underbrace{\varepsilon_i = \sum_{c=1}^C \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} (\nabla \mathcal{L}_i^c - \nabla \mathcal{L}^c)}_{\text{noise}}$

Under label skew, the error term has zero mean, i.e., $\mathbb{E}[\varepsilon_i] = 0$. Furthermore, global class-wise gradients are not worker-specific; they depend only on the global dataset $\mathcal{D}$.

Equation (VII.2) shows that each honest update is a noisy convex combination of global class-wise gradients.

Class-wise gradient estimator

For any selected subgroup $B_1, \dots, B_{K'}$ of workers, the gradients and label proportions used to produce their updates define a linear system from which the global class-wise gradients can be estimated.

Consider a subgroup B of s honest workers indexed by $\{1, \dots, s\}$. By row stacking Equation (VII.2), we obtain

$$\underbrace{\begin{pmatrix} \nabla \mathcal{L}_1 \\ \vdots \\ \nabla \mathcal{L}_s \end{pmatrix}}_{\text{updates } K \times D} = \underbrace{\begin{pmatrix} |\mathcal{D}_1^1|/|\mathcal{D}_1| & \dots & |\mathcal{D}_1^C|/|\mathcal{D}_1| \\ \vdots & & \vdots \\ |\mathcal{D}_s^1|/|\mathcal{D}_s| & \dots & |\mathcal{D}_s^C|/|\mathcal{D}_s| \end{pmatrix}}_{\text{label distributions } K \times C} \underbrace{\begin{pmatrix} \nabla \mathcal{L}^1 \\ \vdots \\ \nabla \mathcal{L}^C \end{pmatrix}}_{\text{global class gradients } C \times D} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_s \end{pmatrix}}_{\text{noise } K \times D}$$

$$\downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow$$

$$U \quad \quad P \quad \quad \mathcal{U} \quad \quad + \quad \quad E$$

where U denotes the update matrix, P the label distribution matrix, $\mathcal{U}$ the global class gradient matrix, and E the noise matrix.

To estimate the global class-wise gradients, we consider the following minimization problem

$$\min_{X \in \mathbb{R}^{C \times D}} \|PX - U\|_F, \quad (\text{VII.3})$$

which seeks the C vectors that best explain the updates according to P . Indeed, we have

$$\|PX - U\|_F^2 = \sum_{i=1}^s \left\| \sum_{c=1}^C \frac{|\mathcal{D}_i^c|}{|\mathcal{D}_i|} (X_{c:} - \nabla \mathcal{L}_i^c) \right\|_2^2,$$

where $X_{1:}, \dots, X_{C:}$ denote the rows of X . When sufficient data are available (i.e., when s is sufficiently large), a natural candidate for X is the matrix of global class-wise gradients $\mathcal{U}$. The solution to Problem (VII.3) is given by the least-squares estimator [26]

$$P^\dagger U \in \underset{X \in \mathbb{R}^{C \times D}}{\operatorname{argmin}} \|PX - U\|_F, \quad (\text{VII.4})$$

where $P^\dagger$ denotes the Moore–Penrose pseudoinverse² of P .

If workers have sufficiently different label distributions (i.e., the label distribution matrix P has full column rank), this estimator is unbiased and, under an additional assumption, optimal among linear estimators (see Proposition 20 in Appendix). Specifically, if the variance of the loss gradient is class-independent, it achieves the minimum variance among all linear unbiased estimators.

²The Moore–Penrose pseudoinverse $A^\dagger$ of a matrix A extends the notion of matrix inversion to non-square or rank-deficient matrices. When A is invertible, the pseudoinverse coincides with the inverse, i.e., $A^\dagger = A^{-1}$.

In HullGuard, for each group A , we sample subgroups $B_1, B_2, \dots, B_{K'}$. For each subgroup B_i , we estimate the global class-wise gradient using the least-squares estimator $\hat{U}_i = P_i^\dagger U_i$, where P_i and U_i denote here the label distribution and the updates matrices corresponding to subgroup B_i .

Variance as a Byzantine detector

The Frobenius variance [94] is defined as

Definition 2 (Frobenius variance). *The variance of a random matrix X with respect to the Frobenius norm is defined as $\sigma^2(X) = \mathbb{E}[\|X - \mathbb{E}[X]\|_F^2]$.*

Once the matrices $\hat{U}_1, \dots, \hat{U}_{K'}$ are computed, we estimate the Frobenius variance of the class-wise gradient estimator as

$$\underbrace{\hat{\sigma}^2(A) = \frac{1}{K' - 1} \sum_{k=1}^{K'} \|\hat{U}_k - \hat{\mu}\|_F^2}_{\text{empirical Frobenius variance of class gradients}}, \quad \text{where } \hat{\mu} = \frac{1}{K'} \sum_{k=1}^{K'} \hat{U}_k$$

The more similar the workers in group A are—where similarity is measured through their class-wise gradients across subgroups—the lower the empirical variance $\hat{\sigma}^2(A)$.

For theoretical purposes, we make the following assumptions: (i) the number of Byzantine workers satisfies $f < N/3$, which is equivalent to $f < h/2$; (ii) as with honest workers, Byzantine updates are also convex combinations of C vectors, and their label distributions are consistent (i.e., Byzantine updates are also derived from their class-wise gradients); (iii) for any subgroup B , the associated label distribution matrix has full column rank. All details are provided in the appendix.

The intuition is the following. The variance measures worker similarity. A group A composed solely of Byzantine workers could exhibit high internal similarity and therefore a low variance. However, since we assume $f < h/2$, Byzantine workers are always in the minority, and no such group can exist. Under the constraint $f < h/2$, the variance does not only capture similarity across workers, but similarity to the majority, i.e., similarity to honest workers. Consequently, as soon as Byzantine workers behave differently from honest ones (from a class-wise gradient perspective), their presence increases the variance.

We formalize this observation by studying the Frobenius variance of the class-wise gradient estimator for a fixed group A , denoted by $\sigma^2(A)$, which we estimate by $\hat{\sigma}^2(A)$. Let h' and f' denote the numbers of honest and Byzantine workers in A . Before proceeding, we introduce the random variables involved in the analysis. To avoid heavy notation, we present realizations of these random variables.

Consider a sampled subgroup $B \subset A$. Let P and Q denote the label distribution matrices associated with the honest and Byzantine workers in B , respectively. Let U and V denote the corresponding updates, defined in the same way as previously. Define

$$\hat{u} = P^\dagger U, \quad \hat{v} = Q^\dagger V, \quad w = (X + Y)^{-1} X, \quad X = P^\top P, \quad \text{and } Y = Q^\top Q.$$

Then $\hat{u}$ is a realization of the random variable $\hat{U}$, which represents the honest class-wise gradient estimator computed from the honest elements of A . Similarly, $\hat{v}$ is a realization of the random variable $\hat{V}$, the corresponding estimator obtained from Byzantine workers.

Regarding w , the matrix $X = P^\top P$ captures the structure of honest workers' label distributions. When these distributions are diverse and cover different classes, X spans the class space well; when they are similar or concentrated on a few classes, X becomes nearly low rank and carries less information. The same interpretation holds for Y for Byzantine workers. Thus, w measures the share of total information $X + Y$

contributed by honest workers, while $I - w = (X + Y)^{-1}Y$ measures the contribution of Byzantine workers. The matrix w is a realization of the random matrix W .

We can now explain how the variance associated with a fixed group A decomposes:

$$\begin{aligned} \sigma^2(A) = & \underbrace{\mathbb{E} \left[\left\| W(\hat{\mathcal{U}} - \mathcal{U}) \right\|_F^2 \right]}_{\text{variance of honest estimator weighted by } W} + \underbrace{\mathbb{E} \left[\left\| (I - W)(\hat{\mathcal{V}} - \mathcal{V}) \right\|_F^2 \right]}_{\text{variance of Byzantine estimator weighted by } (I - W)} \\ & + \underbrace{\mathbb{E} \left[\left\| (W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U}) \right\|_F^2 \right]}_{\text{variance of } W \text{ weighted by } (\mathcal{V} - \mathcal{U}), \text{ the difference between honest and Byzantine global class-wise gradients}}, \end{aligned} \quad (\text{VII.5})$$

where $\mathcal{U}$ and $\mathcal{V}$ denote the class-wise gradient matrices of honest and Byzantine workers, respectively. The first term corresponds to the variance of the honest estimator in A , weighted by the information contained in the honest label distributions through W . The second term is the analogous contribution for Byzantine workers. The last term captures the variability of W , scaled by the difference between the honest and Byzantine global class-wise gradients.

In HullGuard, the score of each worker depends on its membership in groups with low variance. Therefore, to bypass the HullGuard filter, Byzantine workers are incentivized to adopt strategies that reduce the group variance $\sigma^2(A)$ while remaining disruptive. The first term mainly depends on honest workers. The second term can be driven to zero by an appropriate Byzantine strategy. However, reducing the last term is more challenging. Byzantine workers would need either to make the variance of W vanish—which does not directly depend on them—or to match the honest class-wise gradients, which would no longer hinder model convergence.

In summary, obtaining a high worker score requires Byzantine workers to reduce $\sigma^2(A)$. However, achieving this effectively forces their class-wise gradients $\mathcal{V}$ to align with the honest ones $\mathcal{U}$.

Worker score

Once the empirical Frobenius variances $\sigma^2(A_1), \dots, \sigma^2(A_K)$ have been computed, HullGuard assigns each worker i a score $s(i)$ defined as

$$s(i) = \frac{|\{(k, l) : \hat{\sigma}^2(A_k) < \hat{\sigma}^2(A_l) \text{ such that } i \in A_k, i \notin A_l\}|}{|\{k : i \in A_k\}| |\{l : i \notin A_l\}|} \approx \mathbb{P}(\sigma^2(A) < \sigma^2(A') \mid i \in A, i \notin A')$$

A sufficient condition for this score to be theoretically grounded is that the variance $\sigma^2(A)$ increases with the number of Byzantine workers in A . Under some simplifying assumptions—Byzantine label distributions mimic the honest ones, and Byzantine class-wise gradients are sufficiently disruptive compared to the honest ones—we show that this property holds in Appendix.

2 Experiments

To guide the analysis, we address the following research questions:

RQ1 How effective is the HullGuard filter?

RQ2 Does HullGuard improve robustness against Byzantine attacks?

2.1 Experimental Setup

Datasets

We conduct our experiments on the following datasets : CIFAR-10 [58], MNIST [68], Fashion MNIST (FMNIST) [112] and EuroSAT [48]. Training data is divided among the workers using a Dirichlet distribution to generate non-IID local datasets, following the approach of [4, 50], with concentration parameters $\alpha \in \{0.1, 1, 10\}$. For each

Algorithm 7 HullGuard

Require: $\{(p_i, g_i)\}_{i=1}^n$ where $p_i \in \mathbb{R}_+^C$ is the class-distribution vector of worker i and $g_i \in \mathbb{R}^D$ is its gradient

Ensure: Robust aggregated gradient $\tilde{g} \in \mathbb{R}^D$

- 1: **Class attributes:** n (number of workers), h (number of honest workers), K (number of groups), K' (number of sub-subsets for variance estimation), s (sub-subset size), RobustAgg (robust aggregation rule)
- 2: Sample K groups $A_1, \dots, A_K \subset [n]$ uniformly at random, with $|A_k| = h$ for all $k \in [K]$ {Groups may overlap; elements are unique within each group}
- 3: **for** $k = 1$ to K **do**
- 4: Sample K' subgroups $B_{k,1}, \dots, B_{k,K'} \subset A_k$ of size s {Subgroups may overlap; elements are unique within each group}
- 5: **for** $j = 1$ to K' **do**
- 6: $\hat{\mathcal{U}}_{k,j} \leftarrow \text{EstimateClassWiseGradients}(B_{k,j}) \in \mathbb{R}^{C \times D}$
- 7: **end for**
- 8: $\hat{\sigma}^2(A_k) \leftarrow \text{EmpiricalFrobeniusVariance}(\hat{\mathcal{U}}_{k,1}, \dots, \hat{\mathcal{U}}_{k,K'})$
- 9: **end for**
- 10: **for** $i = 1$ to n **do**
- 11: $s(i) \leftarrow \frac{|\{(k, l) : \hat{\sigma}^2(A_k) < \hat{\sigma}^2(A_l) \text{ such that } i \in A_k, i \notin A_l\}|}{|\{k : i \in A_k\}| |\{l : i \notin A_l\}|}$
- 12: **end for**
- 13: $H \leftarrow \text{Top}_h(s(1), \dots, s(n))$ {Select the h workers with the highest scores}
- 14: $\tilde{g} \leftarrow \text{RobustAgg}(\{g_i\}_{i \in H})$ {Apply robust aggregation on the selected workers}
- 15: **return** $\tilde{g}$

dataset, the test set follows the same class distribution as its corresponding training set.

Attacks & Baselines

We consider the following attacks : A Little Is Enough (ALIE) [8], Fall Of Empire (FOE) [113], Mimic [54], Min-Sum [97] and Label Flipping (LF) [3]. We compare our solution to state-of-the-art robust aggregators : Coordinate-Wise Median (CWMed) [116], Coordinate-Wise Trimmed Mean (CwTM) [11], RFA [89] and GAS [73]. We also compare performance with and without pre-aggregation, including Bucketing (BKT) [54] and Nearest Neighbor Mixing (NNM) [4]. Due to limited computational resources, NNM is evaluated only on the CIFAR-10 dataset, where it exhibits trends similar to those observed with the other methods. Results can be found in appendix in Table C.6.

HullGuard requires the label distribution used to compute each update. We consider three Byzantine distributions: uniform, Dirac, and random. The main text reports results for the random case. The others show very similar behavior, indicating that HullGuard is largely invariant to these label distribution attacks. Results are provided in the Appendix.

FL Settings

We consider 60 workers, with $f \in \{8, 14, 20, 26\}$ Byzantine workers. We train a four-layer CNN for CIFAR-10 and EuroSAT, and a two-layer CNN for MNIST and FMNIST (model in Appendix). The models are trained with batch size 32 and momentum $\beta = 0.9$. Momentum has shown strong robustness against Byzantine behavior in both heterogeneous [4] and homogeneous settings [27, 55, 37]. Model performance is evaluated by computing the accuracy averaged over the last 200 training steps.

HullGuard Parameters

We set $K = 1000$ sampled groups and $K' = 5$ sampled subgroups per group, with subgroup size $s = 2C$. In practice, the user chooses an aggregation method applied after

Table VII.1: Mean number of Byzantine workers bypassing the HullGuard filter during training, averaged across all attacks (ALIE, FOE, Mimic, MinSum, LF).

		8	14	20	26
CIFAR10	10	1.34	0.80	0.50	0.59
	1	1.37	0.79	0.41	0.58
	0.1	4.10	2.92	1.90	1.79
MNIST	10	1.64	1.02	0.46	0.69
	1	1.64	1.01	0.76	0.93
	0.1	3.13	2.30	1.59	2.01
FMNIST	10	1.18	0.76	0.48	0.64
	1	1.22	0.99	0.82	1.14
	0.1	3.14	2.55	1.75	2.42
EuroSAT	10	1.25	0.65	0.30	0.41
	1	1.35	0.80	0.46	0.59
	0.1	5.46	3.17	1.78	1.62

the filter. We use Bucketing—with bucket size $s = 5$, corresponding to an estimated 10% fraction of Byzantine workers among the h selected workers—combined with a coordinate-wise median aggregator. In the Appendix, we also evaluate HullGuard with GAS (Table C.1), which exhibits a similar overall trend.

2.2 RQ1: How effective is the HullGuard filter?

Table VII.1 evaluates the effectiveness of the HullGuard filtering stage by reporting the mean number of Byzantine workers that pass the filter during training, averaged across all attacks. Lower values indicate a more effective filter.

Overall, the results show that HullGuard successfully blocks the majority of Byzantine updates across all datasets and configurations. For low and moderate heterogeneity levels ($\alpha = 10$ and $\alpha = 1$), the number of Byzantine workers that bypass the filter remains close to one or below in most settings, indicating that the filter almost completely removes adversarial contributions.

We also observe is that the filtering performance improves as the number of Byzantine workers increases. For instance, on CIFAR10 with $\alpha = 10$, the average number of Byzantine workers passing the filter decreases from 1.34 (8 workers) to 0.59 (26 workers). Similar trends appear across other settings, suggesting that the Frobenius variance of class-wise gradients increases with the proportion of Byzantine workers, making worker scores more discriminative.

When $\alpha = 0.1$, more Byzantine workers occasionally pass the filter. However, even in this highly heterogeneous regime, their number remains well below the total number of Byzantine participants, meaning that the filter still removes most adversarial updates before aggregation.

These results demonstrate that HullGuard’s filter is highly effective at eliminating Byzantine workers prior to aggregation.

2.3 RQ2: Does HullGuard improve robustness against Byzantine attacks?

Table VII.2 compares the final accuracy of state of the art methods face to HullGuard on CIFAR-10 under different attacks, heterogeneity.

Overall, the results show that HullGuard substantially improves robustness in the most challenging scenarios, particularly when the number of workers increases or when attacks strongly degrade the performance of existing aggregators.

First, under the ALIE attack, most baseline aggregators suffer severe performance degradation as the number of workers grows. For example, with $\alpha = 10$, CWMed drops from 80.01 (8 workers) to 17.48 (26 workers). In contrast, HullGuard maintains stable performance around 78 – 80% accuracy regardless the number of Byzantine workers. This leads to very large improvements over the best baseline, reaching gains of 50.66 and 60.39 points for 20 and 26 workers, respectively. Similar patterns appear for $\alpha = 1$ and $\alpha = 0.1$, where HullGuard again prevents the dramatic accuracy collapse observed for other methods.

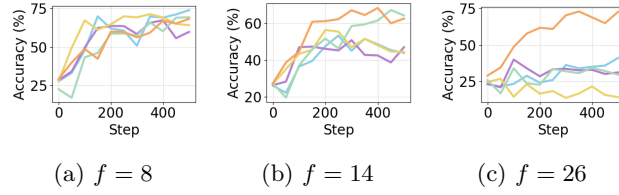


Figure VII.1: Training Accuracy for dataset EuroSAT, under the attack ALIE, with Dirichlet concentration parameter 0.1. Legend : • BKT + CWMed • BKT + CwTM • BKT + RFA • BKT + GAS • HullGuard

Second, under FOE and Mimic attacks, HullGuard also provides significant robustness improvements when attacks become more harmful. For instance, with $\alpha = 10$ and 26 workers, RFA collapses to 3.60% accuracy under FOE, whereas HullGuard maintains 80.40%, producing a gain of 13.35 points over the best baseline. Under Mimic with the same configuration, HullGuard improves accuracy by 23.90 points. Similar gains appear for $\alpha = 1$ and $\alpha = 0.1$, especially for large numbers of workers.

Third, for weaker or less disruptive attacks such as MinSum and LF, existing robust aggregators already maintain strong performance (around 80% accuracy). In these settings HullGuard performs comparably to state-of-the-art methods, with small positive or negative gains typically within a few percentage points.

Finally, HullGuard shows remarkable consistency, never collapsing in contrast to its competitors. Its robustness advantage becomes more pronounced as the difficulty of the setting increases. In particular, the largest improvements consistently appear when the number of Byzantine workers is high (20 and 26) or under stronger attacks, where competing methods often fail while HullGuard maintains stable performance.

Similar trends are observed across the other datasets, confirming that HullGuard consistently improves robustness against strong Byzantine attacks while remaining competitive with state-of-the-art methods when attacks are weaker. This behavior is illustrated in Figure VII.1, which shows training on EuroSAT under high data heterogeneity ($\alpha = 0.1$). Additional results for CIFAR-10, MNIST, Fashion-MNIST, and EuroSAT are reported in the Appendix, in Tables C.2, C.3, C.4, and C.5, respectively.

3 Conclusion

In this paper, we introduced HullGuard, a Byzantine filtering method for federated learning under label-skew heterogeneity. The method leverages a key property of this setting: despite update heterogeneity, honest updates remain noisy convex combinations of shared class-wise gradients. By exploiting this structure, HullGuard detects and filters workers whose updates are inconsistent with those of honest participants before aggregation.

Our experiments show that HullGuard effectively removes Byzantine workers during training and significantly improves robustness under strong attacks, including regimes where existing defenses often fail. HullGuard does not require server-side validation data.

We also provide theoretical insight explaining why the filter is difficult to evade: Byzantine workers that attempt to bypass it must align their class gradients with those of honest workers, which limits their capacity to disrupt training.

Future work will focus on reducing the computational cost of the filtering stage to improve scalability to larger federated systems.

Table VII.2: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (CWMed, CwTM, RFA, GAS with BKT, and HullGuard) on CIFAR-10 for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers $f \in \{8, 14, 20, 26\}$. The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
$\alpha = 10$	CWMed	80.01	64.81	24.43	17.48	78.77	77.08	48.94	29.59	78.37	76.74	62.04	52.41	81.03	81.13	81.04	80.20	77.09	75.72	69.98	61.98
	CwTM	77.69	58.69	22.06	17.71	80.21	78.66	64.38	30.33	80.25	78.15	66.16	52.81	81.29	81.29	81.07	80.67	79.49	77.37	71.26	66.63
	RFA	77.68	61.67	23.32	18.97	80.03	78.16	64.14	3.60	80.03	77.95	55.90	48.17	81.51	81.35	81.39	80.40	79.71	78.71	78.34	75.36
	GAS	75.47	45.12	27.31	18.70	80.27	78.21	76.82	67.05	78.26	74.03	71.61	56.73	80.53	81.28	80.92	80.56	80.58	80.25	80.30	80.36
	HullGuard	80.35	78.58	77.97	79.36	80.26	77.96	77.52	80.40	80.49	78.06	77.24	80.63	80.65	78.71	77.65	80.82	80.39	77.98	77.38	80.05
	Gain	0.34	13.77	50.66	60.39	-0.01	-0.70	0.70	13.35	0.24	-0.09	5.63	23.90	-0.86	-2.64	-3.74	0.15	-0.19	-2.27	-2.92	-0.31
$\alpha = 1$	CWMed	79.24	52.87	18.10	13.03	76.41	73.93	34.44	16.95	75.84	74.20	53.87	42.03	79.18	79.84	79.78	78.80	75.16	72.99	65.10	56.11
	CwTM	76.65	42.06	23.58	12.98	79.17	77.58	57.85	25.86	78.17	76.18	60.23	43.96	79.81	79.42	79.20	79.19	78.04	75.92	69.05	56.77
	RFA	76.08	50.00	25.38	14.34	78.68	77.17	53.68	16.89	77.51	76.30	52.16	38.91	79.89	79.91	79.19	79.19	77.70	76.40	75.66	66.06
	GAS	74.73	31.25	16.49	12.58	78.14	77.06	73.94	56.96	76.73	71.39	67.48	50.09	79.73	80.05	79.24	78.74	78.81	78.40	78.79	77.58
	HullGuard	79.15	76.25	74.94	77.61	78.79	75.36	74.34	79.02	78.91	75.61	74.16	78.54	79.47	76.52	75.11	78.45	79.16	75.53	74.68	79.03
	Gain	-0.09	23.38	49.56	63.27	-0.38	-2.22	0.40	22.06	0.74	-0.69	6.68	28.45	-0.42	-3.53	-4.67	-0.74	0.35	-2.87	-4.11	1.45
$\alpha = 0.1$	CWMed	64.38	29.47	11.08	7.87	60.18	50.06	28.52	10.75	61.97	53.87	29.50	25.47	67.39	68.01	65.66	67.22	57.58	52.71	36.75	38.78
	CwTM	60.29	26.04	12.71	16.68	63.86	53.55	32.05	28.49	63.93	58.12	42.36	27.10	66.82	67.84	64.82	66.33	60.01	55.33	38.01	35.98
	RFA	61.19	34.11	18.72	4.30	64.93	62.47	25.98	7.20	64.46	62.64	37.96	26.92	66.97	67.54	64.97	64.94	61.28	62.18	55.71	51.46
	GAS	58.88	15.70	15.04	4.30	65.01	64.03	50.63	22.53	64.94	58.40	50.12	28.00	66.83	68.34	65.54	66.74	64.09	62.88	57.31	59.89
	HullGuard	64.20	65.80	62.04	55.36	65.07	61.03	56.21	65.36	63.82	64.89	57.22	64.86	66.50	67.51	63.41	66.89	60.85	59.74	54.18	62.35
	Gain	-0.18	31.69	43.32	38.68	0.06	-3.00	5.58	36.87	-1.12	2.25	7.10	36.86	-0.89	-0.83	-2.25	-0.33	-3.24	-3.14	-3.13	2.46

Chapter VIII

Conclusion

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1 Contributions

This thesis studied machine learning vectors to better understand and improve classification systems. The first part focused on model outputs through confusion matrices. The second part focused on local gradients in Byzantine-robust federated learning.

1.1 Confusion Matrices

The first part of this thesis focus on confusion matrices. Although this tool is simple and widely used, its interpretation becomes difficult in heterogeneous settings. In particular, class imbalance in the test set and prediction imbalance can distort matrix values and hide the true similarities between classes.

Chapter II addressed this issue by studying normalization methods for confusion matrices. We introduced bi-normalization, based on Iterative Proportional Fitting, to normalize rows and columns simultaneously. This makes it possible to reduce the influence of distribution bias and better recover class similarities. We also connected confusion matrix normalization to importance sampling and to the geometry of class representations in the latent space of a model. These connections provide a clearer interpretation of what normalization reveal about classifier behavior.

Chapter III extended the confusion matrix beyond the single-label setting. We introduced the Transport-based Confusion Matrix, a formulation based on optimal transport and the principle of maximum entropy. This approach applies to single-label, multi-label, and soft-label classification while preserving the familiar square-matrix structure of the standard confusion matrix. It also unifies several existing formulations and provides a consistent way to analyze errors in more complex classification settings.

Chapter IV addressed the problem of evaluating confusion matrix extensions. Since real datasets do not provide a ground-truth confusion matrix in multi-label and soft-label settings, we introduced ConfusionBench, a synthetic benchmark based on stochastic classifiers with controlled latent confusion structures. This framework makes it possible to quantitatively evaluate whether a confusion matrix extension recovers the true behavior of a classifier. The results show that transport-based approaches are particularly effective for recovering latent confusion patterns in complex classification settings.

Overall, these contributions provide a more general and principled view of confusion matrices. They clarify how normalization affects interpretation, extend the confusion

matrix to broader classification frameworks, and introduce a benchmark for comparing competing extensions.

1.2 Federated Learning

The second part of this thesis studied Byzantine-robust federated learning under label skew. In this setting, honest workers may have different label distributions. As a result, their gradients can be naturally dissimilar, even without any malicious behavior. This weakens many Byzantine-robust aggregation methods, which often rely on the assumption that honest gradients are close to each other.

Chapter VI introduced NorthStar, a worker-side sampling strategy designed to reduce label-induced gradient dissimilarity. Instead of modifying the server or requiring a validation set, NorthStar changes how local minibatches are sampled. By aligning the label distributions used by workers during training, it makes honest gradients more comparable. The theoretical analysis shows that NorthStar removes a label-distribution mismatch term from honest-gradient dissimilarity. The experiments show that NorthStar improves Byzantine robustness, especially under moderate to strong label skew, and that it can be combined with other robustness strategies such as NNM.

Chapter VII introduced HullGuard, a Byzantine filter based on the geometry of class-wise gradients. The main observation is that, under label skew, honest gradients can be viewed as convex combinations of class-wise gradients. HullGuard uses this structure to filter Byzantine updates without requiring server-side validation data. The experiments show that HullGuard removes many Byzantine updates and can improve robustness when combined with existing aggregation rules.

Together, NorthStar and HullGuard address the same difficulty from two complementary perspectives. NorthStar is a lightweight worker-side defense: it acts locally, before gradients are sent to the server, by reducing the dissimilarity among honest updates. HullGuard is a heavier server-side defense: it acts after receiving the updates, by exploiting the convex structure induced by class-wise gradients, but provides stronger robustness guarantees in more adversarial settings.

2 Future Work

Several directions naturally follow from this thesis.

For confusion matrices, future work could extend the use of the Transport-based Confusion Matrix beyond evaluation, for instance to design new loss functions, or model-selection criteria. ConfusionBench could also be extended toward semi-synthetic or real-world settings, where part of the confusion structure is known through expert annotations or controlled perturbations.

For Byzantine-robust federated learning, future work could study NorthStar and HullGuard under broader forms of heterogeneity, such as feature skew, or mixed heterogeneity.

Chapter A

Normalization of Confusion Matrices: Methods and Geometric Interpretations

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The supplementary materials are organized as follows: **Section 1** provides additional results that further support our propositions. **Section 2** describes the Iterative Proportional Fitting (IPF) algorithm and an equivalent formulation known as the RAS algorithm, whose variations with respect to IPF are later exploited. **Section 3** details the overlap metric used in our experiments. **Section 4** presents experimental details, including the multidimensional histogram setup and the model architectures. **Section 5** provides proofs of key properties such as information preservation, idempotence, and class distribution invariance. **Section 6** describes additional weighting schemes mentioned in Subsection Normalization as Importance Sampling. **Section 7** derives and proves the explicit formula for the Geometric Confusion Matrix (GCM). **Section 8** demonstrates an auxiliary lemma used in the preceding sections.

1 Additional Results

Figure 1 presents the remaining results of Experiment 2 (see Subsection Baselines & Experiments) with target matrices GCM_a and GCM_c . We observe trends consistent with those obtained for GCM_r : GCM_a aligns closely with the all normalization (Subfigure 1a), while GCM_c aligns closely with the col normalization (Subfigure 1b). These results further support the geometric interpretation of normalizations.

Figure 2 reports the results for Experiment 1 (see Subsection Baselines & Experiments) using the KL divergence instead of the overlap metric. It shows that bi-normalization still achieves the highest similarity with the confusion matrix under a balanced setting, as expected.

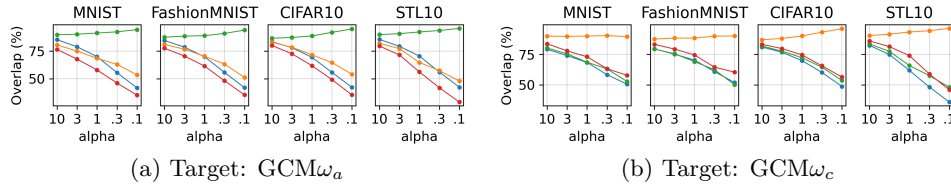


Figure 1: Overlap (%) between target matrices and normalized matrices. Legend: ● bi ● row ● col ● all

2 IPF & RAS Algorithms

This section presents the IPF (Algorithm 8) and RAS (Algorithm 9) procedures.

According to Lemma 2 (Section 8), there exist positive diagonal matrices D^l and D^r such that $\text{bi}(M) = D^l M D^r$. The RAS method estimates D^l and D^r . This algorithm will be used in the following section and is equivalent to Algorithm 8; see [51] for further details.

3 Overlap Metric

This section explains the rationale for normalizing confusion matrices when computing the Overlap metric, and shows its equivalence with the L_1 distance.

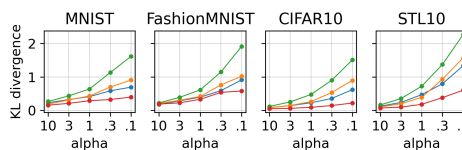


Figure 2: KL divergence (%) between matrix from balanced setting and normalized matrices. Legend: ● bi ● row ● col ● all

Algorithm 8 Iterative Proportional Fitting (IPF)

Require: Initial matrix $M \in \mathbb{R}_{>0}^{C \times C}$, target row sums $u \in \mathbb{R}_{>0}^C$, target column sums $v \in \mathbb{R}_{>0}^C$ such that $u_+ = v_+$, tolerance $\epsilon > 0$, maximum number of steps T

- 1: Initialize $Q^{(0)} \leftarrow M$, $t \leftarrow 0$
- 2: **repeat**
- 3: **for** each row $i = 1$ to C **do**
- 4: $Q_{i,:}^{(t+1)} \leftarrow Q_{i,:}^{(t)} \cdot \frac{u_i}{Q_{i+}^{(t)}}$
- 5: **end for**
- 6: **for** each column $j = 1$ to C **do**
- 7: $Q_{:,j}^{(t+2)} \leftarrow Q_{:,j}^{(t+1)} \cdot \frac{v_j}{Q_{+j}^{(t+1)}}$
- 8: **end for**
- 9: $t \leftarrow t + 2$
- 10: **until** $\|Q_{:,+}^{(t)} - u\|_1 + \|Q_{+,:}^{(t)} - v\|_1 \leq \epsilon$ **or** $t \geq T$
- 11: **return** $\hat{Q} \leftarrow Q^{(t)}$

Algorithm 9 RAS method

Require: Positive matrix $M \in \mathbb{R}_{>0}^{C \times C}$, target row sums $r \in \mathbb{R}_{>0}^C$, target column sums $c \in \mathbb{R}_{>0}^C$ such that $r_+ = c_+$, tolerance $\epsilon > 0$, maximum number of steps T

Initialize $D^{r(0)} \leftarrow I$, $t \leftarrow 0$

repeat

for each $i = 1$ to C **do**

$D_{ii}^{l(t+1)} \leftarrow \frac{r_i}{\sum_j M_{ij} D_{jj}^{r(t)}}$

end for

for each $j = 1$ to C **do**

$D_{jj}^{r(t+2)} \leftarrow \frac{c_j}{\sum_i M_{ij} D_{ii}^{l(t+1)}}$

end for

$t \leftarrow t + 2$

until $\|(D^{l(t)} M D^{r(t)})_{:,+} - r\|_1 + \|(D^{l(t)} M D^{r(t)})_{+,:} - c\|_1 \leq \epsilon$ **or** $t \geq T$

return $\hat{D}^l \leftarrow D^{l(t)}$, $\hat{D}^r \leftarrow D^{r(t)}$

3.1 Why Normalize?

The entries of a confusion matrix are only meaningful relative to each other; absolute values lack interpretability. For example, $P_{ii} = 3$ indicates perfect classification if $P_{i+} = 3$, but poor performance if $P_{i+} = 30$. More generally, for any $\lambda > 0$, P and λP represent the same model behavior.

Comparisons between two matrices are meaningful when they share the same total sum, i.e., $P_{++} = Q_{++}$. For instance, $P_{ii} = 1$ may seem worse than $Q_{ii} = 10$, but if $P_{++} = 3$ and $Q_{++} = 30$, the relative accuracies are identical: $P_{ii}/P_{++} = Q_{ii}/Q_{++}$.

For consistency, we compare normalized confusion matrices: $\text{all}(P) = P/P_{++}$ and $\text{all}(Q) = Q/Q_{++}$. This ensures that behaviors are preserved and enables consistent comparisons.

3.2 Equivalence with L_1 Distance

Let P and Q be two matrices such that $P_{++} = Q_{++} = 1$. Then,

$$\begin{aligned}
\|P - Q\|_1 &= \sum_{ij} |P_{ij} - Q_{ij}| \\
&= \sum_{ij} \max(P_{ij}, Q_{ij}) - \min(P_{ij}, Q_{ij}) \\
&= \sum_{ij: P_{ij} \geq Q_{ij}} P_{ij} - \min(P_{ij}, Q_{ij}) + \sum_{ij: Q_{ij} > P_{ij}} Q_{ij} - \min(P_{ij}, Q_{ij}) \\
&= \sum_{ij} P_{ij} - \min(P_{ij}, Q_{ij}) + \sum_{ij} Q_{ij} - \min(P_{ij}, Q_{ij}) \\
&= 2 - 2 \text{Overlap}(P, Q)
\end{aligned}$$

Thus, the Overlap metric is directly related to the L_1 distance.

4 Experiments Details

This section presents details on the multidimensional histogram setup and the model architectures.

4.1 Histogram in Experiments

First, the projection space is divided into a regular grid of m -dimensional hyperrectangles,¹ which define the histogram bins. We then construct non-overlapping histograms [102] associated with the clusters $\mathcal{C}_k$ and $\hat{\mathcal{C}}_k$ for $k = 1, \dots, C$.

Considering a cluster $\mathcal{C}$ and its unweighted histogram, each bin height represents the number of points from $\mathcal{C}$ that fall within the bin (i.e., hyperrectangle) under consideration. In the weighted case, the bin height instead corresponds to the sum of the point weights, using one of these weighting schemes ω_a , ω_r , ω_c , or ω_b .

In the experiments, we fix the projected space dimensionality to $m = 10$ and specify the bin width accordingly. Following the recommendation of [96] (see Subsection 3.4, Eq. (3.66)), we adopt the optimal bin width $3.5 \sigma_k n^{-1/(2+m)}$, where σ_k denotes the empirical standard deviation along the k -th dimension, n is the sample size, and m is the data dimensionality.

4.2 Model Architectures

We use the following abbreviations to describe our architectures: $L(n)$ denotes a fully connected linear layer with n outputs; R is a ReLU activation; $C(c)$ represents a 2D convolutional layer with c output channels, kernel size 5 for CIFAR datasets and 3 otherwise, padding 0 for CIFAR and 1 otherwise, and stride 1; M denotes 2D max pooling with kernel size 2; B stands for batch normalization; and D represents dropout with a fixed probability of 0.25.

In line with Allouah et al. [4], the model architectures are defined as follows. For MNIST and Fashion-MNIST, the architecture is $(1, 28 \times 28) - C(8) - R - M - C(16) - R - M - L(64) - R - L(10)$. For CIFAR-10 and STL-10, the architecture is $(3, 32 \times 32) - C(64) - R - B - C(64) - R - B - M - D - C(128) - R - B - C(128) - R - B - M - D - L(128) - R - D - L(10)$.

¹In geometry, a hyperrectangle generalizes a rectangle (in 2D) and a rectangular cuboid (in 3D) to higher dimensions [95].

5 Information Preservation, Idempotence & Class Distribution Invariance

This section demonstrates that normalization methods satisfy the properties of information preservation, idempotence, and class distribution invariance.

5.1 Information Preservation for Standards Normalizations

We state that, for a positive confusion matrix M , the normalizations minimize the KL divergence under specific constraints:

$$\begin{aligned} \text{all}(M) &\in \underset{P \in \mathbb{R}_{>0}^{C \times C}}{\text{argmin}} D_{\text{KL}}(P \| M), & \text{row}(M) &\in \underset{P \in \mathbb{R}_{>0}^{C \times C}}{\text{argmin}} D_{\text{KL}}(P \| M), \\ P_{i+} &= \frac{M_{i+}}{M_{++}}, P_{+j} = \frac{M_{+j}}{M_{++}} \quad \forall i, j, & P_{i+} &= 1, P_{+j} = \sum_i \frac{M_{ij}}{M_{i+}} \quad \forall i, j \end{aligned}$$

$$\begin{aligned} \text{col}(M) &\in \underset{P \in \mathbb{R}_{>0}^{C \times C}}{\text{argmin}} D_{\text{KL}}(P \| M) \\ P_{i+} &= \sum_j \frac{M_{ij}}{M_{+j}}, P_{+j} = 1 \quad \forall i, j \end{aligned}$$

We observe that these normalizations can be expressed as the product of diagonal matrices and the original confusion matrix:

$$\text{row}(M) = \text{diag}\left(\frac{1}{M_{1+}}, \dots, \frac{1}{M_{C+}}\right) M I, \quad \text{col}(M) = I M \text{diag}\left(\frac{1}{M_{+1}}, \dots, \frac{1}{M_{+C}}\right), \quad (1)$$

$$\text{and } \text{all}(M) = I M I / M_{++}.$$

According to Lemma 2 (Section 8), these normalization procedures indeed minimize the KL divergence, which concludes the proof.

5.2 Idempotence & Class Distribution Invariance of Bi Normalization

In Proposition 1, we state that the bi-normalization satisfies idempotence,

$$\text{bi} \circ \text{bi}(M) = \text{bi}(M),$$

and class distribution invariance, that is, for any diagonal matrices $A, B \in \mathbb{R}_{>0}^{C \times C}$,

$$\text{bi}(AMB) = \text{bi}(M).$$

We prove these properties in what follows.

Idempotence. By definition, $\text{bi} \circ \text{bi}(M)$ is the solution to the problem

$$\begin{aligned} &\underset{P \in \mathbb{R}_{>0}^{C \times C}}{\text{argmin}} D_{\text{KL}}(P \| \text{bi}(M)) \\ P_{i+} &= P_{+j} = 1 \quad \forall i, j \end{aligned}$$

Moreover, we observe that $I \text{bi}(M) I$ satisfies the marginal constraints. By Lemma 2, $I \text{bi}(M) I$ solves the above problem, and since the solution is unique, we have

$$\text{bi} \circ \text{bi}(M) = I \text{bi}(M) I = \text{bi}(M),$$

which establishes the idempotence property.

Class Distribution Invariance. By Lemma 2, there exist diagonal matrices D^l and D^r such that

$$\text{bi}(AMB) = D^l A M B D^r.$$

By definition of the bi-normalization, the row and column sums satisfy

$$(D^l AMBD^r)_{i+} = (D^l AMBD^r)_{+j} = 1 \quad \forall i, j.$$

Additionally, $D^l A$ and BD^r are diagonal. Therefore, by Lemma 2, $\text{bi}(AMB)$ is the unique solution to

$$\begin{aligned} & \underset{P \in \mathbb{R}_{>0}^{C \times C}}{\text{argmin}} \quad D_{\text{KL}}(P \| M) \\ & P_{i+} = 1, P_{+j} = 1 \quad \forall i, j \end{aligned}$$

This implies $\text{bi}(AMB) = \text{bi}(M)$, completing the proof.

6 Importance Sampling

This section details the weighting schemes mentioned in Subsection Normalization as Importance Sampling.

From the equalities presented in Equation (1) (Subsection 5.1) and by applying the same procedure described in the main part of the paper, it directly follows that

$$\omega_a : (y, \hat{y}) \mapsto \frac{1}{M_{++}} \quad \text{and} \quad \text{all}(M) = \sum_{k=1}^N \omega_a(y_k, \hat{y}_k) E_{y_k \hat{y}_k},$$

and

$$\omega_c : (y, \hat{y}) \mapsto \frac{1}{M_{+\hat{y}}} \quad \text{and} \quad \text{col}(M) = \sum_{k=1}^N \omega_c(y_k, \hat{y}_k) E_{y_k \hat{y}_k}.$$

According to Lemma 2 (Section 8), there exist positive diagonal matrices D^l and D^r such that $\text{bi}(M) = D^l M D^r$, using Algorithm 9 (Section 2), we estimate these matrices by matrices $\hat{D}^l$ and $\hat{D}^r$. It follows

$$\omega_b : (y, \hat{y}) \mapsto \hat{D}_y^l \hat{D}_{\hat{y}}^r \quad \text{and} \quad \text{bi}(M) = \sum_{k=1}^N \omega_b(y_k, \hat{y}_k) E_{y_k \hat{y}_k}.$$

7 Geometric Confusion Matrix

This section establishes the volumes of the various scaled histograms introduced previously, and compares the analytical form of the classical confusion matrix with that of the geometric confusion matrix.

7.1 Normalized Histograms Under Weighting Schemes

In Section Geometric View of Normalized Confusion Matrices, we state that, under specific weighting schemes, some histograms are normalized up to a scaling factor. We prove this statement below.

We first introduce the following lemma:

Lemma 1. *Let $\mathcal{C}$ be a cluster of points in the model's latent space, based on dataset $\mathcal{D}$. Let P be a projection matrix, and let ω be the weighting function. Then,*

$$\lambda(V(\mathcal{C}, \omega)) = r \sum_{(x, y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P\varepsilon(x) \in \mathcal{C}},$$

where λ denotes the Lebesgue measure, $\mathbb{1}$ is the indicator function, and r the volume of hyperrectangles which split the projection space.

Proof. We begin by decomposing the histogram into a union of hyperrectangles which grid the projection space, denoted by $\mathcal{V}$. Since $\mathcal{D}$ contains a finite number of points, the covering set $\mathcal{V}$ is also finite. By construction of the weighted histogram, we have

$$\begin{aligned} V(\mathcal{C}, \omega) &= \bigcup_{v \in \mathcal{V}} (V(\mathcal{C}, \omega) \cap v) \\ &= \bigcup_{v \in \mathcal{V}} \underbrace{r_1^v \times \cdots \times r_m^v}_{\text{bin/hyperrectangle splitting the projection space}} \times \underbrace{\left[0, \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C} \cap v} \right]}_{\text{bin height}} \end{aligned}$$

where $r_1^v, \dots, r_m^v$ denote the intervals describing the sides of each hyperrectangle.

Its Lebesgue measure equals the sum of the measures of the individual hyperrectangles:

$$\lambda(V(\mathcal{C}, \omega)) = \sum_{v \in \mathcal{V}} \lambda(r_1^v \times \cdots \times r_m^v \times \left[0, \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C} \cap v} \right])$$

The Lebesgue measure of a hyperrectangle is the product of its side lengths. Only the heights vary, while the other side lengths are bin-independent. Let $r = \prod_{i=1}^m \lambda(r_i^v)$. Then,

$$\lambda(V(\mathcal{C}, \omega)) = \sum_{v \in \mathcal{V}} r \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C} \cap v} = r \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C}}$$

which concludes the proof. $\square$

According to Lemma 1, we have

$$\begin{aligned} \lambda(V(\mathcal{C}_i, \omega_r)) &= r \sum_{(x,y) \in \mathcal{D}} \omega_r(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C}_i} = r \sum_{(x,y) \in \mathcal{D}} \frac{1}{M_{i+}} \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C}_i} = r, \\ \lambda(V(\hat{\mathcal{C}}_j, \omega_c)) &= r \sum_{(x,y) \in \mathcal{D}} \omega_c(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \hat{\mathcal{C}}_j} = r \sum_{(x,y) \in \mathcal{D}} \frac{1}{M_{+j}} \mathbb{1}_{P_\varepsilon(x) \in \hat{\mathcal{C}}_j} = r, \\ \lambda(V(\mathcal{C}_i, \omega_b)) &= r \sum_{(x,y) \in \mathcal{D}} \omega_b(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C}_i} = r \sum_{(x,y) \in \mathcal{D}} \hat{D}_i^l \hat{D}_y^r \mathbb{1}_{P_\varepsilon(x) \in \mathcal{C}_i} = r \text{bi}(M)_{i+} = r, \\ \lambda(V(\hat{\mathcal{C}}_j, \omega_b)) &= r \sum_{(x,y) \in \mathcal{D}} \omega_b(y, \hat{y}) \mathbb{1}_{P_\varepsilon(x) \in \hat{\mathcal{C}}_j} = r \sum_{(x,y) \in \mathcal{D}} \hat{D}_y^l \hat{D}_j^r \mathbb{1}_{P_\varepsilon(x) \in \hat{\mathcal{C}}_j} = r \text{bi}(M)_{+j} = r. \end{aligned}$$

In that sense, these histograms are normalized using weighting schemes up to a scaling factor r , corresponding to the fixed volume of the hyperrectangles partitioning the projection space.

7.2 Geometric Confusion Matrix

The Geometric Confusion Matrix is defined as

$$\left(\text{GCM}_\omega \right)_{ij} = \lambda \left(V(\mathcal{C}_i, \omega) \cap V(\hat{\mathcal{C}}_j, \omega) \right)$$

We begin by deriving an explicit formula for GCM_ω . Following the same approach as in proof of Lemma 1, and noting that the grid of hyperrectangles partitioning the projection space is shared across all histograms, the intersection of two histograms is

given by:

$$\begin{aligned}
& V(\mathcal{C}_i, \omega) \cap V(\widehat{\mathcal{C}}_j, \omega) \\
&= \bigcup_{v \in \mathcal{V}} r_1^v \times \cdots \times r_m^v \times \left[0, \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P\varepsilon(x) \in \mathcal{C}_i \cap v} \right] \cap \\
&\quad \bigcup_{v \in \mathcal{V}} r_1^v \times \cdots \times r_m^v \times \left[0, \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P\varepsilon(x) \in \widehat{\mathcal{C}}_j \cap v} \right] \\
&= \bigcup_{v \in \mathcal{V}} r_1^v \times \cdots \times r_m^v \times \left[0, \min \left(\sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P\varepsilon(x) \in \mathcal{C}_i \cap v}, \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P\varepsilon(x) \in \widehat{\mathcal{C}}_j \cap v} \right) \right]
\end{aligned}$$

As a result, the (i, j) entry of the geometric confusion matrix becomes:

$$\left(\text{GCM}_{l,p} \right)_{ij} = r \sum_{v \in \mathcal{V}} \min \left(\sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P\varepsilon(x) \in \mathcal{C}_i \cap v}, \sum_{(x,y) \in \mathcal{D}} \omega(y, \hat{y}) \mathbb{1}_{P\varepsilon(x) \in \widehat{\mathcal{C}}_j \cap v} \right)$$

The normalized confusion matrix can also be expressed as a sum over grid cells, analogously to GCM_ω . Let norm denote a normalization operator among all, row, col, and bi, and let ω be its associated weighting scheme. It is straightforward to obtain:

$$\begin{aligned}
\text{norm}(M)_{ij} &= \sum_{(x,y) \in \mathcal{D}} \omega(i, j) \mathbb{1}_{y=i} \mathbb{1}_{\hat{y}=j} \\
&= \sum_{(x,y) \in \mathcal{D}} \omega(i, j) \mathbb{1}_{P\varepsilon(x) \in \mathcal{C}_i \cap \widehat{\mathcal{C}}_j} \\
&= \sum_{v \in \mathcal{V}} \sum_{(x,y) \in \mathcal{D}} \omega(i, j) \mathbb{1}_{P\varepsilon(x) \in \mathcal{C}_i \cap \widehat{\mathcal{C}}_j \cap v}
\end{aligned}$$

This explicit formula for GCM_ω highlights how the geometric confusion matrix incorporates spatial information. The geometric confusion matrix compares in each grid cell the number of points labeled i (regardless of prediction) with the number of points predicted as j (regardless of true label). As a result, in a grid cell containing only points with label i and prediction j , the contribution to GCM_ω matches that of the standard confusion matrix. In contrast, in a cell containing a mix of labels and predictions, the minimum operation produces a contribution that differs from the simple count of points labeled i and predicted as j —that is, from the contribution in the standard confusion matrix.

8 Equivalence Scaling Lemma

This section demonstrates the following lemma:

Lemma 2. *Let M be a positive matrix in $\mathbb{R}_{>0}^{C \times C}$, and r and c , be two positive vectors in $\mathbb{R}_{>0}^C$ such that $r_+ = c_+$. Then, any matrix M^* of the form $M^* = D^l M D^r$, where D^l and D^r are positive diagonal matrices such that*

$$(D^l M D^r)_{i+} = r_i \quad \text{and} \quad (D^l M D^r)_{+j} = c_j \quad \text{for all } i, j,$$

is the unique solution to the following problem:

$$\begin{aligned}
M^* &\in \underset{P \in \mathbb{R}_{>0}^{C \times C}}{\text{argmin}} \quad D_{\text{KL}}(P \| M) \\
P_{i+} &= r_i, P_{+j} = c_j \quad \forall i, j
\end{aligned}$$

Note that this result is already known in a different form, referred to as equivalence scaling (see Section 3. Different approaches to equivalence scaling in [51]). We do not claim to introduce a new result with this lemma; rather, we were unable to find this particular formulation in the literature.

8.1 Problem Reformulation

We use the method of Lagrange multipliers. We define the row constraints for $i = 1, \dots, C$ as:

$$\begin{aligned} h_i : \mathbb{R}_{>0}^{C \times C} &\rightarrow \mathbb{R} \\ P &\mapsto P_{i+} - r_i \end{aligned}$$

and the column constraints for $j = 1, \dots, C$ as:

$$\begin{aligned} g_j : \mathbb{R}_{>0}^{C \times C} &\rightarrow \mathbb{R} \\ P &\mapsto P_{+j} - c_j \end{aligned}$$

Since adding a constant to the objective function does not affect the arguments of the minima, we define $f(P) := D_{\text{KL}}(P \| M) - 1$ to simplify the derivative calculations.

The optimization problem becomes:

$$\begin{aligned} M^* \in \operatorname{argmin}_{P \in \mathbb{R}_{>0}^{C \times C}} f(P) \\ h_i(P) = g_j(P) = 0 \quad \forall i, j \end{aligned}$$

The functions f , h_i , and g_j are continuously differentiable.

8.2 Karush-Kuhn-Tucker Conditions

To find a solution, we apply the Karush-Kuhn-Tucker (KKT) conditions (see *Section 5.5.3: KKT Optimality Conditions* in [12]). These conditions are necessary for a point M^* to be a local minimum when a constraint qualification is satisfied. In our case, the Linearity Constraint Qualification (LCQ) holds, since the constraints h_i and g_j are affine.

Moreover, since f is convex (as shown in a subsection below), and the constraint set is convex, the KKT conditions are also sufficient for optimality [12]. Thus, any point satisfying the KKT conditions is a global minimum.

In other words, any feasible matrix P^* and multipliers μ^* and ν^* satisfying:

$$\nabla_P L(P^*, \mu^*, \nu^*) = 0$$

is a global minimum, where the Lagrangian is given by:

$$L(P, \mu, \nu) = f(P) + \sum_{i=1}^C \mu_i h_i(P) + \sum_{j=1}^C \nu_j g_j(P)$$

Since f is strictly convex (Subsection *Convexity of f*), this minimum is also unique.

8.3 Solution

Let P be an admissible point. We compute the gradient of the Lagrangian²:

$$\nabla_P L(P, \mu, \nu)_{ij} = \left[\ln \left(\frac{P_{ij}}{M_{ij}} \right) + \mu_i + \nu_j \right]_{ij}$$

Setting the gradient to zero gives:

$$\ln \left(\frac{P_{ij}}{M_{ij}} \right) + \mu_i + \nu_j = 0 \quad \Leftrightarrow \quad P_{ij} = \exp(-\mu_i) M_{ij} \exp(-\nu_j)$$

²Though the gradient is a vector, we index it here by (i, j) for clarity.

We define diagonal matrices D^l and D^r with entries:

$$(D^l)_{ii} = \exp(-\mu_i), \quad (D^r)_{jj} = \exp(-\nu_j)$$

Then the solution takes the form:

$$P = D^l M D^r$$

In conclusion, any matrix of the form $D^l M D^r$ that satisfies the given row and column constraints is the unique solution to the minimization problem.

8.4 Convexity of f

Let $P, P' \in \mathbb{R}_{>0}^{C \times C}$ be two distinct matrices, and let $\gamma \in (0, 1)$. The function f is strictly convex if and only if

$$f(\gamma P + (1 - \gamma)P') < \gamma f(P) + (1 - \gamma)f(P').$$

We now prove this inequality. We first show that the KL divergence is strictly convex in P when M is fixed.

$$\begin{aligned} D_{\text{KL}}(\gamma P + (1 - \gamma)P' \| M) &= \sum_{ij} \left(\gamma P_{ij} + (1 - \gamma)P'_{ij} \right) \ln \left(\frac{\gamma P_{ij} + (1 - \gamma)P'_{ij}}{\gamma M_{ij} + (1 - \gamma)M_{ij}} \right) \\ &< \sum_{ij} \gamma P_{ij} \ln \left(\frac{P_{ij}}{M_{ij}} \right) + (1 - \gamma)P'_{ij} \ln \left(\frac{P'_{ij}}{M_{ij}} \right) \\ &= \gamma D_{\text{KL}}(P \| M) + (1 - \gamma)D_{\text{KL}}(P' \| M) \end{aligned}$$

where the strict inequality follows from the log-sum inequality, which becomes strict when $P \neq P'$. This implies that f is strictly convex, which completes the proof.

Chapter A

Unifying Confusion Matrices from Single-Label to Soft-Label

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1 Proof of Proposition 3

We first observe that

$$\sum_{i,j} c(i,j)\pi_{ij} = \sum_{i \neq j} \pi_{ij} = 1 - \text{Tr}(\pi),$$

since π has total mass 1. Therefore, minimizing the transport cost is equivalent to maximizing $\text{Tr}(\pi)$.

For any admissible plan $\pi \in T(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$, the marginal constraints imply

$$\pi_{ii} \leq \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} \quad \text{and} \quad \pi_{ii} \leq \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1},$$

hence

$$\pi_{ii} \leq \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}).$$

Summing over i yields

$$\text{Tr}(\pi) = \sum_{i=1}^C \pi_{ii} \leq \sum_{i=1}^C \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}).$$

It remains to show that this upper bound is achievable. Define $d := \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ and set

$$\pi_{ii} = d_i \quad \text{for all } i.$$

Let the residual vectors be

$$r := \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - d, \quad s := \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - d.$$

By construction, $r, s \geq 0$ and

$$\sum_{i=1}^C r_i = \sum_{i=1}^C s_i.$$

Therefore, there exists a nonnegative matrix $(\pi_{ij})_{i \neq j}$ with row sums r and column sums s , which completes π into a feasible transport plan in $T(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$.

Thus, the upper bound is attained, and we conclude that

$$\max_{\pi \in T(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})} \text{Tr}(\pi) = \sum_{i=1}^C \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}).$$

This proves optimality.

2 Proof of Proposition 4

Let $(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ be a label-prediction pair, and let π be an optimal plan. By Proposition 3, we have

$$\text{diag}(\pi) = \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}).$$

Therefore, the row sums of the off-diagonal terms are

$$\mathbf{u} := \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}),$$

and the column sums are

$$\hat{\mathbf{u}} := \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}).$$

We first note that:

1. $\mathbf{u}$ and $\hat{\mathbf{u}}$ have the same total mass, that is,

$$\mathbf{u}_+ = \hat{\mathbf{u}}_+,$$

2. any non-uniqueness of optimal plans can only arise from the off-diagonal terms, since the diagonal is fixed by optimality.

We now distinguish several cases.

If $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$, then $\mathbf{u} = \hat{\mathbf{u}} = 0$. Hence, all off-diagonal terms vanish, and the optimal plan is unique.

Assume next that only one class is underestimated, that is, there exists some i such that

$$\frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1} < \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \quad \text{and} \quad \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} \leq \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} \quad \text{for all } k \neq i.$$

Then $\mathbf{u}$ vanishes everywhere except at coordinate i . Equivalently, all off-diagonal mass is received by row i . Since $\pi_{jj} = \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1})$ for every j , admissibility implies that, for every $j \neq i$,

$$\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} = \pi_{+j} = \pi_{jj} + \pi_{ij} = \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}) + \pi_{ij},$$

and therefore

$$\pi_{ij} = \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}\right).$$

Thus, every off-diagonal coefficient is uniquely determined, and the optimal plan is unique. The same argument applies when only one class is overestimated.

We now assume that none of the previous cases holds. Then there exist at least two underestimated classes, indexed by $i_1, \dots, i_K$, and at least two overestimated classes, indexed by $j_1, \dots, j_L$, with $K \geq 2$ and $L \geq 2$. Equivalently,

$$\mathbf{u}_{i_k} > 0 \quad \text{for all } k = 1, \dots, K, \quad \hat{\mathbf{u}}_{j_\ell} > 0 \quad \text{for all } \ell = 1, \dots, L.$$

We show that in this case there exist at least two distinct optimal plans.

Since the diagonal part is fixed, the off-diagonal part defines a transport problem with marginals $\mathbf{u}$ and $\hat{\mathbf{u}}$. It is therefore sufficient to construct two distinct admissible couplings between these marginals.

Define a first plan π^a by transporting the mass greedily from the overestimated classes $j_1, j_2, \dots, j_L$ to the underestimated classes $i_1, i_2, \dots, i_K$, in this order. More precisely, one starts with column j_1 and fills rows $i_1, i_2, \dots, i_K$ successively until the mass $\hat{\mathbf{u}}_{j_1}$ is exhausted; one then proceeds similarly with column j_2 , and so on. Define a second plan π^b in the same way, except that the underestimated classes are filled in the reverse order $i_K, i_{K-1}, \dots, i_1$.

Both constructions yield admissible transport plans for the off-diagonal mass, and therefore optimal plans once the diagonal $\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ is added back. Moreover, $\pi^a \neq \pi^b$. We now show that

$$\pi_{i_1 j_1}^a > \pi_{i_1 j_1}^b.$$

By construction of π^a , the mass sent from column j_1 is first assigned to row i_1 , so that

$$\pi_{i_1 j_1}^a = \min(\mathbf{u}_{i_1}, \hat{\mathbf{u}}_{j_1}) > 0.$$

On the other hand, for any $\pi \in T^{\text{opt}}(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$, one has

$$\pi_{ij} \leq \min(\mathbf{u}_i, \hat{\mathbf{u}}_j),$$

since $\mathbf{u}_i$ is the total off-diagonal mass required in row i , and $\hat{\mathbf{u}}_j$ is the total off-diagonal mass available in column j .

We now show that this upper bound is not attained at (i_1, j_1) for π^b .

First, assume by contradiction that

$$\pi_{i_1 j_1}^b = \mathbf{u}_{i_1}.$$

Since π^b fills column j_1 starting from rows $i_K, i_{K-1}, \dots, i_1$, this implies that column j_1 still has available mass when reaching row i_1 . Therefore, all previous rows in the construction order must have sent their entire mass to j_1 , that is,

$$\pi_{i_k j_1}^b = \mathbf{u}_{i_k} \quad \text{for all } k = 1, \dots, K.$$

Hence,

$$\hat{\mathbf{u}}_{j_1} \geq \sum_{k=1}^K \mathbf{u}_{i_k} = \mathbf{u}_+.$$

Since $\mathbf{u}_+ = \hat{\mathbf{u}}_+$, this implies

$$\hat{\mathbf{u}}_{j_1} = \hat{\mathbf{u}}_+ \quad \text{and} \quad \hat{\mathbf{u}}_{j_\ell} = 0 \quad \text{for all } \ell \neq 1,$$

which contradicts the assumption that at least two classes are overestimated. Therefore,

$$\pi_{i_1 j_1}^b < \mathbf{u}_{i_1}.$$

Next, assume by contradiction that

$$\pi_{i_1 j_1}^b = \hat{\mathbf{u}}_{j_1}.$$

During the construction of π^b , column j_1 is filled starting from rows $i_K, i_{K-1}, \dots$. Since j_1 is the first column treated, any row with positive mass encountered before i_1 would contribute strictly positively to column j_1 . Thus, this equality implies

$$\mathbf{u}_{i_k} = 0 \quad \text{for } k = 2, \dots, K,$$

which contradicts the assumption that at least two classes are underestimated. Hence,

$$\pi_{i_1 j_1}^b < \hat{\mathbf{u}}_{j_1}.$$

Combining the two strict inequalities, we obtain

$$\pi_{i_1 j_1}^b < \min(\mathbf{u}_{i_1}, \hat{\mathbf{u}}_{j_1}) = \pi_{i_1 j_1}^a.$$

Thus, $\pi^a \neq \pi^b$, and there exist at least two distinct optimal plans.

Finally, since $T^{\text{opt}}(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ is convex, every convex combination

$$\lambda \pi^a + (1 - \lambda) \pi^b, \quad \lambda \in [0, 1],$$

also belongs to $T^{\text{opt}}(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$. Therefore, there are infinitely many optimal plans.

3 Corollaries of Propositions 3 & 4

As soon as $C \leq 3$, there is either equality or at most one overestimated or one underestimated class. As a result, there is a unique optimal plan by Proposition 4.

By optimality, the row sums of the off-diagonal terms are

$$\mathbf{u} := \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right),$$

and the column sums are

$$\hat{\mathbf{u}} := \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right).$$

By admissibility, any surplus mass from overestimated classes is transported only to underestimated classes.

An optimal plan is diagonal if and only if $\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})_+ = \frac{\mathbf{y}}{\|\mathbf{y}\|_1}_+ = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}_+$, that is, if and only if $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$.

4 Proof of proposition 5

Let's show that the matrix

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \text{diag}\left(\min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)\right) + \frac{\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)\right) \otimes \left(\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)\right)}{\left\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)\right\|_1}$$

is in $T^{\text{opt}}(y, \hat{y})$ and maximises the entropy. Moreover, let's demonstrate that $\pi(\mathbf{y}, \hat{\mathbf{y}})$ becomes

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \text{diag}\left(\min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)\right),$$

when $y = \hat{y}$ by continuous extension.

4.1 Case $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$ and continuous extension

When $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$, there is a unique optimal plan (see Section 3), namely

$$\text{diag}(\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})),$$

which is therefore the unique entropy maximizer.

Now let $(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})$ be such that $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} \neq \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$. We show that, as

$$\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1 \rightarrow 0,$$

the formula in Proposition 5 converges to

$$\pi(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}) = \text{diag}(\min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})),$$

thereby proving that the formula remains valid by continuous extension at $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$.

Set

$$E := \frac{\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\right) \otimes \left(\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\right)}{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1}.$$

It is enough to prove that

$$E \rightarrow 0 \in \mathbb{R}^{C \times C} \quad \text{as} \quad \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1 \rightarrow 0.$$

For every $i, j \in \{1, \dots, C\}$, the entry E_{ij} is

$$E_{ij} = \frac{\left(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1})\right) \left(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1})\right)}{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1}.$$

We prove that $E_{ij} \rightarrow 0$ by deriving an upper bound depending on $\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1$.

First, observe that

$$\begin{aligned} \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}) &= \max(\frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}, 0) \leq \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1, \\ \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1}) &= \max(\frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} - \frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}, 0) \leq \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1. \end{aligned}$$

Hence,

$$0 \leq E_{ij} \leq \frac{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1^2}{\|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1}.$$

Next, we use the identity

$$\begin{aligned} \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\|_1 &= \sum_{k=1}^C \left| \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} \right| \\ &= \sum_{k: \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} < \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}} \left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} \right) + \sum_{k: \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} < \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}} \left(\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} - \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} \right) \\ &= \sum_{k=1}^C \left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}) \right) + \sum_{k=1}^C \left(\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}) \right) \\ &= \|\frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1 + \|\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1})\|_1. \end{aligned}$$

Since

$$\sum_{k=1}^C \left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\right) \right) = \sum_{k=1}^C \left(\frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\right) \right),$$

we obtain

$$\left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} \right\|_1 = 2 \left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1.$$

Substituting this into the previous bound gives

$$0 \leq E_{ij} \leq 2 \left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} \right\|_1.$$

Therefore,

$$E_{ij} \rightarrow 0 \quad \text{as} \quad \left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} \right\|_1 \rightarrow 0,$$

for all $i, j \in \{1, \dots, C\}$. Hence,

$$E \rightarrow 0 \in \mathbb{R}^{C \times C}.$$

We have thus proved that, by continuous extension at $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$, one has

$$\pi\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) = \text{diag}\left(\min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)\right).$$

4.2 Case $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} \neq \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$

Define, for each $k \in \{1, \dots, C\}$,

$$\begin{aligned} d_k &:= \min\left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\right), & r_k &:= \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\right), \\ c_k &:= \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\right). \end{aligned}$$

By construction,

$$\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} = d_k + r_k, \quad \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1} = d_k + c_k, \quad r_k c_k = 0.$$

Moreover,

$$\sum_{k=1}^C r_k = \sum_{k=1}^C c_k =: m > 0,$$

since $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} \neq \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}$.

From admissibility and optimality constraints, any feasible matrix M must satisfy

$$M_{kk} = d_k, \quad \sum_{j:j \neq k} M_{kj} = r_k, \quad \sum_{i:i \neq k} M_{ik} = c_k.$$

Therefore, the problem reduces to maximizing

$$H(M) := - \sum_{i=1}^C \sum_{j=1, j \neq i}^C M_{ij} \ln(M_{ij})$$

over the off-diagonal entries, under

$$M_{ij} \geq 0, \quad \sum_{j \neq i} M_{ij} = r_i, \quad \sum_{i \neq j} M_{ij} = c_j.$$

Define the index sets of underestimated and overestimated classes as

$$A := \{i : r_i > 0\} = \{i : \frac{\mathbf{y}_i}{\|\mathbf{y}\|_1} > \frac{\hat{\mathbf{y}}_i}{\|\hat{\mathbf{y}}\|_1}\}, \quad B := \{j : c_j > 0\} = \{j : \frac{\hat{\mathbf{y}}_j}{\|\hat{\mathbf{y}}\|_1} > \frac{\mathbf{y}_j}{\|\mathbf{y}\|_1}\}.$$

Since $r_k c_k = 0$ for every k , the sets A and B are disjoint. Also, if $i \notin A$, then $r_i = 0$, hence $M_{ij} = 0$ for all $j \neq i$. Likewise, if $j \notin B$, then $c_j = 0$, hence $M_{ij} = 0$ for all $i \neq j$. Thus the only potentially nonzero off-diagonal entries are those with $(i, j) \in A \times B$.

We thus solve

$$\max_{M_{ij} \geq 0} - \sum_{(i,j) \in A \times B} M_{ij} \ln(M_{ij})$$

subject to

$$\sum_{j \in B} M_{ij} = r_i \quad (i \in A), \quad \sum_{i \in A} M_{ij} = c_j \quad (j \in B).$$

The objective is strictly concave on the convex feasible set, so any critical point is the unique maximizer. Introduce Lagrange multipliers $(\alpha_i)_{i \in A}$ and $(\beta_j)_{j \in B}$ and consider

$$\mathcal{L}(M, \alpha, \beta) = - \sum_{(i,j) \in A \times B} M_{ij} \ln(M_{ij}) + \sum_{i \in A} \alpha_i \left(\sum_{j \in B} M_{ij} - r_i \right) + \sum_{j \in B} \beta_j \left(\sum_{i \in A} M_{ij} - c_j \right).$$

For any $(i, j) \in A \times B$, stationarity gives

$$\frac{\partial \mathcal{L}}{\partial M_{ij}} = -(\ln M_{ij} + 1) + \alpha_i + \beta_j = 0,$$

hence

$$M_{ij} = e^{\alpha_i - 1} e^{\beta_j}.$$

Thus M_{ij} factorizes as

$$M_{ij} = a_i b_j.$$

Using the row and column constraints,

$$r_i = \sum_{j \in B} M_{ij} = a_i \sum_{j \in B} b_j, \quad c_j = \sum_{i \in A} M_{ij} = b_j \sum_{i \in A} a_i,$$

leading to

$$M_{ij} = \frac{r_i c_j}{\left(\sum_{k \in A} a_k \right) \left(\sum_{l \in B} b_l \right)} = \frac{r_i c_j}{\lambda}$$

To identify λ , we use the normalization condition $M_{++} = 1$. First, observe that

$$\begin{aligned} m &= \sum_{k \in A} r_k = \sum_{k \in A} \frac{\mathbf{y}_k}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}_k}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_k}{\|\hat{\mathbf{y}}\|_1}\right) = \left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1, \\ m &= \sum_{l \in B} c_l = \sum_{l \in B} \frac{\hat{\mathbf{y}}_l}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}_l}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}_l}{\|\hat{\mathbf{y}}\|_1}\right) = \left\| \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1, \end{aligned}$$

and therefore $\left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1 = \left\| \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1$.

It follows that

$$\begin{aligned} 1 = M_{++} &= \sum_{k=1}^C M_{kk} + \sum_{k \neq l} M_{kl} \\ &= \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)_+ + \sum_{(k,l) \in A \times B} \frac{r_k c_l}{\lambda} \\ &= \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)_+ + \frac{\left(\sum_{k \in A} r_k \right) \left(\sum_{l \in B} c_l \right)}{\lambda} \\ &= \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)_+ + \frac{m^2}{\lambda}. \end{aligned}$$

Since $\left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} \right\|_1 = 1$ and

$$\left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right)_+ \right\|_1 = \left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1 = m,$$

which implies $\lambda = m = \left\| \frac{\mathbf{y}}{\|\mathbf{y}\|_1} - \min\left(\frac{\mathbf{y}}{\|\mathbf{y}\|_1}, \frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1}\right) \right\|_1$, concluding the proof.

5 Proof of Proposition 6

Let $(\mathbf{y}, \hat{\mathbf{y}})$ be two single-label vectors. In this case, $\frac{\mathbf{y}}{\|\mathbf{y}\|_1} = \mathbf{y}$ and $\frac{\hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} = \hat{\mathbf{y}}$, hence

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \text{diag}(\min(\mathbf{y}, \hat{\mathbf{y}})) + \frac{(\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})) \otimes (\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}}))}{\|\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})\|_1}.$$

If $\mathbf{y} = \hat{\mathbf{y}}$ with label and prediction i , then $\mathbf{y} = \hat{\mathbf{y}} = e_i$, where e_i denotes the canonical basis vector. Then

$$\min(\mathbf{y}, \hat{\mathbf{y}}) = e_i, \quad \mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}}) = 0,$$

and therefore

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = \text{diag}(e_i) = E_{ii},$$

which corresponds to the elementary matrix of the single-label confusion matrix.

If $\mathbf{y} \neq \hat{\mathbf{y}}$, assume $\mathbf{y} = e_i$ and $\hat{\mathbf{y}} = e_j$ with $i \neq j$. Then

$$\min(\mathbf{y}, \hat{\mathbf{y}}) = 0, \quad \mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}}) = e_i, \quad \hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}}) = e_j,$$

and $\|\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})\|_1 = 1$. Hence,

$$\pi(\mathbf{y}, \hat{\mathbf{y}}) = e_i \otimes e_j = E_{ij}.$$

In both cases, $\pi(\mathbf{y}, \hat{\mathbf{y}})$ coincides with the elementary matrix of the standard single-label confusion matrix, which concludes the proof.

6 Proof of proposition 7

Let $(\mathbf{y}, \hat{\mathbf{y}})$ be two multi-label vectors. Let Y and $\hat{Y}$ be the sets of present classes in $\mathbf{y}$ and $\hat{\mathbf{y}}$, respectively. The four formulas presented in [62] are presented in set notation; we translate them in vector form:

(i) In the case where $Y = \hat{Y}$, contribution formula is:

$$\text{diag}(\mathbf{y})$$

(ii) In the case where $Y \subsetneq \hat{Y}$, contribution formula is:

$$\frac{\mathbf{y} \otimes (\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}}))}{\|\hat{\mathbf{y}}\|_1} + \text{diag}(\mathbf{y}) \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}$$

(iii) In the case where $\hat{Y} \subsetneq Y$, contribution formula is:

$$\frac{(\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})) \otimes \hat{\mathbf{y}}}{\|\hat{\mathbf{y}}\|_1} + \text{diag}(\hat{\mathbf{y}})$$

(iv) In none of the previous cases, the contribution formula is:

$$\frac{(\mathbf{y} - \min(\mathbf{y}, \hat{\mathbf{y}})) \otimes (\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}}))}{\|\hat{\mathbf{y}} - \min(\mathbf{y}, \hat{\mathbf{y}})\|_1} + \text{diag}(\min(\mathbf{y}, \hat{\mathbf{y}}))$$

It is straightforward to verify that f is homogeneous, in the sense that

$$f(\alpha \mathbf{u}, \alpha \hat{\mathbf{u}}) = \alpha f(\mathbf{u}, \hat{\mathbf{u}})$$

for any $\alpha \geq 0$. Therefore,

$$\begin{aligned} & \|\mathbf{y}\|_1 \pi(\mathbf{y}, \hat{\mathbf{y}}) \\ &= f(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}) \\ &= \text{diag} \left(\min \left(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} \right) \right) + \frac{\left(\mathbf{y} - \min \left(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} \right) \right) \otimes \left(\hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} - \min \left(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} \right) \right)}{\left\| \mathbf{y} - \min \left(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} \right) \right\|_1} \end{aligned}$$

We now compare the proposed expression with the four contribution formulas of [62].

6.1 Case $Y = \hat{Y}$

In this case,

$$\hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} = \mathbf{y},$$

so that

$$\min \left(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} \right) = \mathbf{y}.$$

Hence the outer-product term vanishes, and

$$\|\mathbf{y}\|_1 \pi(\mathbf{y}, \hat{\mathbf{y}}) = \text{diag}(\mathbf{y}),$$

which is exactly the contribution formula in case (i).

6.2 Case $Y \subsetneq \hat{Y}$ and $\hat{Y} \subsetneq Y$

Since $\min(\mathbf{y}, \hat{\mathbf{y}}) = \mathbf{y}$, one has

$$\min \left(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} \right) = \mathbf{y} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1}.$$

Therefore,

$$\text{diag} \left(\min \left(\mathbf{y}, \hat{\mathbf{y}} \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} \right) \right) = \text{diag}(\mathbf{y}) \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1},$$

and the second term simplifies to

$$\frac{\mathbf{y} \otimes (\hat{\mathbf{y}} - \mathbf{y})}{\|\hat{\mathbf{y}}\|_1}.$$

Thus,

$$\|\mathbf{y}\|_1 \pi(\mathbf{y}, \hat{\mathbf{y}}) = \text{diag}(\mathbf{y}) \frac{\|\mathbf{y}\|_1}{\|\hat{\mathbf{y}}\|_1} + \frac{\mathbf{y} \otimes (\hat{\mathbf{y}} - \mathbf{y})}{\|\hat{\mathbf{y}}\|_1},$$

which is exactly the contribution formula in case (ii).

The case $\hat{Y} \subsetneq Y$ is similar to case (ii).

6.3 $Y \not\subset \hat{Y}$ and $\hat{Y} \not\subset Y$

In this case, the obtained expression does *not* coincide with the contribution formula proposed in [62]. Therefore, cases (i)–(iii) are recovered exactly, whereas case (iv) leads to a different formula.

7 Proof of Proposition 8

Silvan-Cardenas and Wang [98] introduce a soft-label confusion matrix defined on the simplex. Let $(\mathbf{y}, \hat{\mathbf{y}})$ be two points in the simplex of dimension C . They define the *MIN-LEAST* and *MIN-MIN* operators to construct the sub-pixel confusion matrix.

The diagonal entries of *SCM* and *TCM* coincide. The off-diagonal entries (i, j) are given as intervals:

$$I(i, j) := \left[\max \left(0, \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j) - \sum_{k \neq i} (\mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k)) \right), \right. \\ \left. \min \left(\mathbf{y}_i - \min(\mathbf{y}_i, \hat{\mathbf{y}}_i), \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j) \right) \right].$$

We first show that, for all $\pi \in T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ and all $i \neq j$, one has $\pi_{ij} \in I(i, j)$. We then prove that the bounds are tight.

7.1 Interval Bounds for Optimal Plan Entries

Let $\pi \in T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ and $i \neq j$.

Upper bound. By admissibility,

$$\pi_{ij} \leq \sum_{k \neq i} \pi_{ik} = \mathbf{y}_i - \min(\mathbf{y}_i, \hat{\mathbf{y}}_i), \quad \pi_{ij} \leq \sum_{k \neq j} \pi_{kj} = \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j).$$

Hence,

$$\pi_{ij} \leq \min \left(\mathbf{y}_i - \min(\mathbf{y}_i, \hat{\mathbf{y}}_i), \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j) \right),$$

which proves the upper bound.

Lower bound. We prove the lower bound by contradiction. Assume

$$\pi_{ij} < \max \left(0, \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j) - \sum_{k \neq i} (\mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k)) \right).$$

Then,

$$\begin{aligned} \sum_{k \neq j} \pi_{kj} &= \pi_{ij} + \sum_{k \neq i, j} \pi_{kj} \\ &\leq \pi_{ij} + \sum_{k \neq i, j} (\mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k)) \\ &< \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j), \end{aligned}$$

which contradicts the column constraint

$$\sum_{k \neq j} \pi_{kj} = \hat{\mathbf{y}}_j - \min(\mathbf{y}_j, \hat{\mathbf{y}}_j).$$

Thus, the lower bound holds.

We conclude that, for all $\pi \in T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$ and all $i \neq j$, one has $\pi_{ij} \in I(i, j)$.

7.2 Existence of Elements Attaining the Bounds

Let $i \neq j$ and

$$r_k := \mathbf{y}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k), \quad c_k := \hat{\mathbf{y}}_k - \min(\mathbf{y}_k, \hat{\mathbf{y}}_k).$$

Then, for any $\pi \in T^{\text{opt}}(\mathbf{y}, \hat{\mathbf{y}})$, the off-diagonal entries form a transportation problem with row sums (r_k) and column sums (c_k) . We now describe constructions attaining the upper and lower bounds.

Upper bound. We maximize the mass at (i, j) by assigning $\min(r_i, c_j)$ to this entry. The remaining mass can then be distributed to satisfy the residual row and column sums, yielding a feasible plan. Hence, the upper bound is attained.

Lower bound. We minimize the mass at (i, j) by assigning the excess mass in class j (quantified by c_j) first to underestimated classes $k \neq i$, and then to class i only if necessary.

If the total underestimation of classes other than i , given by $\sum_{k \neq i} r_k$, is greater than or equal to c_j , then these classes can fully absorb the excess of j . Hence, no mass needs to be assigned from i to j , and

$$\pi_{ij} = 0 = \max \left(0, c_j - \sum_{k \neq i} r_k \right).$$

If the total underestimation of classes other than i is strictly smaller than c_j , then the remaining excess must be assigned to class i . Therefore,

$$\pi_{ij} = c_j - \sum_{k \neq i} r_k = \max \left(0, c_j - \sum_{k \neq i} r_k \right).$$

Chapter B

A Local Sampling Strategy

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1 Results Averaged over Aggregators

This section reports results averaged over CWMed, CwTM, and RFA, both with and without NNM, as shown in Tables B.3, B.2, and B.1.

2 Proofs

2.1 Proof of Proposition 10

For each honest worker i ,

$$\begin{aligned}
 g_i - g^\star &= \sum_{c=1}^C q_{i,c} g_i^{(c)} - \sum_{c=1}^C q_c g^{(c)} \\
 &= \sum_{c=1}^C q_{i,c} \left(g_i^{(c)} - g^{(c)} \right) + \sum_{c=1}^C (q_{i,c} - q_c) g^{(c)}.
 \end{aligned}$$

Define

$$A_i = \sum_{c=1}^C q_{i,c} \left(g_i^{(c)} - g^{(c)} \right), \quad B_i = \sum_{c=1}^C (q_{i,c} - q_c) g^{(c)}.$$

Then $g_i - g^\star = A_i + B_i$, and therefore

$$\|g_i - g^\star\|^2 = \|A_i\|^2 + \|B_i\|^2 + 2\langle A_i, B_i \rangle.$$

We now show that the cross term has zero expectation. Since $\mathbf{q}_i$ depends only on the class counts and the class counts are independent of the observations, the class-wise estimators are conditionally unbiased:

$$\mathbb{E}[g_i^{(c)} \mid \mathbf{q}_i] = g^{(c)}.$$

It follows that

$$\mathbb{E}[g_i^{(c)} - g^{(c)} \mid \mathbf{q}_i] = 0.$$

	ALIE					Mimic					MinSum					FOE					LF					
	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	
10	CE	98.66	98.64	98.30	97.89	95.79	98.66	98.59	98.33	97.93	97.65	98.55	98.16	97.19	95.98	86.48	98.41	97.84	97.58	97.04	95.26	98.67	98.76	98.53	98.20	97.10
	NS	98.67	98.47	98.27	97.91	95.79	98.66	98.61	98.34	97.94	97.50	98.66	98.04	97.00	95.39	93.20	98.49	97.71	97.17	96.16	95.18	98.71	98.69	98.56	97.72	96.08
	gain	0.01	-0.18	-0.04	0.01	0.00	0.01	0.02	0.01	0.00	-0.15	0.11	-0.12	-0.18	-0.59	6.72	0.08	-0.13	-0.41	-0.87	-0.08	0.04	-0.07	0.03	-0.49	-1.02
	CE	98.70	98.69	98.67	98.67	98.62	98.73	98.71	98.59	98.28	97.78	98.53	98.27	97.90	97.65	96.34	98.53	98.27	97.90	97.65	96.34	98.53	98.27	97.90	97.65	96.34
1	agg + NNM	98.70	98.69	98.74	98.51	98.56	98.73	98.71	98.59	98.28	97.78	98.53	98.27	97.90	97.65	96.34	98.53	98.27	97.90	97.65	96.34	98.53	98.27	97.90	97.65	96.34
	gain	-0.00	0.00	0.07	-0.15	-0.06	0.01	0.02	0.01	0.00	-0.15	0.11	-0.12	-0.18	-0.59	6.72	0.08	-0.13	-0.41	-0.87	-0.08	0.04	-0.07	0.03	-0.49	-1.02
	CE	98.47	98.32	97.63	96.00	80.90	98.56	98.40	98.17	97.65	96.08	98.20	97.81	96.93	93.79	69.93	98.24	97.70	96.38	91.36	50.72	98.46	98.47	98.07	97.31	94.44
	NS	98.53	98.29	98.20	97.44	93.68	98.57	98.35	98.14	97.81	96.97	98.49	98.00	97.13	96.15	94.10	98.38	97.72	96.95	95.65	92.06	98.56	98.57	98.14	97.33	92.88
0.1	agg	0.06	-0.03	0.56	1.44	12.78	0.01	-0.05	-0.02	0.17	0.89	0.29	0.19	0.20	2.36	24.17	0.14	0.02	0.57	4.30	41.34	0.11	0.10	0.08	0.02	-1.56
	CE	98.71	98.72	98.54	98.44	98.65	98.71	98.72	98.54	98.44	98.65	98.60	98.49	98.40	98.15	97.43	98.36	97.92	97.46	94.94	89.03	98.36	97.92	97.46	94.94	89.03
	NS	98.55	98.60	98.57	98.48	98.44	98.60	98.49	98.40	98.15	97.43	98.36	97.92	97.46	94.94	89.03	98.17	98.23	97.97	97.48	96.34	98.17	98.23	97.97	97.48	96.34
	gain	-0.16	-0.12	0.03	0.04	-0.21	0.01	-0.05	-0.02	0.17	0.89	0.29	0.19	0.20	2.36	24.17	-0.19	0.31	0.51	2.55	7.32	0.11	0.10	0.08	0.02	-1.56
0.1	CE	86.24	93.09	83.42	37.10	9.84	90.94	89.63	76.00	70.69	53.72	88.78	50.94	38.76	32.30	22.02	86.07	67.25	48.19	40.08	22.04	93.38	91.15	85.41	66.58	67.11
	NS	97.56	97.09	93.13	67.78	28.82	97.50	95.47	94.21	89.70	67.83	96.73	93.52	86.42	75.60	51.27	96.84	92.99	81.68	69.17	37.76	97.66	97.34	95.64	91.81	85.37
	gain	11.32	4.00	9.71	30.69	18.98	6.56	5.84	18.22	19.01	14.10	7.95	42.58	47.66	43.30	29.25	10.77	25.74	33.49	29.09	15.73	4.28	6.19	10.23	25.23	18.26
	CE	98.67	98.35	95.96	84.61	86.18	98.67	98.35	95.96	84.61	86.18	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73
0.1	agg + NNM	98.51	98.29	98.11	98.09	98.23	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73
	NS	98.51	98.29	98.11	98.09	98.23	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73
	gain	-0.16	-0.06	2.15	13.48	12.06	-0.16	-0.06	2.15	13.48	12.06	-0.16	-0.06	2.15	13.48	12.06	-0.16	-0.06	2.15	13.48	12.06	-0.16	-0.06	2.15	13.48	12.06
	CE	98.67	98.35	95.96	84.61	86.18	98.67	98.35	95.96	84.61	86.18	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73	98.51	98.31	98.07	97.23	89.73

Table B.1: MNIST. Mean accuracy (%) over the last 200 steps for $\overline{\text{agg}}$ (average over CWMed, CwTM, and RFA without NNM) and $\overline{\text{agg}} + \text{NNM}$ (same with NNM). We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

		ALIE						Mimic						MinSum						FOE						LF					
		2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26	2	8	14	20	26					
10	CE	86.75	86.07	82.69	80.03	73.39	86.71	86.58	85.58	84.74	82.27	85.99	83.68	82.12	77.97	69.08	85.62	83.86	82.22	79.47	73.98	86.61	86.17	84.48	82.76	78.70					
	NS	87.04	86.54	84.82	81.21	75.27	87.12	87.10	86.06	85.21	82.62	86.83	84.93	83.46	79.53	77.72	86.31	83.95	82.61	80.41	76.52	87.05	86.83	86.04	85.04	82.33					
	gain	0.29	0.48	2.13	1.18	1.88	0.41	0.52	0.48	0.48	0.35	0.84	1.26	1.35	1.56	8.63	0.69	0.08	0.39	0.95	2.54	0.44	0.67	1.55	2.28	3.63					
	CE	86.82	86.57	86.69	86.84	86.73	86.81	86.82	86.84	86.73	86.73	86.81	86.82	86.84	86.73	86.73	86.81	86.82	86.84	86.73	86.73	86.81	86.82	86.84	86.73	86.73					
	agg + NNM	87.11	86.93	86.93	86.82	86.68	87.06	87.08	86.31	85.75	83.15	86.81	86.82	86.84	86.73	86.73	86.81	86.82	86.84	86.73	86.73	86.81	86.82	86.84	86.73	86.73					
1	gain	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05					
	CE	85.05	82.38	77.32	70.25	49.68	85.28	84.48	83.58	80.83	77.49	83.68	79.49	75.50	69.19	52.15	83.69	79.74	75.13	71.82	52.03	85.13	83.88	81.40	74.05	68.31					
	NS	86.92	85.48	82.87	78.90	69.26	86.71	86.31	85.80	84.46	82.37	86.12	84.35	81.88	79.47	74.69	86.00	84.01	81.70	79.18	74.80	86.83	86.46	85.57	83.45	79.22					
	gain	1.87	3.10	5.55	8.64	19.57	1.43	1.83	2.22	3.63	4.87	2.44	4.86	6.38	10.28	22.54	2.32	4.26	6.57	7.35	22.77	1.70	2.58	4.16	9.40	10.91					
	agg + NNM	86.68	85.92	85.14	84.65	84.57	86.93	86.77	86.20	85.62	84.03	86.93	86.77	86.20	85.62	84.03	86.93	86.77	86.20	85.62	84.03	86.93	86.77	86.20	85.62	84.03					
0.1	gain	0.35	0.80	1.13	1.60	1.31	0.35	0.80	1.13	1.60	1.31	0.35	0.80	1.13	1.60	1.31	0.35	0.80	1.13	1.60	1.31	0.35	0.80	1.13	1.60	1.31					
	CE	77.44	70.69	59.25	24.75	9.93	78.21	73.99	67.50	56.45	42.78	71.13	42.24	27.94	33.44	19.14	70.96	50.10	31.46	33.88	17.01	78.68	76.61	72.33	66.12	56.55					
	NS	80.07	76.56	70.56	55.66	30.83	81.17	78.11	76.32	67.92	51.93	79.31	71.44	69.59	61.78	36.70	78.72	73.31	67.99	51.88	28.93	79.80	82.92	79.21	75.23	72.21					
	gain	2.63	5.87	11.31	30.91	20.90	2.96	4.12	8.82	11.47	9.15	8.18	29.20	41.65	28.34	17.56	7.76	23.20	36.53	18.00	11.93	1.12	6.31	6.88	9.11	15.66					
	agg + NNM	86.14	81.53	80.11	78.01	63.55	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27					
	gain	-0.05	3.45	3.45	3.61	22.70	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05					
	CE	86.14	81.53	80.11	78.01	63.55	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27					
	NS	86.09	84.98	83.56	81.63	86.25	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27					
	gain	-0.05	3.45	3.45	3.61	22.70	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05	0.28	0.36	0.24	-0.02	-0.05					
	agg + NNM	86.14	81.53	80.11	78.01	63.55	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27	86.42	86.42	86.12	77.29	72.27					

Table B.2: Fashion-MNIST. Mean accuracy (%) over the last 200 steps for $\overline{\text{agg}}$ (average over CWMed, CwTM, and RFA without NNМ) and $\overline{\text{agg}} + \text{NNM}$ (same with NNМ). We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

	ALIE						Mimic						MinSum						FOE						LF					
	2	8	14	20	26		2	8	14	20	26		2	8	14	20	26		2	8	14	20	26		2	8	14	20	26	
10	CE	73.45	66.58	56.51	45.32	31.93	73.86	72.48	68.97	64.54	57.70		72.47	62.90	53.72	30.60	15.58		70.76	61.30	54.30	46.06	32.23		73.86	72.97	71.25	68.98	63.64	
	NS	74.02	67.08	56.62	44.11	36.13	74.55	73.04	69.64	64.88	59.18		72.90	64.64	56.32	43.14	21.91		71.21	62.02	56.27	44.18	23.08		74.55	73.46	72.16	69.11	64.34	
	gain	0.58	0.50	0.11	-1.22	4.21	0.69	0.57	0.67	0.34	1.48		0.43	1.75	2.60	12.55	6.32		0.45	0.71	1.97	-1.88	-9.15		0.70	0.49	0.91	0.13	0.70	
	agg + NNM	73.77	72.38	68.82	68.11	67.68	74.89	73.78	72.05	69.82	63.77		72.90	64.64	56.32	43.14	21.91		71.19	67.45	62.50	57.29	49.23		74.89	73.78	72.05	69.82	63.77	
1	CE	74.86	72.21	68.58	65.19	65.33	74.89	73.78	72.05	69.82	63.77		72.90	64.64	56.32	43.14	21.91		72.07	67.98	63.47	57.42	47.27		74.89	73.78	72.05	69.82	63.77	
	NS	71.97	63.41	53.42	41.72	29.87	72.90	70.99	67.58	61.45	53.58		71.09	61.38	54.52	39.86	16.82		69.11	60.96	54.81	44.02	19.37		72.98	71.59	70.55	68.32	63.63	
	gain	0.78	2.61	6.74	10.96	7.67	1.29	1.27	2.68	4.99	4.84		3.82	4.33	14.97	11.78	2.52		2.51	2.97	7.03	7.57	-5.11		1.66	2.34	3.14	4.24	7.74	
	agg + NNM	71.93	66.66	61.36	57.60	66.13	72.90	71.61	71.01	68.31	61.17		72.90	71.61	71.01	68.31	61.17		69.05	67.79	66.77	61.56	52.05		72.90	71.61	71.01	68.31	61.17	
0.1	CE	50.61	31.26	22.13	19.48	10.68	55.76	53.83	43.98	33.20	28.39		51.74	31.36	17.78	16.45	13.91		51.46	35.21	20.61	17.32	14.65		54.72	52.55	48.42	43.68	41.69	
	NS	60.53	42.99	30.16	24.37	12.38	60.38	59.02	53.66	42.36	36.75		56.37	39.70	32.13	23.08	20.16		55.59	41.58	29.15	18.40	7.38		58.30	54.84	54.00	49.64	46.44	
	gain	9.92	11.73	8.03	4.89	1.69	4.62	5.19	9.68	9.16	8.36		4.63	8.34	14.35	6.63	6.25		4.13	6.37	8.54	1.08	-7.27		3.58	2.29	5.57	5.97	4.75	
	agg + NNM	55.51	45.33	34.44	31.14	23.08	63.36	63.32	63.11	57.68	48.94		63.36	63.32	63.11	57.68	48.94		55.97	49.11	40.24	34.27	20.85		63.36	63.32	63.11	57.68	48.94	
	gain	8.40	13.93	20.92	16.74	24.82	8.40	13.93	20.92	16.74	24.82		8.40	13.93	20.92	16.74	24.82		5.02	6.56	9.32	5.19	-5.32		5.02	6.56	9.32	5.19	-5.32	

Table B.3: CIFAR-10. Mean accuracy (%) over the last 200 steps for $\overline{\text{agg}}$ (average over CWMed, CwTM, and RFA without NNM) and $\text{agg} + \text{NNM}$ (same with NNM). We compare cross-entropy only (CE) and cross-entropy with NorthStar (NS) across $\alpha \in \{10, 1, 0.1\}$, attacks, and Byzantine counts $f \in \{2, 8, 14, 20, 26\}$.

Hence

$$\mathbb{E}[A_i \mid \mathbf{q}_i] = \sum_{c=1}^C q_{i,c} \mathbb{E}[g_i^{(c)} - g^{(c)} \mid \mathbf{q}_i] = 0.$$

Moreover, B_i is a function of $\mathbf{q}_i$. Therefore,

$$\begin{aligned} \mathbb{E}[\langle A_i, B_i \rangle] &= \mathbb{E}[\mathbb{E}[\langle A_i, B_i \rangle \mid \mathbf{q}_i]] \\ &= \mathbb{E}[\langle \mathbb{E}[A_i \mid \mathbf{q}_i], B_i \rangle] \\ &= 0. \end{aligned}$$

Since honest workers are i.i.d. and the assumed second-moment conditions ensure integrability, the strong law of large numbers gives

$$\frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \|g_i - g^*\|^2 \xrightarrow[|\mathcal{H}| \rightarrow \infty]{\text{a.s.}} \mathbb{E}[\|A_i\|^2] + \mathbb{E}[\|B_i\|^2].$$

Substituting the definitions of A_i and B_i completes the proof.

2.2 Proof of Proposition 11

For each honest worker i ,

$$g_{\text{NS } i} - g_{\text{NS}}^* = \sum_{c=1}^C p_c \left(g_{\text{NS } i}^{(c)} - g^{(c)} \right).$$

Taking squared norms and applying the strong law of large numbers gives the result.

2.3 Proof of Proposition 12

By definition,

$$g_{\text{NS}}^* = \sum_{c=1}^C p_c g^{(c)}, \quad g^* = \sum_{c=1}^C q_c g^{(c)}.$$

If $\mathbf{p} = \mathbf{q}$, then $p_c = q_c$ for every class c , and therefore $g_{\text{NS}}^* = g^*$.

Chapter C

Inside the Convex Hull of Gradients

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1 Additionnal Experiments

2 Variance Decomposition of the Gradient Estimator

Let $P \in \mathbb{R}^{h' \times C}$ and $U \in \mathbb{R}^{h' \times D}$ denote the label distribution and update matrices of the honest workers in A , respectively. Similarly, let $Q \in \mathbb{R}^{f' \times C}$, $V \in \mathbb{R}^{f' \times D}$ denote the corresponding matrices for the Byzantine workers. Let $\mathcal{U} \in \mathbb{R}^{C \times D}$ denote the expected class-wise gradients over the global dataset $\mathcal{D}$.

To model the random selection of a subgroup $B \subseteq A$ with s workers, we introduce two diagonal matrices $S = \text{diag}(s_1, \dots, s_{h'})$ and $T = \text{diag}(t_1, \dots, t_{f'})$, whose diagonal entries are identically distributed Bernoulli variables. These variables indicate whether a worker is selected in the subgroup. The subgroup size is fixed by the constraint $\text{Tr}(S + T) = s$.

The following label distribution and update matrices represent the workers in subgroup B ; rows corresponding to workers not selected in B are set to zero. We denote by $\hat{\mathcal{G}}$ the global class-wise gradient estimator; zero rows have no effect on these estimators:

$$\underbrace{\begin{pmatrix} SP \\ TQ \end{pmatrix}}_{\text{label distributions } h \times C}, \quad \underbrace{\begin{pmatrix} SU \\ TV \end{pmatrix}}_{\text{updates } h \times D}, \quad \hat{\mathcal{G}} = \underbrace{\begin{pmatrix} SP \\ TQ \end{pmatrix}^\dagger \begin{pmatrix} SU \\ TV \end{pmatrix}}_{\text{global class gradient estimator } C \times D},$$

For any realizations of S and T , we assume that the label distribution matrix has full column rank.

Then, it holds

$$\begin{aligned}
\hat{\mathcal{G}} &= \begin{pmatrix} SP \\ TQ \end{pmatrix}^\dagger \begin{pmatrix} SU \\ TV \end{pmatrix} \\
&= \left(\begin{pmatrix} SP \\ TQ \end{pmatrix}^\top \begin{pmatrix} SP \\ TQ \end{pmatrix} \right)^{-1} \begin{pmatrix} SP \\ TQ \end{pmatrix}^\top \begin{pmatrix} SU \\ TV \end{pmatrix} \\
&= \left(P^\top SP + Q^\top TQ \right)^{-1} \left(P^\top SU + Q^\top TV \right) \\
&= \underbrace{\left(P^\top SP + Q^\top TQ \right)^{-1} P^\top}_{\tilde{P}} SU + \underbrace{\left(P^\top SP + Q^\top TQ \right)^{-1} Q^\top}_{\tilde{Q}} TV \\
&= \tilde{P}U + \tilde{Q}V,
\end{aligned}$$

Considering the Frobenius variance of $\hat{\mathcal{G}}$, Proposition 15 yields

$$\begin{aligned}
\sigma^2(\hat{\mathcal{G}}) &= \mathbb{E} \left[\left\| \hat{\mathcal{G}} - \mathbb{E}[\hat{\mathcal{G}}] \right\|_F^2 \right] \\
&= \mathbb{E} \left[\left\| \tilde{P}(U - \mathbb{E}[U]) \right\|_F^2 \right] + \mathbb{E} \left[\left\| \tilde{Q}(V - \mathbb{E}[V]) \right\|_F^2 \right] + \mathbb{E} \left[\left\| (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU) \right\|_F^2 \right]
\end{aligned}$$

We assume that Byzantine updates are convex combinations of C vectors consistent with their broadcast label distributions. Let $\mathcal{V} \in \mathbb{R}^{C \times D}$ denote the Byzantine analogue of the class-wise gradient matrix $\mathcal{U}$. Consequently, $\mathbb{E}[V] = Q\mathcal{V}$.

Let $\hat{\mathcal{U}}$ denote the class-wise gradient estimator constructed from the honest workers selected by S , and $\hat{\mathcal{V}}$ the corresponding estimator constructed from the Byzantine workers selected by T , defined as

$$\hat{\mathcal{U}} = (SP)^\dagger SU, \quad \hat{\mathcal{V}} = (TQ)^\dagger TV$$

Let W be the matrix defined as $W = \tilde{P}P$.

Then, Proposition 18 gives

$$\begin{aligned}
\sigma^2(\hat{\mathcal{G}}) &= \mathbb{E} \left[\left\| W(\hat{\mathcal{U}} - \mathcal{U}) \right\|_F^2 \right] + \mathbb{E} \left[\left\| (I - W)(\hat{\mathcal{V}} - \mathcal{V}) \right\|_F^2 \right] + \mathbb{E} \left[\left\| (W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U}) \right\|_F^2 \right],
\end{aligned}$$

with W equals to

$$\begin{aligned}
W &= (X + Y)^{-1}X, \\
I - W &= (X + Y)^{-1}Y, \\
\text{where } X &= P^\top SP, \quad Y = Q^\top TQ.
\end{aligned}$$

The matrices W and $I - W$ measures the share of total information $X + Y$ contributed by honest workers, while $I - W = (X + Y)^{-1}Y$ measures the contribution of Byzantine workers. Additionally, W and $I - W$ are diagonalisable and their singular values lie in the interval $[0, 1]$.

We now consider the following simplification, which does not hold in practice but helps clarify the equations: (i) We assume that Byzantine label distributions perfectly mimic the honest ones, i.e., for any realizations of S and T we have $Y = \lambda X$. This implies, $\lambda = \text{Tr}(Y)/\text{Tr}(X)$, and

$$W = (X + Y)^{-1}X = \frac{1}{1 + \lambda}I = \frac{\text{Tr}(X)}{s}I, \quad I - W = \frac{\lambda}{1 + \lambda}I = \frac{\text{Tr}(Y)}{s}I.$$

Assume moreover that (ii) $\text{Tr}(X)$ is independent of $\hat{\mathcal{U}}$.

Substituting this into the previous expression for $\sigma^2(\hat{\mathcal{G}})$, and neglecting the second term, which corresponds to the variance of the Byzantine estimator and can be made zero by an appropriate Byzantine strategy, we obtain

$$\begin{aligned}\sigma^2(\hat{\mathcal{G}}) &= \mathbb{E} \left[\left(\frac{\text{Tr}(X)}{s} \right)^2 \right] \mathbb{E} [\|\hat{\mathcal{U}} - \mathcal{U}\|_F^2] + \mathbb{E} \left[\left(\frac{\text{Tr}(X)}{s} - \mathbb{E} \left[\frac{\text{Tr}(X)}{s} \right] \right)^2 \right] \|\mathcal{V} - \mathcal{U}\|_F^2 \\ &= \frac{\mathbb{E}[\text{Tr}(X)^2]}{s^2} \mathbb{E} [\|\hat{\mathcal{U}} - \mathcal{U}\|_F^2] + \frac{\text{Var}(\text{Tr}(X))}{s^2} \|\mathcal{V} - \mathcal{U}\|_F^2.\end{aligned}$$

In practice, the mismatch term $\|\mathcal{V} - \mathcal{U}\|_F^2$ is often much larger than the honest estimation error $\mathbb{E}[\|\hat{\mathcal{U}} - \mathcal{U}\|_F^2]$. Under this assumption, the behavior of $\sigma^2(\hat{\mathcal{G}})$ is primarily driven by the second term. Consequently, increasing h' tends to reduce the variance whenever it decreases $\text{Var}(\text{Tr}(X))$. In this regime, the worker score strategy is theoretically grounded by Proposition 19 in Appendix.

3 Theoretical Analysis

Proposition 13.

$$(\tilde{P} - \mathbb{E}[\tilde{P}])\mathbb{E}[U] = -(\tilde{Q} - \mathbb{E}[\tilde{Q}])QU$$

Proof. It holds that $\mathbb{E}[U] = PU$. Since P is deterministic, we obtain

$$\begin{aligned}(\tilde{P} - \mathbb{E}[\tilde{P}])\mathbb{E}[U] &= (\tilde{P} - \mathbb{E}[\tilde{P}])PU \\ &= (\tilde{P}P - \mathbb{E}[\tilde{P}P])U\end{aligned}$$

Observing that $\tilde{P}P = I - \tilde{Q}Q$, it follows that

$$\begin{aligned}(\tilde{P}P - \mathbb{E}[\tilde{P}P])U &= (I - \tilde{Q}Q - \mathbb{E}[I - \tilde{Q}Q])U \\ &= -(\tilde{Q} - \mathbb{E}[\tilde{Q}])QU\end{aligned}$$

□

□

Proposition 14.

$$\hat{\mathcal{G}} - \mathbb{E}[\hat{\mathcal{G}}] = \tilde{P}(U - \mathbb{E}[U]) + \tilde{Q}(V - \mathbb{E}[V]) + (\tilde{Q} - \mathbb{E}[\tilde{Q}])((\mathbb{E}[V] - QU))$$

Proof. Since S and T are independent of U and V , it follows that $\tilde{P}$ and $\tilde{Q}$ are also independent of U and V . It holds that

$$\mathbb{E}[\hat{\mathcal{G}}] = \mathbb{E}[\tilde{P}U + \tilde{Q}V] = \mathbb{E}[\tilde{P}]\mathbb{E}[U] + \mathbb{E}[\tilde{Q}]\mathbb{E}[V]$$

Injecting previous result in $\hat{\mathcal{G}} - \mathbb{E}[\hat{\mathcal{G}}]$, it holds that

$$\begin{aligned}\hat{\mathcal{G}} - \mathbb{E}[\hat{\mathcal{G}}] &= \tilde{P}U + \tilde{Q}V - \mathbb{E}[\tilde{P}]\mathbb{E}[U] - \mathbb{E}[\tilde{Q}]\mathbb{E}[V] \\ &= \tilde{P}(U - \mathbb{E}[U]) + \tilde{Q}(V - \mathbb{E}[V]) + (\tilde{P} - \mathbb{E}[\tilde{P}])\mathbb{E}[U] + (\tilde{Q} - \mathbb{E}[\tilde{Q}])\mathbb{E}[V]\end{aligned}$$

According to Proposition 13, it holds

$$(\tilde{P} - \mathbb{E}[\tilde{P}])\mathbb{E}[U] = -(\tilde{Q} - \mathbb{E}[\tilde{Q}])QU,$$

which leads to

$$\hat{\mathcal{G}} - \mathbb{E}[\hat{\mathcal{G}}] = \tilde{P}(U - \mathbb{E}[U]) + \tilde{Q}(V - \mathbb{E}[V]) + (\tilde{Q} - \mathbb{E}[\tilde{Q}])((\mathbb{E}[V] - QU))$$

□

□

Proposition 15.

$$\sigma^2(\hat{\mathcal{G}}) = \mathbb{E} \left[\left\| \tilde{P}(U - \mathbb{E}[U]) \right\|_F^2 \right] + \mathbb{E} \left[\left\| \tilde{Q}(V - \mathbb{E}[V]) \right\|_F^2 \right] + \mathbb{E} \left[\left\| (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU) \right\|_F^2 \right]$$

Proof. According to Proposition 14, it holds

$$\sigma^2(\hat{\mathcal{G}}) - \mathbb{E}[\sigma^2(\hat{\mathcal{G}})] = \tilde{P}(U - \mathbb{E}[U]) + \tilde{Q}(V - \mathbb{E}[V]) + (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU),$$

using this, we obtain

$$\begin{aligned} & \mathbb{E} \left[\left\| \hat{\mathcal{G}} - \mathbb{E}[\hat{\mathcal{G}}] \right\|_F^2 \right] \\ &= \mathbb{E} \left[\left\| \tilde{P}(U - \mathbb{E}[U]) + \tilde{Q}(V - \mathbb{E}[V]) + (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU) \right\|_F^2 \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\left\| \tilde{P}(U - \mathbb{E}[U]) + \tilde{Q}(V - \mathbb{E}[V]) + (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU) \right\|_F^2 \mid S, T \right] \right], \end{aligned}$$

where the last equality follows from the law of total expectation. Inside the conditional expectation, $\tilde{P}$ and $\tilde{Q}$ are deterministic. It holds that

$$\begin{aligned} & \mathbb{E} \left[\left\| \tilde{P}(U - \mathbb{E}[U]) + \tilde{Q}(V - \mathbb{E}[V]) + (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU) \right\|_F^2 \mid S, T \right] \\ &= \mathbb{E} \left[\left\| \tilde{P}(U - \mathbb{E}[U]) \right\|_F^2 \mid S, T \right] + \mathbb{E} \left[\left\| \tilde{Q}(V - \mathbb{E}[V]) \right\|_F^2 \mid S, T \right] \\ & \quad + \left\| (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU) \right\|_F^2, \end{aligned}$$

where we used that, conditional on (S, T) , the random matrices $\tilde{P}(U - \mathbb{E}[U])$ and $\tilde{Q}(V - \mathbb{E}[V])$ are centered and independent, and are also independent of $(\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU)$, which is deterministic given (S, T) . Therefore,

$$\begin{aligned} & \mathbb{E} \left[\left\| \hat{\mathcal{G}} - \mathbb{E}[\hat{\mathcal{G}}] \right\|_F^2 \right] \\ &= \mathbb{E} \left[\left\| \tilde{P}(U - \mathbb{E}[U]) \right\|_F^2 \right] + \mathbb{E} \left[\left\| \tilde{Q}(V - \mathbb{E}[V]) \right\|_F^2 \right] + \mathbb{E} \left[\left\| (\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - QU) \right\|_F^2 \right] \end{aligned}$$

□

□

Proposition 16.

$$\tilde{P}P(SP)^\dagger = \tilde{P}, \quad \tilde{Q}Q(TQ)^\dagger = \tilde{Q}$$

Proof.

$$\begin{aligned} \tilde{P}P(SP)^\dagger &= (P^\top SP + Q^\top TQ)^{-1} P^\top SP(SP)^\dagger \\ &= (P^\top SP + Q^\top TQ)^{-1} (SP)^\top SP(SP)^\dagger \end{aligned}$$

It holds $M^\top = M^\top MM^\dagger$ for any matrix M . Applying this with $M = SP$ gives

$$\begin{aligned} \tilde{P}P(SP)^\dagger &= (P^\top SP + Q^\top TQ)^{-1} (SP)^\top \\ &= (P^\top SP + Q^\top TQ)^{-1} P^\top S \\ &= \tilde{P} \end{aligned}$$

Similarly,

$$\begin{aligned}
\tilde{Q}Q(TQ)^\dagger &= (P^\top SP + Q^\top TQ)^{-1} Q^\top TQ(TQ)^\dagger \\
&= (P^\top SP + Q^\top TQ)^{-1} (TQ)^\top TQ(TQ)^\dagger \\
&= (P^\top SP + Q^\top TQ)^{-1} (TQ)^\top \\
&= (P^\top SP + Q^\top TQ)^{-1} Q^\top T \\
&= \tilde{Q}
\end{aligned}$$

□

□

Proposition 17.

$$\tilde{P}(U - \mathbb{E}[U]) = W(\hat{\mathcal{U}} - \mathcal{U}).$$

Proof. We have $\mathbb{E}[U] = P\mathcal{U}$. Hence, $\tilde{P}\mathbb{E}[U] = \tilde{P}P\mathcal{U} = W\mathcal{U}$.

By Proposition 16, $\tilde{P} = W(SP)^+$. Moreover, since $\tilde{P} = \tilde{P}S$, we obtain

$$\tilde{P}U = \tilde{P}SU = W(SP)^+SU = W\hat{\mathcal{U}}.$$

Combining the two identities,

$$\tilde{P}(U - \mathbb{E}[U]) = \tilde{P}U - \tilde{P}\mathbb{E}[U] = W\hat{\mathcal{U}} - W\mathcal{U} = W(\hat{\mathcal{U}} - \mathcal{U}).$$

□

□

Proposition 18. *Under Byzantine convex combination assumption,*

$$\begin{aligned}
&\sigma^2[\hat{\mathcal{G}}] \\
&= \mathbb{E}\left[\left\|W(\hat{\mathcal{U}} - \mathcal{U})\right\|_F^2\right] + \mathbb{E}\left[\left\|(I - W)(\hat{\mathcal{V}} - \mathcal{V})\right\|_F^2\right] + \mathbb{E}\left[\left\|(W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U})\right\|_F^2\right]
\end{aligned}$$

Proof. Under Byzantine convex combination assumption, $\mathbb{E}[V] = Q\mathcal{V}$.

Moreover, applying the reasoning of Proposition 17 and using the identity $\tilde{Q}Q = (I - W)$, we obtain $\tilde{Q}(V - \mathbb{E}[V]) = (I - W)(\hat{\mathcal{V}} - \mathcal{V})$.

Proposition 15 gives

$$\begin{aligned}
&\mathbb{E}\left[\left\|\hat{\mathcal{U}}(\mathfrak{g}) - \mathbb{E}[\hat{\mathcal{U}}(\mathfrak{g})]\right\|_F^2\right] \\
&= \mathbb{E}\left[\left\|\tilde{P}(U - \mathbb{E}[U])\right\|_F^2\right] + \mathbb{E}\left[\left\|\tilde{Q}(V - \mathbb{E}[V])\right\|_F^2\right] + \mathbb{E}\left[\left\|(\tilde{Q} - \mathbb{E}[\tilde{Q}])(\mathbb{E}[V] - Q\mathcal{U})\right\|_F^2\right]
\end{aligned}$$

Proposition 17 gives $\tilde{P}(U - \mathbb{E}[U]) = W(\hat{\mathcal{U}} - \mathcal{U})$.

Combining all these results, we get:

$$\begin{aligned}
&\sigma^2[\hat{\mathcal{U}}(\mathfrak{g})] \\
&= \mathbb{E}\left[\left\|W(\hat{\mathcal{U}} - \mathcal{U})\right\|_F^2\right] + \mathbb{E}\left[\left\|(I - W)(\hat{\mathcal{V}} - \mathcal{V})\right\|_F^2\right] + \mathbb{E}\left[\left\|(W - \mathbb{E}[W])(\mathcal{V} - \mathcal{U})\right\|_F^2\right]
\end{aligned}$$

□

□

Proposition 19. *Assume that each candidate is either good or bad. Let there be h good candidates and f bad candidates with $f < N/3$, $N = h + f$. Teams are formed by sampling h candidates. Assume that the time achieved by a team depends only on the proportion of good rowers in the boat.*

Then, for any good candidate i and bad candidate j , $\text{Score}(i) > \text{Score}(j)$.

Proof. Fix a good candidate i and a bad candidate j . Couple the random teams used to define $\text{Score}(i)$ and $\text{Score}(j)$ as follows: sample the same set of $h - 1$ other candidates in both cases, excluding i and j . Then compare the two teams

$$T_i = i \cup A \quad \text{and} \quad T_j = j \cup A,$$

where A is any subset of size $h - 1$ of the remaining $N - 2$ candidates.

The two teams differ only in that T_i contains the good candidate i whereas T_j contains the bad candidate j . Hence T_i has exactly one more good candidate than T_j . Since the team time depends only on the proportion of good candidates, and improves when this proportion increases, the team containing i always achieves a strictly better time than the team containing j .

Therefore, for every realization of A , the score contribution of i is strictly larger than that of j . Taking expectation over A yields

$$\text{Score}(i) > \text{Score}(j).$$

This concludes the proof. $\square$

$\square$

Proposition 20. Let $\varepsilon_i = p_i \mathcal{U}_i - p_i \mathcal{U}$ and

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_h \end{pmatrix} \in \mathbb{R}^{h \times D}.$$

Then $(\varepsilon_i)_{i=1}^h$ are independent random vectors and, for all $i = 1, \dots, h$,

$$\begin{aligned} \mathbb{E}[\varepsilon_i] &= 0, \\ \mathbb{V}(\varepsilon_{id}) &= \frac{1}{s^2} \sum_{c=1}^C |\mathcal{D}_i^c| \mathbb{V}[\nabla \ell(\Phi(x), c)_d], \end{aligned}$$

where $|\mathcal{D}_i| = s$ is the number of samples used to compute the local update (typically the batch size), and is the same for all workers.

In addition, if the variance is independent of the class, i.e., $\mathbb{V}[\nabla \ell(\Phi(x), c)_d] = \sigma_d^2$ for all c , then

$$\mathbb{V}(\varepsilon_{id}) = \frac{\sigma_d^2}{s}.$$

In particular, the noise variance is the same across workers, which corresponds to a homoscedasticity assumption.

Proof. The null expectation follows is straightforward. Moreover, since the observations are independent and each error term is computed from a disjoint set of observations, the vectors ε_i are independent. We now compute the variance.

For notational simplicity, denote the column vector $(\mathcal{U}_i)_d$ by q . Note that $\mathbb{E}(q) = \mathbb{E}((\mathcal{U}_i)_d) = \mathcal{U}_d$. We then have

$$\begin{aligned} \mathbb{V}(\varepsilon_{id}) &= \mathbb{E}[\varepsilon_{id}^2] = \mathbb{E}[(p_i q - p_i \mathbb{E}(q))^2] \\ &= \mathbb{E}[(p_i (q - \mathbb{E}(q)))^2] \\ &= \mathbb{E}[p_i (q - \mathbb{E}(q))(q - \mathbb{E}(q))^\top p_i^\top] \\ &= p_i \mathbb{E}[(q - \mathbb{E}(q))(q - \mathbb{E}(q))^\top] p_i^\top \\ &= p_i \text{Cov}(q) p_i^\top. \end{aligned}$$

We now study $\text{Cov}(q)$. Note that components of q are independent. As a result,

$$\begin{aligned} \text{Cov}(q) &= \text{diag}(\mathbb{V}[q_1], \dots, \mathbb{V}[q_C]) \\ &= \text{diag}(\mathbb{V}[(u_i^1)_d], \dots, \mathbb{V}[(u_i^C)_d]) \end{aligned}$$

Let's now study the component variance:

$$\begin{aligned}
\mathbb{V}[(u_i^c)_d] &= \mathbb{V}\left[\frac{1}{|\mathcal{D}_i^c|} \sum_{(x,y) \in \mathcal{D}_i^c} \nabla \ell(\Phi(x), c)_d\right] \\
&= \frac{1}{|\mathcal{D}_i^c|^2} \mathbb{V}\left[\sum_{(x,y) \in \mathcal{D}_i^c} \nabla \ell(\Phi(x), c)_d\right] \\
&= \frac{1}{|\mathcal{D}_i^c|^2} \sum_{(x,y) \in \mathcal{D}_i^c} \mathbb{V}[\nabla \ell(\Phi(x), c)_d] \\
&= \frac{1}{|\mathcal{D}_i^c|} \mathbb{V}[\nabla \ell(\Phi(x), c)_d]
\end{aligned}$$

Finally, compiling the previous results, we obtain

$$\begin{aligned}
\mathbb{V}(\varepsilon_{id}) &= p_i \text{Cov}(q) p_i^\top \\
&= \sum_{c=1}^C (p_i)_c \text{Cov}(q)_{cc} (p_i)_c \\
&= \sum_{c=1}^C (p_i)_c^2 \frac{1}{|\mathcal{D}_i^c|} \mathbb{V}[\nabla \ell(\Phi(x), c)_d].
\end{aligned}$$

Using $(p_i)_c = |\mathcal{D}_i^c|/|\mathcal{D}_i|$, we further get

$$\begin{aligned}
\mathbb{V}(\varepsilon_{id}) &= \sum_{c=1}^C \frac{|\mathcal{D}_i^c|^2}{|\mathcal{D}_i|^2} \frac{1}{|\mathcal{D}_i^c|} \mathbb{V}[\nabla \ell(\Phi(x), c)_d] \\
&= \frac{1}{|\mathcal{D}_i|^2} \sum_{c=1}^C |\mathcal{D}_i^c| \mathbb{V}[\nabla \ell(\Phi(x), c)_d].
\end{aligned}$$

All worker updates are based on the same number of samples, i.e., $|\mathcal{D}_i| = s$. Additionally, assuming that the variance is independent of the class, i.e., $\mathbb{V}[\nabla \ell(\Phi(x), c)_d] = \sigma_d^2$ for all c , it holds

$$\mathbb{E}[\varepsilon_{id}^2] = \frac{1}{|\mathcal{D}_i|^2} \sum_{c=1}^C |\mathcal{D}_i^c| \sigma_d^2 = \frac{\sigma_d^2}{|\mathcal{D}_i|} = \frac{\sigma_d^2}{s},$$

ending the proof. $\square$

$\square$

Table C.1: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard + GAS) on CIFAR-10 for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers $f \in \{8, 14, 20, 26\}$. The Gain row reports the accuracy improvement of HullGuard paired with GAS robust aggregator over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
10	CWMed	80.01	64.81	24.43	17.48	78.77	77.08	48.94	29.59	78.37	76.74	62.04	52.41	81.03	81.13	81.04	80.20	77.09	75.72	69.98	61.98
	CwTM	77.69	58.69	22.06	17.71	80.21	78.66	64.38	30.33	80.25	78.15	66.16	52.81	81.29	81.29	81.07	80.67	79.49	77.37	71.26	66.63
	RFA	77.68	61.67	23.32	18.97	80.03	78.16	64.14	3.60	80.03	77.95	55.90	48.17	81.51	81.35	81.39	80.40	79.71	78.71	78.34	75.36
	GAS	75.47	45.12	27.31	18.70	80.27	78.21	76.82	67.05	78.26	74.03	71.61	56.73	80.53	81.28	80.92	80.56	80.58	80.25	80.30	80.36
	HullGas	80.01	80.33	80.20	79.89	80.12	80.09	80.20	79.87	80.09	80.42	80.24	80.15	80.20	80.62	80.52	80.46	80.25	80.38	80.38	80.38
	Gain	0.00	15.52	52.89	60.92	-0.15	1.43	3.38	12.82	-0.16	2.27	8.63	23.42	-1.31	-0.73	-0.87	-0.21	-0.33	0.13	0.08	0.02
1	CWMed	79.24	52.87	18.10	13.03	76.41	73.93	34.44	16.95	75.84	74.20	53.87	42.03	79.18	79.84	79.78	78.80	75.16	72.99	65.10	56.11
	CwTM	76.65	42.06	23.58	12.98	79.17	77.58	57.85	25.86	78.17	76.18	60.23	43.96	79.81	79.42	79.20	79.19	78.04	75.92	69.05	56.77
	RFA	76.08	50.00	25.38	14.34	78.68	77.17	53.68	16.89	77.51	76.30	52.16	38.91	79.89	79.91	79.19	79.19	77.70	76.40	75.66	66.06
	GAS	74.73	31.25	16.49	12.58	78.14	77.06	73.94	56.96	76.73	71.39	67.48	50.09	79.73	80.05	79.24	78.74	78.81	78.40	78.79	77.58
	HullGas	78.40	79.05	78.60	78.55	79.01	78.68	78.94	78.11	78.49	78.99	78.39	77.88	79.15	78.70	78.65	78.16	78.63	79.10	79.23	78.34
	Gain	-0.84	26.18	53.22	64.21	-0.16	1.10	5.00	21.15	0.32	2.69	10.91	27.79	-0.74	-1.35	-1.13	-1.03	-0.18	0.70	0.44	0.76
0.1	CWMed	64.38	29.47	11.08	7.87	60.18	50.06	28.52	10.75	61.97	53.87	29.50	25.47	67.39	68.01	65.66	67.22	57.58	52.71	36.75	38.78
	CwTM	60.29	26.04	12.71	16.68	63.86	53.55	32.05	28.49	63.93	58.12	42.36	27.10	66.82	67.84	64.82	66.33	60.01	55.33	38.01	35.98
	RFA	61.19	34.11	18.72	4.30	64.93	62.47	25.98	7.20	64.46	62.64	37.96	26.92	66.97	67.54	64.97	64.94	61.28	62.18	55.71	51.46
	GAS	58.88	15.70	15.04	4.30	65.01	64.03	50.63	22.53	64.94	58.40	50.12	28.00	66.83	68.34	65.54	66.74	64.09	62.88	57.31	59.89
	HullGas	63.71	63.31	61.79	65.94	65.32	65.10	64.11	66.64	65.07	66.06	63.47	66.23	66.25	65.61	65.21	66.12	63.98	65.56	64.40	64.49
	Gain	-0.67	29.20	43.07	49.26	0.31	1.07	13.48	38.15	0.13	3.42	13.35	38.23	-1.14	-2.73	-0.45	-1.10	-0.11	2.68	7.09	4.60

Table C.2: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Dirac (D), Uniform (U)) on CIFAR-10 for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

	ALIE				FOE				Mimic				MinSum				LF			
	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
CWMed	80.01	64.81	24.43	17.48	78.77	77.08	48.94	29.59	78.37	76.74	62.04	52.41	81.03	81.13	81.04	80.20	77.09	75.72	69.98	61.98
CwTM	77.69	58.69	22.06	17.71	80.21	78.66	64.38	30.33	80.25	78.15	66.16	52.81	81.29	81.29	81.07	80.67	79.49	77.37	71.26	66.63
10 RFA	77.68	61.67	23.32	18.97	80.03	78.16	64.14	3.60	80.03	77.95	55.90	48.17	81.51	81.35	81.39	80.40	79.71	78.71	78.34	75.36
GAS	75.47	45.12	27.31	18.70	80.27	78.21	76.82	67.05	78.26	74.03	71.61	56.73	80.53	81.28	80.92	80.56	80.58	80.25	80.30	80.36
HullGuard + D	80.32	78.45	77.90	79.47	80.26	77.96	77.52	80.40	80.49	78.06	77.24	80.63	80.65	78.71	77.65	80.82	80.39	77.98	77.38	80.05
Gain	0.31	13.64	50.59	60.50	-0.01	-0.70	0.70	13.35	0.24	-0.09	5.63	23.90	-0.86	-2.64	-3.74	0.15	-0.19	-2.27	-2.92	-0.31
HullGuard + U	80.35	78.58	77.97	79.36	80.26	77.96	77.52	80.40	80.49	78.06	77.24	80.63	80.65	78.71	77.65	80.82	80.39	77.98	77.38	80.05
Gain	0.34	13.77	50.66	60.39	-0.01	-0.70	0.70	13.35	0.24	-0.09	5.63	23.90	-0.86	-2.64	-3.74	0.15	-0.19	-2.27	-2.92	-0.31
CWMed	79.24	52.87	18.10	13.03	76.41	73.93	34.44	16.95	75.84	74.20	53.87	42.03	79.18	79.84	79.78	78.80	75.16	72.99	65.10	56.11
CwTM	76.65	42.06	23.58	12.98	79.17	77.58	57.85	25.86	78.17	76.18	60.23	43.96	79.81	79.42	79.20	79.19	78.04	75.92	69.05	56.77
1 RFA	76.08	50.00	25.38	14.34	78.68	77.17	53.68	16.89	77.51	76.30	52.16	38.91	79.89	79.91	79.19	79.19	77.70	76.40	75.66	66.06
GAS	74.73	31.25	16.49	12.58	78.14	77.06	73.94	56.96	76.73	71.39	67.48	50.09	79.73	80.05	79.24	78.74	78.81	78.40	78.79	77.58
HullGuard + D	79.04	77.03	75.52	78.05	78.79	75.36	74.34	79.02	78.91	75.61	74.16	78.54	79.47	76.52	75.11	78.45	79.16	75.53	74.68	79.03
Gain	-0.20	24.16	50.14	63.71	-0.38	-2.22	0.40	22.06	0.74	-0.69	6.68	28.45	-0.42	-3.53	-4.67	-0.74	0.35	-2.87	-4.11	1.45
HullGuard + U	79.15	76.25	74.94	77.61	78.79	75.36	74.34	79.02	78.91	75.61	74.16	78.54	79.47	76.52	75.11	78.45	79.16	75.53	74.68	79.03
Gain	-0.09	23.38	49.56	63.27	-0.38	-2.22	0.40	22.06	0.74	-0.69	6.68	28.45	-0.42	-3.53	-4.67	-0.74	0.35	-2.87	-4.11	1.45
CWMed	64.38	29.47	11.08	7.87	60.18	50.06	28.52	10.75	61.97	53.87	29.50	25.47	67.39	68.01	65.66	67.22	57.58	52.71	36.75	38.78
CwTM	60.29	26.04	12.71	16.68	63.86	53.55	32.05	28.49	63.93	58.12	42.36	27.10	66.82	67.84	64.82	66.33	60.01	55.33	38.01	35.98
0.1 RFA	61.19	34.11	18.72	4.30	64.93	62.47	25.98	7.20	64.46	62.64	37.96	26.92	66.97	67.54	64.97	64.94	61.28	62.18	55.71	51.46
GAS	58.88	15.70	15.04	4.30	65.01	64.03	50.63	22.53	64.94	58.40	50.12	28.00	66.83	68.34	65.54	66.74	64.09	62.88	57.31	59.89
HullGuard + D	63.00	65.49	63.41	54.75	65.07	61.03	56.21	65.36	63.82	64.89	57.22	64.86	66.50	67.51	63.41	66.89	60.85	59.74	54.18	62.35
Gain	-1.38	31.38	44.69	38.07	0.06	-3.00	5.58	36.87	-1.12	2.25	7.10	36.86	-0.89	-0.83	-2.25	-0.33	-3.24	-3.14	-3.13	2.46
HullGuard + U	64.20	65.80	62.04	55.36	65.07	61.03	56.21	65.36	63.82	64.89	57.22	64.86	66.50	67.51	63.41	66.89	60.85	59.74	54.18	62.35
Gain	-0.18	31.69	43.32	38.68	0.06	-3.00	5.58	36.87	-1.12	2.25	7.10	36.86	-0.89	-0.83	-2.25	-0.33	-3.24	-3.14	-3.13	2.46

Table C.3: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Random (R), Dirac (D), Uniform (U)) on MNIST for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
10	CWMed	98.74	98.53	96.98	93.72	98.69	98.32	96.17	95.35	98.71	98.50	97.60	97.10	99.01	99.04	98.79	99.03	98.41	97.91	97.40	95.09
	CwTM	98.68	98.48	96.90	91.95	98.66	98.31	96.44	94.44	98.72	98.48	97.78	96.98	98.96	99.04	98.96	99.04	98.45	98.09	97.06	95.33
	RFA	98.75	98.54	97.19	91.57	98.72	98.56	90.49	75.45	98.76	98.61	98.04	97.45	99.02	99.04	98.98	99.08	98.86	98.65	98.77	98.69
	GAS	98.65	98.38	97.76	89.90	98.67	98.36	98.10	96.53	98.64	98.42	98.36	97.27	98.92	99.11	99.06	99.10	98.78	98.81	98.77	98.77
	HullGuard + R	98.69	98.77	98.74	98.76	98.72	98.76	98.68	98.83	98.76	98.77	98.70	98.80	98.78	98.75	98.75	98.86	98.70	98.83	98.68	98.82
	Gain	-0.06	0.23	0.98	5.04	0.00	0.20	0.58	2.30	0.00	0.16	0.34	1.35	-0.24	-0.36	-0.31	-0.24	-0.16	0.02	-0.09	0.05
	HullGuard + D	98.79	98.86	98.68	98.75	98.72	98.76	98.68	98.83	98.76	98.77	98.70	98.80	98.78	98.75	98.75	98.86	98.70	98.83	98.68	98.82
	Gain	0.04	0.32	0.92	5.03	0.00	0.20	0.58	2.30	0.00	0.16	0.34	1.35	-0.24	-0.36	-0.31	-0.24	-0.16	0.02	-0.09	0.05
	HullGuard + U	98.69	98.77	98.74	98.76	98.72	98.76	98.68	98.83	98.76	98.77	98.70	98.80	98.78	98.75	98.75	98.86	98.70	98.83	98.68	98.82
	Gain	-0.06	0.23	0.98	5.04	0.00	0.20	0.58	2.30	0.00	0.16	0.34	1.35	-0.24	-0.36	-0.31	-0.24	-0.16	0.02	-0.09	0.05
1	CWMed	98.68	98.43	94.02	84.86	98.63	98.29	93.81	87.95	98.69	98.38	97.30	95.30	99.10	99.06	98.85	99.00	98.64	98.08	96.81	90.93
	CwTM	98.58	98.27	95.37	74.34	98.66	98.26	94.60	89.35	98.62	98.39	97.81	96.00	99.07	98.98	99.11	98.98	98.70	98.25	97.13	92.27
	RFA	98.73	98.40	95.91	75.08	98.71	98.43	75.20	16.51	98.63	98.64	97.89	96.35	99.08	99.09	99.11	98.93	98.82	98.68	98.64	98.54
	GAS	98.65	98.09	96.54	63.32	98.60	98.07	97.30	87.58	98.51	98.17	98.32	96.54	99.00	98.96	99.03	98.37	98.82	98.50	98.77	98.78
	HullGuard + R	98.80	98.71	98.65	98.63	98.71	98.69	98.58	98.70	98.71	98.78	98.75	98.75	98.82	98.83	98.84	98.86	98.75	98.69	98.67	98.68
	Gain	0.07	0.28	2.11	13.77	0.00	0.26	1.28	9.35	0.02	0.14	0.43	2.21	-0.28	-0.26	-0.27	-0.14	-0.07	0.01	-0.10	-0.10
	HullGuard + D	98.70	98.66	98.78	98.66	98.71	98.69	98.58	98.70	98.71	98.78	98.75	98.75	98.82	98.83	98.84	98.86	98.75	98.69	98.67	98.68
	Gain	-0.03	0.23	2.24	13.80	0.00	0.26	1.28	9.35	0.02	0.14	0.43	2.21	-0.28	-0.26	-0.27	-0.14	-0.07	0.01	-0.10	-0.10
	HullGuard + U	98.80	98.71	98.65	98.63	98.71	98.69	98.58	98.70	98.71	98.78	98.75	98.75	98.82	98.83	98.84	98.86	98.75	98.69	98.67	98.68
	Gain	0.07	0.28	2.11	13.77	0.00	0.26	1.28	9.35	0.02	0.14	0.43	2.21	-0.28	-0.26	-0.27	-0.14	-0.07	0.01	-0.10	-0.10
0.1	CWMed	97.85	92.21	44.80	9.34	96.71	85.36	54.08	27.68	97.25	89.20	62.27	39.76	99.08	99.15	98.67	97.86	97.07	89.16	53.11	45.52
	CwTM	98.00	92.60	30.47	11.37	97.46	88.38	52.56	28.99	97.43	91.69	59.70	52.91	98.95	99.07	98.95	98.75	97.43	93.46	50.55	48.89
	RFA	98.57	98.05	20.95	8.76	98.69	98.12	18.99	4.66	98.62	98.43	94.89	83.34	99.08	99.02	98.98	98.01	98.74	98.59	98.40	97.08
	GAS	98.51	96.61	74.34	8.76	98.21	87.63	80.72	58.06	98.08	96.60	86.73	87.73	98.79	98.97	98.96	98.32	98.51	98.35	98.57	98.56
	HullGuard + R	98.52	98.59	98.49	97.96	98.33	98.10	98.32	98.56	98.63	98.23	98.26	98.37	98.86	98.85	98.84	98.92	98.47	98.18	98.04	98.18
	Gain	-0.05	0.54	24.15	86.59	-0.36	-0.02	17.60	40.50	0.01	-0.20	3.37	10.64	-0.22	-0.30	-0.14	0.17	-0.27	-0.41	-0.53	-0.38
	HullGuard + D	98.37	98.55	98.45	98.04	98.33	98.10	98.32	98.56	98.63	98.23	98.26	98.37	98.86	98.85	98.84	98.92	98.47	98.18	98.04	98.18
	Gain	-0.20	0.50	24.11	86.67	-0.36	-0.02	17.60	40.50	0.01	-0.20	3.37	10.64	-0.22	-0.30	-0.14	0.17	-0.27	-0.41	-0.53	-0.38
	HullGuard + U	98.52	98.59	98.49	97.96	98.33	98.10	98.32	98.56	98.63	98.23	98.26	98.37	98.86	98.85	98.84	98.92	98.47	98.18	98.04	98.18
	Gain	-0.05	0.54	24.15	86.59	-0.36	-0.02	17.60	40.50	0.01	-0.20	3.37	10.64	-0.22	-0.30	-0.14	0.17	-0.27	-0.41	-0.53	-0.38

Table C.4: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Random (R), Dirac (D), Uniform (U)) on Fashion-MNIST for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				LF			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
10	CWMed	86.74	85.79	77.74	67.27	86.13	85.39	78.27	74.68	86.69	85.59	82.74	80.46	88.43	88.75	88.58	88.89	85.36	83.12	80.71	72.55
	CwTM	86.78	85.36	77.42	67.19	86.44	85.58	79.19	75.33	86.85	85.61	82.87	80.27	88.40	88.91	88.59	88.48	86.01	83.89	80.30	74.04
	RFA	86.93	85.64	78.37	68.64	86.44	85.30	68.44	57.40	86.73	86.01	82.81	79.93	88.62	88.87	88.89	88.36	86.78	86.47	86.50	86.20
	GAS	86.17	83.80	80.52	68.20	85.61	84.70	83.60	76.41	85.79	84.62	83.60	80.18	88.27	88.56	88.88	87.76	86.32	86.51	87.05	87.10
	HullGuard + R	86.82	86.72	86.20	86.63	86.80	86.34	85.99	86.63	86.75	86.22	86.07	87.02	87.05	86.60	86.29	87.24	86.98	86.40	86.26	86.67
	Gain	-0.11	0.93	5.68	17.99	0.36	0.76	2.39	10.22	-0.10	0.21	2.47	6.56	-1.57	-2.31	-2.60	-1.65	0.20	-0.11	-0.79	-0.43
	HullGuard + D	86.73	86.52	86.01	86.71	86.80	86.34	85.99	86.63	86.75	86.22	86.07	87.02	87.05	86.60	86.29	87.24	86.98	86.40	86.26	86.67
	Gain	-0.20	0.73	b	18.07	0.36	0.76	2.39	10.22	-0.10	0.21	2.47	6.56	-1.57	-2.31	-2.60	-1.65	0.20	-0.11	-0.79	-0.43
	HullGuard + U	86.82	86.72	86.20	86.63	86.80	86.34	85.99	86.63	86.75	86.22	86.07	87.02	87.05	86.60	86.29	87.24	86.98	86.40	86.26	86.67
	Gain	-0.11	0.93	5.68	17.99	0.36	0.76	2.39	10.22	-0.10	0.21	2.47	6.56	-1.57	-2.31	-2.60	-1.65	0.20	-0.11	-0.79	-0.43
1	CWMed	85.48	82.39	64.55	52.87	84.09	82.38	70.56	57.29	85.15	85.01	79.91	76.29	88.76	88.73	87.84	87.30	83.42	82.39	70.14	56.97
	CwTM	84.87	81.13	70.45	52.58	85.02	82.58	71.28	60.34	85.79	85.08	80.72	77.87	89.05	88.80	86.87	87.63	84.60	83.29	72.42	58.29
	RFA	86.31	83.64	71.44	47.99	85.95	84.12	55.02	32.98	86.33	85.83	80.58	77.12	89.15	88.79	88.69	87.82	86.24	86.04	86.17	83.02
	GAS	85.22	81.07	73.19	25.32	84.14	82.86	82.24	66.62	85.29	83.64	81.77	78.06	87.75	88.24	87.31	87.76	85.29	85.70	86.93	6.70
	HullGuard + R	85.82	85.10	85.49	84.61	85.86	85.07	84.69	85.34	85.83	84.90	84.50	85.11	86.33	86.23	85.60	86.63	85.75	84.50	84.94	85.16
	Gain	-0.49	1.46	12.30	31.74	-0.09	0.95	2.45	18.72	-0.50	-0.93	2.73	7.05	-2.82	-2.57	-3.09	-1.19	-0.49	-1.54	-1.99	2.14
	HullGuard + D	85.72	84.85	85.27	85.10	85.86	85.07	84.69	85.34	85.83	84.90	84.50	85.11	86.33	86.23	85.60	86.63	85.75	84.50	84.94	85.16
	Gain	-0.59	1.21	12.08	32.23	-0.09	0.95	2.45	18.72	-0.50	-0.93	2.73	7.05	-2.82	-2.57	-3.09	-1.19	-0.49	-1.54	-1.99	2.14
	HullGuard + U	85.82	85.10	85.49	84.61	85.86	85.07	84.69	85.34	85.83	84.90	84.50	85.11	86.33	86.23	85.60	86.63	85.75	84.50	84.94	85.16
	Gain	-0.49	1.46	12.30	31.74	-0.09	0.95	2.45	18.72	-0.50	-0.93	2.73	7.05	-2.82	-2.57	-3.09	-1.19	-0.49	-1.54	-1.99	2.14
0.1	CWMed	79.48	67.62	33.43	11.37	74.80	70.70	41.86	17.54	78.08	70.93	52.70	35.25	89.69	90.11	88.87	18.01	79.79	70.67	56.59	47.07
	CwTM	80.11	64.21	36.92	11.37	76.89	71.63	30.20	19.28	79.31	69.49	48.51	39.55	89.82	90.02	89.70	88.90	78.50	71.96	58.08	54.44
	RFA	84.89	79.52	28.49	8.79	85.37	83.05	16.35	9.16	85.75	84.38	72.93	56.46	90.23	90.36	88.55	88.31	86.18	86.73	85.04	77.86
	GAS	81.48	72.51	15.27	8.72	82.00	79.94	73.16	45.16	80.62	77.05	74.16	60.98	87.97	88.01	87.33	18.01	83.60	83.62	86.51	85.85
	HullGuard + R	83.12	84.32	83.93	81.97	83.70	82.06	83.25	85.06	83.48	83.37	83.76	83.20	87.16	87.34	86.71	89.95	82.54	80.66	82.65	82.72
	Gain	-1.77	4.80	47.01	70.60	-1.67	-0.99	10.09	39.90	-2.27	-1.01	9.60	22.22	-3.07	-3.02	-2.99	1.05	-3.64	-6.07	-3.86	-3.13
	HullGuard + D	82.09	84.07	84.19	81.55	83.70	82.06	83.25	85.06	83.48	83.37	83.76	83.20	87.16	87.34	86.71	89.95	82.54	80.66	82.65	82.72
	Gain	-2.80	4.55	47.27	70.18	-1.67	-0.99	10.09	39.90	-2.27	-1.01	9.60	22.22	-3.07	-3.02	-2.99	1.05	-3.64	-6.07	-3.86	-3.13
	HullGuard + U	83.12	84.32	83.93	81.97	83.70	82.06	83.25	85.06	83.48	83.37	83.76	83.20	87.16	87.34	86.71	89.95	82.54	80.66	82.65	82.72
	Gain	-1.77	4.80	47.01	70.60	-1.67	-0.99	10.09	39.90	-2.27	-1.01	9.60	22.22	-3.07	-3.02	-2.99	1.05	-3.64	-6.07	-3.86	-3.13

Table C.5: Mean Accuracy (%) comparison over last 200 training steps between robust aggregators (BKT + CWMed, BKT + CwTM, BKT + RFA, BKT + GAS, HullGuard with byzantine distributions Random (R), Dirac (D), Uniform (U)) on EuroSAT for different concentration parameter $\alpha \in \{10, 1, 0.1\}$ under several Byzantine attacks (ALIE, FOE, Mimic, MinSum, LF) and different numbers of workers (8, 14, 20, 26). The Gain row reports the accuracy improvement of HullGuard over the best competing method for each configuration. A negative gain indicates that HullGuard is being outperformed by another method.

		ALIE				FOE				Mimic				MinSum				L			
		8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26	8	14	20	26
10	CWMed	87.68	81.64	42.79	27.40	88.05	87.59	69.50	59.99	83.13	81.21	70.46	35.71	88.58	84.83	80.23	61.23	85.99	83.89	82.49	75.87
	CwTM	87.42	80.82	60.22	25.67	88.57	87.53	73.33	59.07	86.19	83.41	72.19	37.02	85.99	85.63	80.03	44.44	87.19	86.62	82.97	76.17
	RFA	88.13	82.41	64.61	29.55	88.79	87.17	71.70	23.65	86.82	83.60	70.80	36.58	86.27	84.50	81.13	30.98	88.18	88.08	88.33	87.96
	GAS	84.84	75.70	67.08	27.30	84.58	84.07	85.51	77.11	83.45	75.53	71.70	31.18	85.05	78.01	75.59	10.94	86.26	82.09	87.65	85.04
	HullGuard + R	87.04	86.09	83.34	86.04	87.97	85.41	83.36	87.06	87.88	85.05	83.42	86.48	87.17	85.85	84.57	84.13	87.39	85.88	83.93	87.16
	Gain	-1.09	3.68	16.26	56.49	-0.82	-2.18	-2.15	9.95	1.06	1.45	11.23	49.46	-1.41	0.22	3.44	22.90	-0.79	-2.20	-4.40	-0.80
	HullGuard + D	87.04	86.09	83.34	86.04	87.97	85.41	83.36	87.06	87.88	85.05	83.42	86.48	87.17	85.85	84.57	84.13	87.39	85.88	83.93	87.16
	Gain	-1.09	3.68	16.26	56.49	-0.82	-2.18	-2.15	9.95	1.06	1.45	11.23	49.46	-1.41	0.22	3.44	22.90	-0.79	-2.20	-4.40	-0.80
	HullGuard + U	87.04	86.09	83.34	86.04	87.97	85.41	83.36	87.06	87.88	85.05	83.42	86.48	87.17	85.85	84.57	84.13	87.39	85.88	83.93	87.16
	Gain	-1.09	3.68	16.26	56.49	-0.82	-2.18	-2.15	9.95	1.06	1.45	11.23	49.46	-1.41	0.22	3.44	22.90	-0.79	-2.20	-4.40	-0.80
1	CWMed	84.83	75.57	25.14	10.24	82.61	78.37	57.08	28.37	80.30	77.71	57.58	32.09	83.66	80.22	11.96	11.24	81.58	79.19	73.39	62.92
	CwTM	84.28	71.68	33.18	18.98	84.89	81.08	69.88	42.31	82.38	80.55	62.38	43.45	84.29	82.71	36.45	11.24	81.94	80.28	74.91	65.62
	RFA	85.53	71.21	44.90	17.80	86.09	83.79	56.93	33.88	83.00	81.12	64.11	29.77	82.98	79.84	27.54	11.24	84.19	84.60	81.19	76.53
	GAS	79.39	62.60	53.47	9.96	78.52	78.44	84.07	70.36	77.58	70.92	69.62	33.45	78.64	21.82	11.96	11.36	78.47	76.77	83.52	37.06
	HullGuard + R	84.39	79.85	82.49	81.52	82.21	79.50	79.21	83.92	82.95	79.46	79.44	83.34	85.15	81.76	80.16	82.46	82.18	79.54	76.58	81.06
	Gain	-1.14	4.28	29.02	62.54	-3.88	-4.29	-4.86	13.56	-0.05	-1.66	9.82	39.89	0.86	-0.95	43.71	71.10	-2.01	-5.06	-6.94	4.53
	HullGuard + D	84.39	79.85	82.49	81.52	82.21	79.50	79.21	83.92	82.95	79.46	79.44	83.34	85.15	81.76	80.16	82.46	82.18	79.54	76.58	81.06
	Gain	-1.14	4.28	29.02	62.54	-3.88	-4.29	-4.86	13.56	-0.05	-1.66	9.82	39.89	0.86	-0.95	43.71	71.10	-2.01	-5.06	-6.94	4.53
	HullGuard + U	84.39	79.85	82.49	81.52	82.21	79.50	79.21	83.92	82.95	79.46	79.44	83.34	85.15	81.76	80.16	82.46	82.18	79.54	76.58	81.06
	Gain	-1.14	4.28	29.02	62.54	-3.88	-4.29	-4.86	13.56	-0.05	-1.66	9.82	39.89	0.86	-0.95	43.71	71.10	-2.01	-5.06	-6.94	4.53
0.1	CWMed	60.65	28.80	10.68	15.33	68.52	58.76	31.89	22.90	59.11	58.12	25.41	14.57	72.13	73.24	64.59	15.33	66.92	46.91	35.05	36.38
	CwTM	62.67	26.77	14.79	13.33	60.53	58.62	41.35	20.27	61.55	59.56	27.31	22.01	68.69	66.10	41.50	15.01	61.38	44.37	29.38	32.02
	RFA	60.04	26.95	10.26	4.81	70.33	70.70	40.68	11.58	73.16	73.70	31.85	17.30	74.40	73.00	16.45	15.33	67.76	47.12	46.06	16.26
	GAS	58.20	22.28	16.02	2.21	59.59	64.45	48.87	30.20	62.01	43.06	25.78	14.19	62.44	14.33	16.45	15.33	64.35	62.24	63.43	31.65
	HullGuard + R	67.58	72.64	71.22	52.91	68.54	70.12	64.82	66.26	65.66	67.32	60.94	67.14	69.05	70.49	72.43	70.54	63.20	64.57	56.78	70.04
	Gain	4.91	43.84	55.20	37.58	-1.79	-0.58	15.95	36.06	-7.50	-6.38	29.09	45.13	-5.35	-2.75	7.84	55.21	-4.56	2.33	-6.65	33.66
	HullGuard + D	67.58	72.64	71.22	52.91	68.54	70.12	64.82	66.26	65.66	67.32	60.94	67.14	69.05	70.49	72.43	70.54	63.20	64.57	56.78	70.04
	Gain	4.91	43.84	55.20	37.58	-1.79	-0.58	15.95	36.06	-7.50	-6.38	29.09	45.13	-5.35	-2.75	7.84	55.21	-4.56	2.33	-6.65	33.66
	HullGuard + U	67.58	72.64	71.22	52.91	68.54	70.12	64.82	66.26	65.66	67.32	60.94	67.14	69.05	70.49	72.43	70.54	63.20	64.57	56.78	70.04
	Gain	4.91	43.84	55.20	37.58	-1.79	-0.58	15.95	36.06	-7.50	-6.38	29.09	45.13	-5.35	-2.75	7.84	55.21	-4.56	2.33	-6.65	33.66

Table C.6: Mean training accuracy comparison over the last 200 steps for aggregators NNM + CWMed, NNM + CwTM, NNM + RFA, NNM + GAS and HullGuard, for concentration parameters 10, 1, 0.1, using dataset CIFAR-10, against ALIE attack with byzantine number 8, 14, 20, 29. Settings for HullGuard are the ones described in section 4.1.

			ALIE				
			8	14	22	26	
CIFAR10	10	CWMed	NNM	77.78	59.87	19.66	80.36
		CwTM	NNM	77.74	60.29	26.38	80.09
		RFA	NNM	78.28	59.05	25.47	73.90
		GAS	NNM	77.72	59.76	23.99	80.71
		HullGuard		80.35	78.58	77.97	79.36
	1	CWMed	NNM	76.78	50.82	15.56	28.05
		CwTM	NNM	76.14	52.13	20.11	21.96
		RFA	NNM	76.64	51.68	18.20	18.92
		GAS	NNM	76.71	50.67	19.30	76.89
		HullGuard		79.15	76.25	74.94	77.61
	0.1	CWMed	NNM	61.31	31.75	9.44	25.26
		CwTM	NNM	61.13	30.15	12.05	27.50
		RFA	NNM	61.64	31.89	11.90	24.45
		GAS	NNM	60.89	30.97	13.27	47.23
		HullGuard		64.20	65.80	62.04	55.36

FOLIO ADMINISTRATIF

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NOM : **ERBANI**

DATE de SOUTENANCE : 17/07/2026

Prénoms : **Johan**

TITRE : **Understanding Machine Learning Vectors: From Confusion Matrices to Byzantine-Robust Federated Learning**

NATURE : **Doctorat**

Numéro d'ordre :

École Doctorale : **InfoMaths**

Spécialité : **Informatique**

RÉSUMÉ :

Ce travail étudie la classification dans des cadres complexes à partir de vecteurs issus de l'apprentissage automatique, en particulier les sorties de modèles et les gradients. Il se divise en deux parties : la première porte sur les matrices de confusion, afin de caractériser le comportement des modèles à partir de leurs prédictions ; la seconde concerne l'apprentissage fédéré robuste, où l'objectif est de défendre un système distribué à partir des mises à jour locales. La matrice de confusion est un outil essentiel pour analyser les erreurs d'un modèle au niveau des classes. Contrairement aux métriques scalaires telles que l'accuracy, elle fournit une description détaillée du comportement du modèle à partir de paires étiquette-prédiction. Cependant, dans les cadres hétérogènes, où les distributions d'étiquettes sont déséquilibrées, ses entrées sont biaisées et peuvent masquer les similarités réelles entre classes. De plus, les cadres multi-labels et à soft-labels capturent mieux la complexité des données réelles, mais la généralisation de la matrice de confusion à ces contextes reste non triviale.

Plusieurs généralisations ont ainsi été proposées. Elles sont généralement analysées sur des jeux de données de référence, soit pour interpréter le comportement des modèles, soit pour être comparées qualitativement à d'autres formulations. Toutefois, ces comparaisons manquent d'un critère quantitatif clair, ce qui rend difficile le choix d'une méthode en pratique. De plus, aucune approche existante ne fournit une formulation unifiée pour les cadres mono-label, multi-labels et à soft-labels.

Nos principales contributions dans ce domaine sont : (i) proposer une nouvelle normalisation de la matrice de confusion afin de mieux retrouver les similarités inter-classes dans les contextes hétérogènes ; (ii) établir des liens entre normalisation, importance sampling et représentations latentes des classes ; (iii) introduire une généralisation fondée sur le transport optimal, couvrant les cadres multi-labels et à soft-labels tout en unifiant plusieurs approches existantes ; (iv) proposer un cadre expérimental pour évaluer si ces extensions retrouvent correctement la confusion réelle du modèle.

La seconde partie porte sur l'apprentissage fédéré robuste. Dans ce cadre, plusieurs clients entraînent collectivement un modèle sans partager leurs données. À chaque round, chaque client calcule une mise à jour locale, par exemple un gradient, que le serveur agrège avant de renvoyer le modèle actualisé aux clients.

Des clients défectueux ou malveillants, dits byzantins, peuvent perturber l'apprentissage. Les méthodes robustes cherchent souvent à écarter les gradients atypiques, en supposant que les gradients byzantins s'écartent davantage des gradients honnêtes que les gradients honnêtes ne s'écartent entre eux. Cependant, l'hétérogénéité des données, notamment le déséquilibre de labels entre clients, peut aussi désaligner les mises à jour honnêtes. Ces dernières peuvent alors être perçues comme des valeurs aberrantes, ce qui affaiblit les stratégies de défense classiques.

Nos principales contributions dans ce domaine sont : (i) introduire une défense légère, fondée sur l'échantillonnage local, ne nécessitant pas de données de validation ; (ii) proposer un filtre byzantin sans données de validation, reposant sur l'observation que les mises à jour honnêtes se situent dans l'enveloppe convexe des gradients propres à chaque classe.

MOTS-CLÉS : **Matrice de confusion, Normalisation, Multi-label, Federated Learning, Byzantine Robustness, Label Skew**

Laboratoire(s) de recherche : **Laboratoire d'InfoRmatique en Image et Systèmes d'information (LIRIS)**

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Président du Jury : ???

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